

A Compact Paravector Form of Space-Time Algebra

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Abstract—A notation-light paravector presentation is developed for the eight-real-dimensional algebra commonly called the real Pauli algebra or the three-dimensional real Clifford algebra. The complex structure is supplied by a central pseudoscalar, so every algebra element is written as one complex scalar plus one complex three-vector. The determinant quadratic form, spatial rotations, Lorentz boosts, relativistic kinematics, Maxwell electrodynamics, Klein–Gordon and Dirac factorizations, and the Pauli limit are derived within this single product calculus. The construction is presented as a self-contained reformulation of established flat-space physics, not as a proposal for new dynamics. Particular care is given to the distinct transformation laws for events, fields, and wave functions and to the projected one-column carrier required for one Dirac field.

Index Terms—space-time algebra, geometric algebra, Lorentz transformations, Maxwell electrodynamics, Dirac equation, Pauli equation.

Introduction

A compact paravector language is developed for several standard parts of flat-space physics. Three anticommuting unit elements σ_n are taken as generators. An eight-real-dimensional associative algebra is spanned by their products. In established terminology, this algebra is called the real Pauli algebra or the three-dimensional real Clifford algebra. A faithful representation is supplied by two-by-two complex matrices, and the same information is carried by the even products in the usual flat space-time algebra. These names are provided only as context; every symbol and operation used in the calculations is defined below.

By use of the central pseudoscalar $i = \sigma_1\sigma_2\sigma_3$, with $i^2 = -1$, every algebra element is written uniquely as

$$Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}, \quad S \in \mathbb{C}, \quad \mathbf{V} \in \mathbb{C}^3.$$

Thus one complex scalar part and one complex three-vector part are assigned to each element. Multiplication is always taken to be the product defined from the sigma basis, not componentwise multiplication of those parts. The calculations are kept short without suppression of their physical content by retaining visible scalar/vector and real/imaginary pieces.

The presentation is organized by three involutions: sigma conjugation $(\cdot)^*$, the Hermitian operation $(\cdot)^H$, and their composition $(\cdot)^{*H}$. Intrinsically, sigma-product order is reversed by H ; in the two-by-two matrix realization it is conjugate transpose. The operation H is used in the positive norm, whereas $ZZ^{*H} = S^2 - \mathbf{V}^2$. This product is defined below as the determinant; on real events it is the Minkowski interval. Rotations, boosts, relativistic kinematics, Maxwell theory, and relativistic wave equations are then obtained from the same calculus.

The physical results derived here are established results expressed in deliberately economical notation. No new interaction or change to the underlying physics is proposed; the common algebraic carrier of familiar constructions is instead exhibited explicitly. The presentation is informed by the STA and paravector literature of Hestenes, Baylis, and Doran–Lasenby [12–14], with historical roots in Hamilton’s quaternions and Clifford’s geometric algebra [1, 3].

Ordinary three-vectors are denoted by bold symbols. In natural units, the reduced Planck constant and the speed of light are set equal to 1. Rationalized Heaviside–Lorentz electromagnetic units and metric signature $(+, -, -, -)$ are used throughout. The product formula for scalar coefficients is extended directly to complex-valued functions; when differential operators occur, their displayed order and the Leibniz rule are part of the definition. In an approximation, terms of order x^n or smaller in the stated limiting parameter are denoted by $O(x^n)$.

Reading Guide

1. The principal definitions and results are collected on the Equation Reference Sheet on the next page in the order of their construction.
2. The identities, sigma basis, three involutions, complex paravectors, extraction functions, vector and paravector products, positive norm, adjugate, determinant, inverse, projectors, elementary functions, transform laws, quaternion correspondence, and matrix realizations are defined in Section 1.
3. Real paravectors are specialized to events, and intervals, velocity, acceleration, energy, momentum, action, and the nonrelativistic limit are developed in Section 2.
4. The event law is applied to Lorentz boosts, length contraction, momentum, and boost–rotor compositions in Section 3, where it is contrasted with the field similarity law.
5. Spatial and paravector derivatives, complex fields, the wave operator, and differentiation rules for noncommuting products are defined in Section 4.
6. Fields and potentials, all four Maxwell equations, boost laws, gauge freedom, field and potential wave equations, the Lorentz force, energy and flux, and the full linear stress-energy map are developed in Section 5.
7. Scalar waves, the positive- and negative-frequency split, minimal electromagnetic coupling, the ordered Klein–Gordon factorization, the projected two-block Dirac field, its conserved pairing and gauge law, the squared Dirac system, and the controlled Pauli and helicity reductions are developed in Section 6.
8. The established results are separated from explicitly labeled open exten-

sions and points requiring sharper foundations in Sections 7 and 8.

9. Independent symbolic checks and standard correspondences are recorded

in Appendix A; historical context is provided by the references without any change of notation.

Equation Reference Sheet (Summary)

Natural units: reduced Planck constant = 1, speed of light = 1. Signature: (+, −, −, −). All space at fixed time is covered by unmarked spatial integrals with measure $dr_1 dr_2 dr_3$.

Name	Formula	Ref.
Custom functions, maps, and identities		
Identity elements	$Z + 0 = Z, \quad Z1 = 1Z = Z$	(1)
Complex paravector	$Z = (a + ib) + (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma} = S + \mathbf{V} \cdot \boldsymbol{\sigma} = X + iY$	(17)–(18)
Real and img functions	$\text{real}(Z) = X, \quad \text{img}(Z) = Y$	(19), (20)
Scalar and vector functions	$\text{scalar}(Z) = S, \quad \text{vector}(Z) = \mathbf{V}$	(21), (22)
Axis functions	$\text{par}(\mathbf{A}) = (\mathbf{A} \cdot \mathbf{u})\mathbf{u}, \quad \text{prp}(\mathbf{A}) = \mathbf{A} - \text{par}(\mathbf{A}), \quad \mathbf{u}^2 = 1$	(34), (35)
Order functions	$\text{com}(Z, Z') = ZZ' - Z'Z, \quad \text{acom}(Z, Z') = ZZ' + Z'Z$	(38), (39)
Adjugate function	$\text{adj}(Z) = Z^{*\text{H}} = S - \mathbf{V} \cdot \boldsymbol{\sigma}, \quad Z^{\text{H}} = \text{adj}(Z)^* = \text{adj}(Z^*)$	(49), (50)
Determinant function	$\det(Z) = \text{adj}(Z)Z = Z \text{adj}(Z) = S^2 - \mathbf{V}^2$	(52), (53)
Inverse	$Z^{-1} = \text{adj}(Z) / \det(Z), \quad \det(Z) \neq 0$	(60)
Matrix functions	$\mathbf{I} = \{1, 0, 0, 1\}, \quad \text{tr}(\{a, b, c, d\}) = a + d, \quad \text{diag}(A, B) = \{A, 0, 0, B\},$ $\{a, b, c, d\}^T = \{a, c, b, d\}, \quad \mathbf{M}^c = \text{entrywise conjugate}, \quad \mathbf{M}^{cT} = (\mathbf{M}^c)^T$	(102)–(108)
Auxiliary block function	$\mathbf{W}(Z) = \{0, Z; \text{adj}(Z), 0\}$	(120)
Algebraic progression		
Pseudoscalar	$i = \sigma_1 \sigma_2 \sigma_3, \quad i^2 = -1, \quad i\sigma_n = \sigma_n i$	(2)
Sigma product	$\sigma_n \sigma_m = \delta_{nm} + \sum_{k=1}^3 \epsilon_{nmk} i \sigma_k$	(3)
Sigma/H operations	$XY^* = X^* Y^*, \quad X Y^{\text{H}} = Y^{\text{H}} X^{\text{H}}, \quad X Y^{*\text{H}} = Y^{\text{H}} X^{*\text{H}}$	(6)
Involution actions	$Z^* = (a - ib) + (-\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \quad Z^{\text{H}} = (a - ib) + (\mathbf{A} - i\mathbf{B}) \cdot \boldsymbol{\sigma}, \quad Z^{*\text{H}} = S - \mathbf{V} \cdot \boldsymbol{\sigma}$	(24)–(26)
Matrix bridge: $* \neq c$	$\mathbf{T} = \{0, 1, -1, 0\}, \quad \mathbf{M}^* = \mathbf{T} \mathbf{M}^c \mathbf{T}^{-1}, \quad \mathbf{M}^{\text{H}} = \mathbf{M}^{cT}, \quad \mathbf{M}^{*\text{H}} = \text{adj}(\mathbf{M}) = \mathbf{T} \mathbf{M}^T \mathbf{T}^{-1}$	(115), (116)
Vector product	$(\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) = \mathbf{A} \cdot \mathbf{B} + i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}$	(36)
Paravector product	$ZZ' = SS' + \mathbf{V} \cdot \mathbf{V}' + (S\mathbf{V}' + S'\mathbf{V} + i\mathbf{V} \times \mathbf{V}') \cdot \boldsymbol{\sigma}$	(37)
Positive norms	$ S ^2 = SS^*, \quad \ \mathbf{V}\ ^2 = \mathbf{V} \cdot \mathbf{V}^*, \quad \ Z\ ^2 = \text{scalar}(ZZ^{\text{H}}) = S ^2 + \ \mathbf{V}\ ^2$	(45)–(47)
Determinant products	$\det(ZZ') = \det(Z)\det(Z'), \quad \det(\text{adj}(Z)) = \det(Z)$	(57), (56)
Axis projectors	$\Pi(\pm) = \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \quad Z(\pm) = \Pi(\pm)Z, \quad Z = Z(+) + Z(-)$	(65), (66)
Ladder function	$a(\pm) = \frac{1}{2}(\sigma_1 \pm i\sigma_2), \quad a(\pm)^2 = 0, \quad \text{com}(\sigma_3, a(\pm)) = \pm 2a(\pm)$	(68)
Functional calculus	$f(\mathbf{u} \cdot \boldsymbol{\sigma}) = f(+)\Pi(+)+f(-)\Pi(-)$	(67)
Vector exponential	$\exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}) = \cosh(\theta) + \sinh(\theta)\mathbf{u} \cdot \boldsymbol{\sigma}$	(75)
Bivector exponential	$\exp(\theta \mathbf{u} \cdot i\boldsymbol{\sigma}) = \cos(\theta) + \sin(\theta)\mathbf{u} \cdot i\boldsymbol{\sigma}$	(78)
Paravector logarithm	$\log(Y) = \log(\lambda(Y, +))\Pi(+) + \log(\lambda(Y, -))\Pi(-)$	(77)
Transforms and kinematics		
Similarity transport	$TZ' = ZT \quad \text{implies} \quad Z' = T^{-1}ZT$	(80), (81)
Event congruence	$X' = T^{*\text{H}}XT^*, \quad \det(T) = 1$	(84)
Active rotor	$R = \exp(\theta \mathbf{u} \cdot i\boldsymbol{\sigma}/2), \quad X' = R^{\text{H}}XR$	(79), (88)
Passive event boost	$L = \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2), \quad X' = L^*XL^*$	(76), (89)
Passive field boost	$F' = L^*FL$	(154)
Space-time interval	$ds^2 = \det(dX) = dt^2 - d\mathbf{r}^2$	(123)
Velocity and momentum	$U = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}), \quad mU = E + \mathbf{P} \cdot \boldsymbol{\sigma}$	(124), (128)
Acceleration	$A = dU/ds, \quad \text{scalar}(AU^*) = 0$	(129), (130)
Differential and electromagnetic fields		
Spatial derivatives	$\partial_r = \{\partial_1, \partial_2, \partial_3\}, \quad \partial_n = \partial/\partial r_n, \quad \partial_r^2 = \partial_1^2 + \partial_2^2 + \partial_3^2$	(138), (139)
Paravector derivatives	$\partial = \partial_t + \boldsymbol{\sigma} \cdot \partial_r, \quad \partial^* = \partial_t - \boldsymbol{\sigma} \cdot \partial_r$	(140), (141)
Wave operator	$\square = \partial\partial^* = \partial^*\partial = \partial_t^2 - \partial^2$	(142)
Ordered product rule	$\partial(ZZ') = (\partial Z)Z' + Z(\partial Z') + \sum_n \text{com}(\sigma_n, Z) \partial_n Z'$	(144)
Field and potential	$F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \quad \Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$	(146), (147)
Potential gradient	$\partial\Phi = S + F^*, \quad S = \partial_t V + \partial_r \cdot \mathbf{A}$	(148), (149)
Maxwell equation	$\partial F = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}$	(151)
Maxwell components	$\partial_r \cdot \mathbf{E} = \rho, \quad \partial_r \cdot \mathbf{B} = 0, \quad \partial_r \times \mathbf{B} = \mathbf{J} + \partial_t \mathbf{E}, \quad \partial_r \times \mathbf{E} = -\partial_t \mathbf{B}$	(152), (153)
Gauge functions	$\Phi(\lambda) = \Phi + \partial^* \lambda, \quad F(\lambda) = F$	(158), (161)
Lorentz force	$m dU/ds = q \text{real}(FU)$	(162)
Energy and flux	$T = \frac{1}{2} F F^{\text{H}} = \frac{1}{2} (\mathbf{E}^2 + \mathbf{B}^2) + (\mathbf{E} \times \mathbf{B}) \cdot \boldsymbol{\sigma}$	(164)
Full stress-energy function	$T(a) = \frac{1}{2} F a F^{\text{H}}, \quad a \in \mathbb{R} + \mathbb{R}^3, \quad T = T(1)$	(169)
Relativistic wave equations		
Frequency split	$\Omega = \sqrt{m^2 - \partial^2}, \quad (\partial_t \pm i\Omega)\psi(\pm) = 0$	(173), (175)
Covariant energy/momentum	$E = i\partial_t - qV, \quad \mathbf{P} = -i\partial_r - q\mathbf{A}$	(176), (177)
Klein–Gordon	$(-E^2 + \mathbf{P}^2 + m^2)\psi = 0$	(178)
Ordered factors	$(E + \boldsymbol{\sigma} \cdot \mathbf{P})(E - \boldsymbol{\sigma} \cdot \mathbf{P}) = E^2 - \mathbf{P}^2 - iqF$	(185)
Dirac system	$(\mathbf{W}(i\partial - q\Phi^*) - m\mathbf{I})\Psi = 0$	(195)
Fixed-column carrier	$\Pi(\text{column}) = \frac{1}{2}(1 + \sigma_3), \quad \Psi_n = \Psi_n \Pi(\text{column}), \quad n \in \{1, 2\}$	(191), (192)
Dirac carrier transform	$\Psi'_1(X') = T^{\text{H}}\Psi_1(X), \quad \Psi'_2(X') = T^{-1}\Psi_2(X)$	(198), (199)
Squared blocks	$(-E^2 + \mathbf{P}^2 + m^2 + iqF^*)\Psi_1 = 0, \quad (-E^2 + \mathbf{P}^2 + m^2 + iqF)\Psi_2 = 0$	(201)
Dirac current	$J(\Psi) = 2\{\Psi_2 \Psi_2^{\text{H}} + (\Psi_1 \Psi_1^{\text{H}})^*\} = \rho(\Psi) + \mathbf{j}(\Psi) \cdot \boldsymbol{\sigma}, \quad \partial_t \rho(\Psi) + \partial_r \cdot \mathbf{j}(\Psi) = 0$	(206)
Klein–Gordon pairing	$\rho(\phi, \psi) = \{\phi^* E \psi + (E\phi)^* \psi\}/(2m), \quad \langle \phi \psi \rangle = \iiint \rho(\phi, \psi) dr_1 dr_2 dr_3$	(187), (190)
	$\langle \phi \psi \rangle = 2 \iiint \text{scalar}(\phi^{\text{H}} \psi) dr_1 dr_2 dr_3,$	
Projected-wave pairings	$\langle \Psi(\text{left}) \Psi(\text{right}) \rangle = 2 \iiint \sum_{n=1}^2 \text{scalar}(\Psi_n(\text{left})^{\text{H}} \Psi_n(\text{right})) dr_1 dr_2 dr_3$	(209), (210)
Leading Pauli limit	$E\phi(\text{large}) = (\mathbf{P}^2 - q\mathbf{B} \cdot \boldsymbol{\sigma})\phi(\text{large})/(2m) + O(m^{-2})$	(220)
Pauli spin split	$\phi(\text{large}, \pm) = \Pi(\pm)\phi(\text{large}), \quad i\partial_t \phi(\text{large}, \pm) = \{\mathbf{P}^2 \mp q\ \mathbf{B}\ \}\phi(\text{large}, \pm)/(2m) + qV\phi(\text{large}, \pm)$	(65), (227), (229)
Free energy/axis split	$\mathcal{H}\{\Psi_1; \Psi_2\} = \{-(\mathbf{p} \cdot \boldsymbol{\sigma})\Psi_1 + m\Psi_2; m\Psi_1 + (\mathbf{p} \cdot \boldsymbol{\sigma})\Psi_2\}, \quad E(\mathbf{p}) = \sqrt{m^2 + \mathbf{p}^2},$ $\Psi(e) = \frac{1}{2}\{\Psi + e\mathcal{H}\Psi/E(\mathbf{p})\}, \quad \Psi(e, s) = \{\Pi(s)\Psi_1(e); \Pi(s)\Psi_2(e)\}$	(230)–(238)

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1 Algebraic Foundations

The algebra is fixed before physical meanings are assigned to particular elements. The product is associative and unital. All coefficients in the master product formula are taken to be central complex scalars unless an ordered differential-operator calculation is stated explicitly.

1.1 Basis and pseudoscalar

The element 0 is the additive identity and 1 is the multiplicative identity:

$$Z + 0 = 0 + Z = Z, \quad Z1 = 1Z = Z. \quad (1)$$

The two identities must not be confused: an inverse is defined by a product equal to 1, whereas the absence of a multiplicative inverse is indicated by a vanishing determinant.

For $n \in \{1, 2, 3\}$, the algebra is generated over $\mathbb{R}$ by three basis elements σ_n . Scalars, vectors, two-generator elements, and a three-generator pseudoscalar are generated by their products. The resulting eight-real-dimensional algebra is commonly called the real Pauli algebra or the three-dimensional real Clifford algebra. The basis relations are

$$\begin{aligned} \sigma_n^2 &= 1, & \sigma_n \sigma_m &= -\sigma_m \sigma_n \quad (n \neq m), \\ i &:= \sigma_1 \sigma_2 \sigma_3, & i^2 &= -1. \end{aligned} \quad (2)$$

The pseudoscalar classification is established by $i\sigma_n = \sigma_n i$, and the scalar -1 is obtained by squaring i :

$$\begin{aligned} (\sigma_1 \sigma_2 \sigma_3)^2 &= \sigma_1 \sigma_2 (\sigma_3 \sigma_1) \sigma_2 \sigma_3, \\ &= -\sigma_1 (\sigma_2 \sigma_1) (\sigma_3 \sigma_2) \sigma_3, \\ &= -\sigma_1^2 \sigma_2^2 \sigma_3^2, \\ &= -1. \end{aligned}$$

With the Kronecker delta and Levi-Civita symbol,

$$\sigma_n \sigma_m = \delta_{nm} + \sum_{k=1}^3 \epsilon_{nmk} i \sigma_k, \quad (3)$$

where

$$\begin{aligned} \delta_{nm} &:= \begin{cases} 1, & n = m, \\ 0, & n \neq m, \end{cases} \\ \epsilon_{nmk} &:= \begin{cases} +1, & (n, m, k) \text{ even permutation of } (1, 2, 3), \\ -1, & (n, m, k) \text{ odd permutation of } (1, 2, 3), \\ 0, & \text{otherwise.} \end{cases} \end{aligned}$$

1.2 The Conjugate and Hermitian Operators

The sigma-conjugate operator is defined by

$$(\cdot)^* := \text{flip the signs of } \sigma_n. \quad (4)$$

The Hermitian operator is defined by

$$(\cdot)^H := \text{reverse the order of multiplied } \sigma_n. \quad (5)$$

The name Hermitian is justified by the matrix realization developed in Section 1.13.1: there H is represented by ordinary entrywise complex conjugation followed by transpose. Within the intrinsic algebra it is defined without matrices, solely by reversing sigma-product order. The star is a real-linear automorphism and H is a real-linear anti-automorphism. Because the central pseudoscalar i is sent to $-i$ by each operation, both are antilinear with respect to the internal complex structure. Their composition is a complex-linear anti-automorphism. In operator form,

$$\begin{aligned} (XY)^* &= X^* Y^*, & (XY)^H &= Y^H X^H, \\ X^{*H} &= (X^H)^* = (X^*)^H, & (X^*)^* &= (X^H)^H = X. \end{aligned} \quad (6)$$

When applied to the generators and the pseudoscalar, these rules are reduced to

$$\sigma_n^* = -\sigma_n, \quad \sigma_n^H = \sigma_n, \quad \sigma_n^{*H} = -\sigma_n, \quad (7)$$

$$i^* = -i, \quad i^H = -i, \quad i^{*H} = i. \quad (8)$$

For a two-generator bivector pair with $n \neq m$,

$$\begin{aligned} (\sigma_n \sigma_m)^* &= \sigma_n^* \sigma_m^* = (-\sigma_n)(-\sigma_m) = \sigma_n \sigma_m, \\ (\sigma_n \sigma_m)^H &= \sigma_m^H \sigma_n^H = \sigma_m \sigma_n = -\sigma_n \sigma_m, \\ (\sigma_n \sigma_m)^{*H} &= (\sigma_m \sigma_n)^* = \sigma_m^* \sigma_n^* \\ &= (-\sigma_m)(-\sigma_n) = \sigma_m \sigma_n = -\sigma_n \sigma_m. \end{aligned} \quad (9)$$

The same sign pattern on the sigma-number-two elements is recorded in Eq. (9). These are the standard intrinsic involutions of the real Pauli algebra. No assertion that every Z^H is itself a Hermitian object is implied by the symbol H ; rather, a represented matrix is Hermitian precisely when $Z^H = Z$. In the conventional even space-time interpretation, the role of reversing space-time products is assigned to the combined operation $(\cdot)^{*H}$, rather than to $(\cdot)^H$ alone [12, 14]. No additional notation or rule is introduced by this contextual statement.

The sigma number of a reduced basis product is defined as the number of sigma generators that remain after simplification. Sigma numbers 0, 1, 2, and 3 are assigned respectively to 1, σ_n , $\sigma_n \sigma_m$ with $n \neq m$, and i . Any unreduced product with sigma number larger than 3 is simplified to one of these four cases. The basis elements and operator signs are summarized by sigma number in Tables 1 and 2.

Table 1. Sigma basis elements by sigma number.

sigma number	representative basis elements
0 (scalar)	1
1 (vector)	$\sigma_1, \sigma_2, \sigma_3$
2 (bivector)	$\sigma_2 \sigma_3, \sigma_3 \sigma_1, \sigma_1 \sigma_2$
3 (pseudoscalar)	$\sigma_1 \sigma_2 \sigma_3$

Table 2. Operator signs by sigma number.

sigma number	$(\cdot)^*$	$(\cdot)^H$	$(\cdot)^{*H}$
0 (scalar)	+	+	+
1 (vector)	−	+	−
2 (bivector)	+	−	−
3 (pseudoscalar)	−	−	+

1.3 Generic Complex Paravector

The three generators are collected into the sigma-vector

$$\sigma := \{\sigma_1, \sigma_2, \sigma_3\}, \quad (10)$$

which is a vector of algebraic objects rather than a vector of scalar components. The associated bivector triple is obtained by multiplication by the pseudoscalar:

$$i\sigma := \{i\sigma_1, i\sigma_2, i\sigma_3\} = \{\sigma_2\sigma_3, \sigma_3\sigma_1, \sigma_1\sigma_2\}, \quad (11)$$

because $i = \sigma_1\sigma_2\sigma_3$ and each product $i\sigma_n$ is reduced to the complementary ordered two-generator product. The canonical sigma-number-two basis built from the generators is therefore recorded in Eq. (11). The descriptor $\mathbb{R} + \mathbb{R}^3$ is used for one real scalar plus one real three-vector, and $\mathbb{C} + \mathbb{C}^3$ is used for one complex scalar plus one complex three-vector. This decomposition is recorded by the plus sign in a type descriptor; no second multiplication rule is introduced. With this basis fixed, for real scalars $a, b \in \mathbb{R}$ and real vectors $\mathbf{A}, \mathbf{B} \in \mathbb{R}^3$, with $\mathbf{A} \cdot \sigma = A_1\sigma_1 + A_2\sigma_2 + A_3\sigma_3$, the real paravectors $X, Y \in \mathbb{R} + \mathbb{R}^3$ are defined by

$$X := a + \mathbf{A} \cdot \sigma, \quad (12)$$

$$Y := b + \mathbf{B} \cdot \sigma. \quad (13)$$

The complex scalar and vector components are defined first:

$$S := a + ib \in \mathbb{C}. \quad (14)$$

$$\mathbf{V} := \mathbf{A} + i\mathbf{B} \in \mathbb{C}^3. \quad (15)$$

Every algebra element can then be defined as a complex paravector $Z \in \mathbb{C} + \mathbb{C}^3$, written as a real paravector plus an imaginary paravector:

$$Z := X + iY. \quad (16)$$

The equivalent Cartesian and scalar–vector forms are

$$Z = (a + ib) + (\mathbf{A} + i\mathbf{B}) \cdot \sigma. \quad (17)$$

$$Z = S + \mathbf{V} \cdot \sigma. \quad (18)$$

On coefficient objects, the induced actions of either basic involution are $S^* = S^H = a - ib$ and $\mathbf{V}^* = \mathbf{V}^H = \mathbf{A} - i\mathbf{B}$. When the full algebra term $\mathbf{V} \cdot \sigma$ is acted upon by an involution, its action on the sigma basis is included as well, according to Eqs. (6) and (7). The real function is defined by

$$\text{real}(Z) := X = \frac{Z + Z^H}{2}. \quad (19)$$

The imaginary function is defined by

$$\text{img}(Z) := Y = \frac{Z - Z^H}{2i}. \quad (20)$$

The scalar function is defined by

$$\text{scalar}(Z) := S = \frac{Z + Z^{*H}}{2}. \quad (21)$$

The vector function is defined by

$$\text{vector}(Z) := \mathbf{V}. \quad (22)$$

The algebra element carried by that vector is recovered as

$$\text{vector}(Z) \cdot \sigma = \frac{Z - Z^{*H}}{2}. \quad (23)$$

The actions of the conjugate and Hermitian operators on the generic complex paravector are

$$Z^* = (a - ib) + (-\mathbf{A} + i\mathbf{B}) \cdot \sigma, \quad (24)$$

$$Z^H = X - iY = (a - ib) + (\mathbf{A} - i\mathbf{B}) \cdot \sigma, \quad (25)$$

$$Z^{*H} = (a + ib) - (\mathbf{A} + i\mathbf{B}) \cdot \sigma. \quad (26)$$

For the real paravector X from Eq. (12), these are reduced to

$$X^* = a - \mathbf{A} \cdot \sigma, \quad (27)$$

$$X^H = X = a + \mathbf{A} \cdot \sigma, \quad (28)$$

$$X^{*H} = a - \mathbf{A} \cdot \sigma = X^*. \quad (29)$$

Equations (24)–(26) are the generic formulas used below. In particular, conjugation of the complex coefficients is already encoded by H ; no overbar notation is needed. The matrix-only superscript c , introduced later, is therefore not a fourth intrinsic operation. Equations (27)–(29) are the real specialization. The real and complex paravector pieces and extractors used throughout the later algebra are fixed by Eqs. (12)–(22). The type assignments are recorded in Table 3, the recovery formulas for the extractor definitions are recorded in Table 4, and the same split is recorded in components in Table 5.

Table 3. Type assignments for the generic complex paravector notation.

Type name	Underlying set	Example
real scalar	$\mathbb{R}$	a, b
real vector	$\mathbb{R}^3$	$\mathbf{A}, \mathbf{B}$
complex scalar	$\mathbb{C}$	S
complex vector	$\mathbb{C}^3$	$\mathbf{V}$
real paravector	$\mathbb{R} + \mathbb{R}^3$	X, Y
complex paravector	$\mathbb{C} + \mathbb{C}^3$	Z

Table 4. Components of Z .

scalar:	$S = (Z + Z^{*H})/2$
vector:	$\mathbf{V} \cdot \sigma = (Z - Z^{*H})/2$
real:	$X = (Z + Z^H)/2$
img:	$Y = (Z - Z^H)/(2i)$

Table 5. Real and imaginary scalar/vector parts of Z .

real scalar:	$a = (S + S^H)/2$ $= (X + X^*)/2$
img scalar:	$b = (S - S^H)/(2i)$ $= (Y + Y^*)/2$
real vector:	$\mathbf{A} \cdot \boldsymbol{\sigma} = (\mathbf{V} + \mathbf{V}^H) \cdot \boldsymbol{\sigma}/2$ $= (X - X^*)/2$
img vector:	$\mathbf{B} \cdot \boldsymbol{\sigma} = (\mathbf{V} - \mathbf{V}^H) \cdot \boldsymbol{\sigma}/(2i)$ $= (Y - Y^*)/2$

1.4 Dot products and cross products

For real vectors $\mathbf{A}, \mathbf{B} \in \mathbb{R}^3$, the dot product is defined by

$$\begin{aligned} \mathbf{A} \cdot \mathbf{B} &:= \sum_{n=1}^3 \sum_{m=1}^3 \delta_{nm} A_n B_m \\ &= A_1 B_1 + A_2 B_2 + A_3 B_3. \end{aligned} \quad (30)$$

The square of a vector is defined by

$$\mathbf{A}^2 := \mathbf{A} \cdot \mathbf{A}. \quad (31)$$

For the cross product, the Levi-Civita form is

$$(\mathbf{A} \times \mathbf{B})_k := \sum_{n=1}^3 \sum_{m=1}^3 \epsilon_{nmk} A_n B_m, \quad (32)$$

from which the vector is obtained:

$$\mathbf{A} \times \mathbf{B} = \begin{Bmatrix} A_2 B_3 - A_3 B_2, \\ A_3 B_1 - A_1 B_3, \\ A_1 B_2 - A_2 B_1 \end{Bmatrix}. \quad (33)$$

Dot products are commutative, whereas cross products are anti-symmetric:

$$\begin{aligned} \mathbf{A} \cdot \mathbf{B} &= +\mathbf{B} \cdot \mathbf{A}, \\ \mathbf{A} \times \mathbf{B} &= -\mathbf{B} \times \mathbf{A}. \end{aligned}$$

For a fixed unit vector $\mathbf{u}$, where $\mathbf{u}^2 = 1$, the parallel component function is defined by

$$\text{par}(\mathbf{A}) := (\mathbf{A} \cdot \mathbf{u})\mathbf{u}. \quad (34)$$

The perpendicular component function is defined by

$$\text{prp}(\mathbf{A}) := \mathbf{A} - \text{par}(\mathbf{A}). \quad (35)$$

With Eqs. (2) and (3),

$$\begin{aligned} (\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) &= \sum_{n=1}^3 \sum_{m=1}^3 A_n B_m \sigma_n \sigma_m \\ &= \sum_{n=1}^3 A_n B_n \end{aligned}$$

$$\begin{aligned} &+ \sum_{n < m} (A_n B_m - A_m B_n) \sigma_n \sigma_m \\ &= A_1 B_1 + A_2 B_2 + A_3 B_3 \\ &+ (A_2 B_3 - A_3 B_2) i \sigma_1 \\ &- (A_1 B_3 - A_3 B_1) i \sigma_2 \\ &+ (A_1 B_2 - A_2 B_1) i \sigma_3. \end{aligned}$$

This can be efficiently expressed in terms of the dot product and cross product as

$$(\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) = \mathbf{A} \cdot \mathbf{B} + i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (36)$$

The same product is recorded in component form in Table 6. The dot and cross products are extended complex-bilinearly to $\mathbb{C}^3$. Formula (36) may also be used for commuting scalar-valued functions. A separate ordered calculation is required for differential-operator coefficients because everything to their right is differentiated according to the Leibniz rule.

Only the antisymmetric term is flipped by the reversed order:

$$\begin{aligned} (\mathbf{B} \cdot \boldsymbol{\sigma})(\mathbf{A} \cdot \boldsymbol{\sigma}) &= \mathbf{A} \cdot \mathbf{B} - i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \\ (\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) + (\mathbf{B} \cdot \boldsymbol{\sigma})(\mathbf{A} \cdot \boldsymbol{\sigma}) &= 2\mathbf{A} \cdot \mathbf{B}, \\ (\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) - (\mathbf{B} \cdot \boldsymbol{\sigma})(\mathbf{A} \cdot \boldsymbol{\sigma}) &= 2i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}. \end{aligned}$$

The following identities are obtained immediately from the same product:

$$\begin{aligned} (\mathbf{A} \cdot \boldsymbol{\sigma})^2 &= \mathbf{A}^2, \\ (\mathbf{A} \cdot \boldsymbol{\sigma})^{2n} &= (\mathbf{A}^2)^n, \\ (\mathbf{A} \cdot \boldsymbol{\sigma})^{2n+1} &= (\mathbf{A}^2)^n (\mathbf{A} \cdot \boldsymbol{\sigma}), \\ (\mathbf{A} \cdot \boldsymbol{\sigma})\boldsymbol{\sigma} &= \mathbf{A} + i\mathbf{A} \times \boldsymbol{\sigma}. \end{aligned}$$

These are the basic sigma-product identities from which the later paravector formulas are built.

1.5 Products

With the scalar/vector split for real vectors fixed by Eq. (36), the fully complex product is obtained as the master formula from which the simpler cases are recovered by restriction.

For Z from Eq. (18) and $Z' = S' + \mathbf{V}' \cdot \boldsymbol{\sigma}$ defined analogously to Eqs. (18) and (15),

$$ZZ' = (SS' + \mathbf{V} \cdot \mathbf{V}') + (S\mathbf{V}' + S'\mathbf{V} + i(\mathbf{V} \times \mathbf{V}')) \cdot \boldsymbol{\sigma}. \quad (37)$$

The generic complex paravector and its master multiplication rule are fixed by Eqs. (18) and (37): pure scalar, real vector, real paravector, and mixed products are all obtained by setting the unused parts to zero.

1.5.1 Order functions

For arbitrary algebra elements A and B , the commutator function is defined by

$$\text{com}(A, B) := AB - BA. \quad (38)$$

The anticommutator function is defined independently by

$$\text{acom}(A, B) := AB + BA. \quad (39)$$

1.5.2 Products of real scalars, vectors, and paravectors

For the real paravectors X and Y from Eqs. (12) and (13), Eq. (36) gives

$$\begin{aligned} XY &= (a + \mathbf{A} \cdot \boldsymbol{\sigma})(b + \mathbf{B} \cdot \boldsymbol{\sigma}) \\ &= (ab + \mathbf{A} \cdot \mathbf{B}) + (a\mathbf{B} + b\mathbf{A} + i(\mathbf{A} \times \mathbf{B})) \cdot \boldsymbol{\sigma}, \\ YX &= (ab + \mathbf{A} \cdot \mathbf{B}) + (a\mathbf{B} + b\mathbf{A} - i(\mathbf{A} \times \mathbf{B})) \cdot \boldsymbol{\sigma} \\ &= (XY)^H. \end{aligned}$$

The same product is given in component form in Table 7. The scalar part is symmetric, while the sign of the imaginary-vector part is changed by the order.

Two immediate consequences are

$$\begin{aligned} \text{com}(X, Y) &= 2i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \\ XY \in \mathbb{R} + \mathbb{R}^3 &\quad \text{if and only if } \mathbf{A} \times \mathbf{B} = \mathbf{0}. \end{aligned}$$

Thus exact commutation of two real paravectors is obtained precisely when their vector parts are parallel.

The antisymmetric term is removed automatically by setting $Y = X$:

$$X^2 = a^2 + \mathbf{A}^2 + 2a\mathbf{A} \cdot \boldsymbol{\sigma}.$$

If $X^2 = b + \mathbf{B} \cdot \boldsymbol{\sigma}$, the following relations are obtained by matching scalar and vector parts:

$$\begin{aligned} b &= a^2 + \mathbf{A}^2, \\ \mathbf{B} &= 2a\mathbf{A}, \end{aligned}$$

hence

$$\begin{aligned} a^2 &= \frac{1}{2} \left(b \pm \sqrt{b^2 - \mathbf{B}^2} \right), \\ \mathbf{A}^2 &= b - a^2. \end{aligned}$$

These equations are necessary, but the direction equation $\mathbf{B} = 2a\mathbf{A}$ must also be retained. When $\mathbf{B} \neq \mathbf{0}$, real paravector roots exist precisely when $b \geq \|\mathbf{B}\|$. For either choice of the inner sign, either sign of the resulting nonzero a may be chosen, with

$$\mathbf{A} = \frac{\mathbf{B}}{2a}.$$

When $\mathbf{B} = \mathbf{0}$, a real root requires $b \geq 0$; the scalar roots are $X = \pm\sqrt{b}$, and the degenerate pure-vector roots satisfy $a = 0$ and $\mathbf{A}^2 = b$. Thus the displayed quadratic relation is a branch condition, rather than by itself a complete square-root formula.

1.5.3 Products of complex paravectors

The master product law (37) is the complex extension of the real formulas. When Z is written as in Eqs. (18)–(17) and $Z' = X' + iY' = S' + \mathbf{V}' \cdot \boldsymbol{\sigma}$, every specialization is obtained by setting the unused scalar or vector parts to zero. The quantities S, S' and the components of $\mathbf{V}, \mathbf{V}'$ are central complex scalars. Scalar coefficients may therefore be passed through one another and through the sigma generators. The full component decomposition

of ZZ' is given in Table 8, that of Z^2 is given in Table 9, and the order-sensitive products built from the conjugate and Hermitian operations are recorded in Tables 10–15.

Within this finite-dimensional scalar-coefficient algebra, the scalar part is cyclic:

$$\text{scalar}(ZZ') = \text{scalar}(Z'Z).$$

In particular, if $Z' = MZ^{*H}$, then

$$\begin{aligned} \text{scalar}(ZMZ^{*H}) &= \text{scalar}(MZ^{*H}Z) \\ &= \text{scalar}(M) (Z^{*H}Z), \end{aligned}$$

because $Z^{*H}Z$ is scalar.

The order-sensitive variants are divided into three algebraic types:

ZZ^* is a real scalar plus a complex vector,

ZZ^H is a real scalar plus a real vector,

ZZ^{*H} is a complex scalar.

The positive norms, determinant forms, inverses, and transformation formulas are supplied by those three cases.

If a component is instead a differential or another noncommuting operator, the ordered expansion must be used before any simplification: in general $\mathbf{V}S' \neq S'\mathbf{V}$, Leibniz terms may occur, and cyclicity of the algebraic scalar projection alone does not justify cycling the operator coefficients. The displayed ordering is therefore required in operator applications, which cannot be obtained by automatic substitution into the central-coefficient formulas.

The basic product decompositions and order-dependent component expansions are collected in the following tables.

Table 6. Components of $(\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma})$.

real scalar:	$\mathbf{A} \cdot \mathbf{B}$
img scalar:	0
real vector:	$\mathbf{0}$
img vector:	$\mathbf{A} \times \mathbf{B}$

Table 7. Components of XY .

real scalar:	$ab + \mathbf{A} \cdot \mathbf{B}$
img scalar:	0
real vector:	$a\mathbf{B} + b\mathbf{A}$
img vector:	$\mathbf{A} \times \mathbf{B}$

The full component expansions for the generic product ZZ' and the special case Z^2 are collected in Tables 8 and 9.

Table 8. Components of ZZ' .

real scalar:	$aa' - bb' + \mathbf{A} \cdot \mathbf{A}' - \mathbf{B} \cdot \mathbf{B}'$
img scalar:	$ab' + ba' + \mathbf{A} \cdot \mathbf{B}' + \mathbf{B} \cdot \mathbf{A}'$
real vector:	$a\mathbf{A}' - b\mathbf{B}' + a'\mathbf{A} - b'\mathbf{B} - \mathbf{A} \times \mathbf{B}' - \mathbf{B} \times \mathbf{A}'$
img vector:	$a\mathbf{B}' + b\mathbf{A}' + a'\mathbf{B} + b'\mathbf{A} + \mathbf{A} \times \mathbf{A}' - \mathbf{B} \times \mathbf{B}'$

Table 9. Components of Z^2 .

real scalar:	$a^2 - b^2 + \mathbf{A}^2 - \mathbf{B}^2$
img scalar:	$2ab + 2\mathbf{A} \cdot \mathbf{B}$
real vector:	$2(a\mathbf{A} - b\mathbf{B})$
img vector:	$2(a\mathbf{B} + b\mathbf{A})$

The order dependence of ZZ^* and Z^*Z is made explicit in Tables 10 and 11.

Table 10. Components of ZZ^* .

real scalar:	$a^2 + b^2 - \mathbf{A}^2 - \mathbf{B}^2$	
img scalar:	0	
real vector:	$-2\mathbf{A} \times \mathbf{B}$	← differs
img vector:	$2(a\mathbf{B} - b\mathbf{A})$	

Table 11. Components of Z^*Z .

real scalar:	$a^2 + b^2 - \mathbf{A}^2 - \mathbf{B}^2$	
img scalar:	0	
real vector:	$2\mathbf{A} \times \mathbf{B}$	← differs
img vector:	$2(a\mathbf{B} - b\mathbf{A})$	

Products involving the Hermitian operation are treated analogously in Tables 12 and 13.

Table 12. Components of ZZ^H .

real scalar:	$a^2 + b^2 + \mathbf{A}^2 + \mathbf{B}^2$	
img scalar:	0	
real vector:	$2(a\mathbf{A} + b\mathbf{B} + \mathbf{A} \times \mathbf{B})$	← differs
img vector:	$\mathbf{0}$	

Table 13. Components of $Z^H Z$.

real scalar:	$a^2 + b^2 + \mathbf{A}^2 + \mathbf{B}^2$	
img scalar:	0	
real vector:	$2(a\mathbf{A} + b\mathbf{B} - \mathbf{A} \times \mathbf{B})$	← differs
img vector:	$\mathbf{0}$	

The collapse of products formed with the star- H operation to purely scalar and imaginary-scalar parts is shown in Tables 14 and 15.

Table 14. Components of ZZ^{*H} .

real scalar:	$a^2 - b^2 - \mathbf{A}^2 + \mathbf{B}^2$
img scalar:	$2ab - 2\mathbf{A} \cdot \mathbf{B}$
real vector:	$\mathbf{0}$
img vector:	$\mathbf{0}$

Table 15. Components of $Z^{*H} Z$.

real scalar:	$a^2 - b^2 - \mathbf{A}^2 + \mathbf{B}^2$
img scalar:	$2ab - 2\mathbf{A} \cdot \mathbf{B}$
real vector:	$\mathbf{0}$
img vector:	$\mathbf{0}$

1.5.4 Order dependence and commutators

The order-dependent part of a product is represented by the commutator function defined in Eq. (38). For generic complex paravectors $Z, Z' \in \mathbb{C} + \mathbb{C}^3$,

$$\text{com}(Z, Z') = 2i(\mathbf{V} \times \mathbf{V}') \cdot \boldsymbol{\sigma} \quad (40)$$

Equation (40) is the compact commutator law. In real and imaginary vector components, the following expansion is obtained:

$$\begin{aligned} \text{com}(Z, Z') = & -2(\mathbf{A} \times \mathbf{B}' + \mathbf{B} \times \mathbf{A}') \cdot \boldsymbol{\sigma} \\ & + 2i(\mathbf{A} \times \mathbf{A}' - \mathbf{B} \times \mathbf{B}') \cdot \boldsymbol{\sigma}. \end{aligned}$$

The real-paravector commutator

$$\text{com}(X, Y) = 2i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}$$

is the corresponding specialization. In every case the scalar terms are canceled, so the cross-product part of the multiplication law is isolated by the commutator. The following identity is obtained by contracting the basis commutators with a second vector:

$$\sum_{n=1}^3 B_n \text{com}(\mathbf{A} \cdot \boldsymbol{\sigma}, \sigma_n) = \text{com}(\mathbf{A} \cdot \boldsymbol{\sigma}, \mathbf{B} \cdot \boldsymbol{\sigma}) = 2i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (41)$$

This is the component bridge between the vector cross-product law and the commutator correction in the paravector Leibniz rule.

The symmetric companion is the anticommutator from Eq. (39). For pure vectors,

$$\begin{aligned} \text{acom}(\mathbf{A} \cdot \boldsymbol{\sigma}, \mathbf{B} \cdot \boldsymbol{\sigma}) &= (\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) \\ &\quad + (\mathbf{B} \cdot \boldsymbol{\sigma})(\mathbf{A} \cdot \boldsymbol{\sigma}) \\ &= 2\mathbf{A} \cdot \mathbf{B}. \end{aligned} \quad (42)$$

Thus the symmetric dot-product part is isolated by $\text{acom}(\cdot, \cdot)$, just as the antisymmetric cross-product part is isolated by $\text{com}(\cdot, \cdot)$.

More generally, for generic complex paravectors $Z, Z' \in \mathbb{C} + \mathbb{C}^3$, Eq. (39) gives

$$\text{acom}(Z, Z') = ZZ' + Z'Z. \quad (43)$$

The anticommutator is therefore

$$\text{acom}(Z, Z') = 2(SS' + \mathbf{V} \cdot \mathbf{V}') + 2(S\mathbf{V}' + S'\mathbf{V}) \cdot \boldsymbol{\sigma}. \quad (44)$$

The cross-product contribution to the anticommutator is canceled identically. Only the symmetric scalar and vector couplings are retained.

The same order dependence is already present in the sigma basis itself. Commutation of the pseudoscalar with the generators is expressed by

$$\begin{aligned} \text{com}(i, \sigma_n) &= 0, \\ i\boldsymbol{\sigma} &= \boldsymbol{\sigma}i = \{\sigma_2\sigma_3, \sigma_3\sigma_1, \sigma_1\sigma_2\}, \end{aligned}$$

while the following relation is satisfied by distinct sigma generators:

$$\text{com}(\sigma_n, \sigma_m) = 2\sigma_n\sigma_m = 2 \sum_{k=1}^3 \epsilon_{nmk} i\sigma_k \quad (n \neq m).$$

Thus the oriented sigma-number-two basis is exactly the commutator closure of the sigma-number-one generators.

The same product types are displayed in full component form in Tables 12–15. Their positive-norm and determinant-form consequences are fixed by Eqs. (47), (55), and (53); no additional quadratic definitions are required.

1.6 Positive Norm, Adjugate, and Determinant

Two distinct quadratic constructions are used: a positive coefficient norm and a generally complex determinant. For a complex scalar $S = a + ib \in \mathbb{C}$, the scalar norm is defined by

$$\begin{aligned} |S| &:= \sqrt{SS^*}. \\ |S|^2 &= SS^* \\ &= a^2 + b^2. \end{aligned} \quad (45)$$

If $S \in \mathbb{R}$, then the ordinary absolute value $|S| = \sqrt{S^2}$ is obtained from Eq. (45). For a complex vector $\mathbf{V} = \mathbf{A} + i\mathbf{B} \in \mathbb{C}^3$, the vector norm is defined by

$$\begin{aligned} \|\mathbf{V}\| &:= \sqrt{\mathbf{V} \cdot \mathbf{V}^*}. \\ \|\mathbf{V}\|^2 &= \mathbf{V} \cdot \mathbf{V}^* \\ &= \mathbf{A}^2 + \mathbf{B}^2. \end{aligned} \quad (46)$$

The quadratic product $\mathbf{V}^2 = \mathbf{V} \cdot \mathbf{V}$ is not the same object as the positive norm square $\mathbf{V} \cdot \mathbf{V}^*$. For the complex paravector Z from Eqs. (18)–(17), the paravector norm is defined by

$$\|Z\| := \sqrt{|S|^2 + \|\mathbf{V}\|^2}. \quad (47)$$

Its square is reduced directly to the coefficient norm:

$$\begin{aligned} \|Z\|^2 &= |S|^2 + \|\mathbf{V}\|^2 \\ &= a^2 + b^2 + \mathbf{A}^2 + \mathbf{B}^2 \\ &= \text{scalar}(ZZ^H) = \text{scalar}(Z^H Z). \end{aligned} \quad (48)$$

This is a nonnegative real norm: it is the sum of the squared magnitudes of the scalar and vector coefficients.

The determinant is obtained through construction of the multiplicative inverse. For $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$, the adjugate is obtained by reversing the sign of the vector part:

$$\text{adj}(Z) := Z^{*H} = S - \mathbf{V} \cdot \boldsymbol{\sigma}. \quad (49)$$

By commutativity of the two basic involutions and involutivity of each, the following useful consistency chain is also obtained from Eq. (49):

$$Z^H = (Z^{*H})^* = (Z^*)^{*H} = \text{adj}(Z)^* = \text{adj}(Z^*). \quad (50)$$

In the compact superscript notation, this is $Z^H = (Z^{*H})^* = (Z^*)^{*H} = \text{adj}(Z)^* = \text{adj}(Z^*)$. Because S is central and $(\mathbf{V} \cdot \boldsymbol{\sigma})^2 = \mathbf{V}^2$, the vector terms are canceled by multiplication by the adjugate:

$$\begin{aligned} Z \text{adj}(Z) &= (S + \mathbf{V} \cdot \boldsymbol{\sigma})(S - \mathbf{V} \cdot \boldsymbol{\sigma}) \\ &= S^2 - \mathbf{V}^2 \\ &= \text{adj}(Z) Z. \end{aligned} \quad (51)$$

The remaining central scalar is exactly the divisor required to normalize the product to the multiplicative identity 1. The determinant function is therefore defined by

$$\det(Z) := \text{adj}(Z) Z. \quad (52)$$

Equation (51) shows that the order of the two factors does not matter. The determinant is reduced to

$$\det(Z) = Z \text{adj}(Z) = S^2 - \mathbf{V}^2. \quad (53)$$

In real and imaginary components, the same result is

$$\det(Z) = (a^2 - b^2 - \mathbf{A}^2 + \mathbf{B}^2) + 2i(ab - \mathbf{A} \cdot \mathbf{B}). \quad (54)$$

In the two-by-two complex matrix realization, $\text{adj}(Z)$ is the ordinary matrix adjugate and Eq. (51) is represented by $\det(Z) \mathbf{I}$. The designation of the scalar in Eq. (53) as the determinant is justified by this representation. No square root is included in its definition.

For the real paravector X from Eq. (12), $X^{*H} = X^*$, and the determinant is restricted to the Minkowski quadratic form

$$\begin{aligned} \det(X) &= a^2 - \mathbf{A}^2 \\ &= XX^* = X^* X. \end{aligned} \quad (55)$$

Product order is reversed by the adjugate, so for complex paravectors Z and Z' ,

$$\begin{aligned} \text{adj}(\text{adj}(Z)) &= Z, \\ \text{adj}(ZZ') &= \text{adj}(Z') \text{adj}(Z), \\ \det(\text{adj}(Z)) &= \det(Z). \end{aligned} \quad (56)$$

Consequently, multiplicativity of the determinant is obtained without any commutativity assumption:

$$\begin{aligned} \det(ZZ') &= ZZ' \text{adj}(Z') \text{adj}(Z) \\ &= Z \det(Z') \text{adj}(Z) \\ &= \det(Z') \det(Z) \\ &= \det(Z) \det(Z') = \det(Z'Z). \end{aligned} \quad (57)$$

Under pure phase multiplication, the positive norm and determinant are transformed as

$$\begin{aligned} \|\exp(i\theta)Z\|^2 &= \|Z\|^2, \\ \det(\exp(i\theta)Z) &= \exp(2i\theta) \det(Z). \end{aligned}$$

More generally, under complex scalar multiplication by $r \exp(i\theta)$ with $r \in \mathbb{R}$, the two quadratic constructions are scaled differently:

$$\|r \exp(i\theta)Z\|^2 = r^2 \|Z\|^2, \quad (58)$$

$$\det(r \exp(i\theta)Z) = r^2 \exp(2i\theta) \det(Z). \quad (59)$$

The positive norm is phase-insensitive, while the phase square is retained by the determinant. The term “Minkowski metric” below is reserved for the real-paravector restriction (55) and its polarization.

1.7 Inverse formulas

The inverse construction is completed by Eqs. (51) and (53). For $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$, when $\det(Z) \neq 0$, the inverse is defined by

$$Z^{-1} := \frac{\text{adj}(Z)}{\det(Z)}. \quad (60)$$

The equivalent explicit forms are

$$Z^{-1} = \frac{Z^{*\text{H}}}{ZZ^{*\text{H}}} = \frac{Z^{*\text{H}}}{Z^{*\text{H}}Z} = \frac{S - \mathbf{V} \cdot \boldsymbol{\sigma}}{S^2 - \mathbf{V}^2}. \quad (61)$$

If $\det(Z) = 0$, the inverse does not exist.

By differentiation of the identity $Z^{-1}Z = 1$, the following result is obtained:

$$d(Z^{-1}) = -Z^{-1}(dZ)Z^{-1}. \quad (62)$$

For $Z' = S' + \mathbf{V}' \cdot \boldsymbol{\sigma}$, if $\mathbf{V} \times \mathbf{V}' = \mathbf{0}$, then

$$Z^{-1}Z' = Z'Z^{-1}. \quad (63)$$

1.8 Supporting Algebraic Derivations

The explicit sigma-basis and operator calculations, together with one generic map decomposition, are recorded in the supporting derivations without repetition of the product development already collected above.

1.8.1 Basis-Element Classification

The four sigma numbers and corresponding basis-element types are listed in Table 1. The signs acquired under the conjugate, Hermitian, and combined star- H operations are listed in Table 2.

1.8.2 Complex Paravector-Valued Maps

A generic complex paravector-valued map is formed from real scalar and vector functions without imposing any physical interpretation:

$$Z(t, \mathbf{r}) := (a(t, \mathbf{r}) + i b(t, \mathbf{r})) + (\mathbf{A}(t, \mathbf{r}) + i \mathbf{B}(t, \mathbf{r})) \cdot \boldsymbol{\sigma}. \quad (64)$$

Its domain and the coefficient-function types are

$$\begin{aligned} a, b &: \mathbb{R} \times \mathbb{R}^3 \rightarrow \mathbb{R}, \\ \mathbf{A}, \mathbf{B} &: \mathbb{R} \times \mathbb{R}^3 \rightarrow \mathbb{R}^3, \\ Z &: \mathbb{R} \times \mathbb{R}^3 \rightarrow \mathbb{C} + \mathbb{C}^3. \end{aligned}$$

1.9 Projector Algebra

Projectors are introduced next so that the later function formulas and ladder constructions can be organized. Once Eqs. (3) and (36) are fixed, the cleanest route from simple unit-direction elements to more general paravector decompositions is supplied by the projector calculus.

If $\mathbf{A} \in \mathbb{R}^3$ is written as $\mathbf{A} = \theta \mathbf{u}$ with $\theta = \|\mathbf{A}\|$, $\mathbf{u} \in \mathbb{R}^3$, and $\mathbf{u}^2 = 1$, the idempotent function is defined by

$$\Pi(\pm) := \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}). \quad (65)$$

The corresponding left-projected part of any algebra element Z is defined by

$$Z(\pm) := \Pi(\pm)Z. \quad (66)$$

The identities $\Pi(\pm)^2 = \Pi(\pm)$, $\Pi(+)\Pi(-) = 0$, $\Pi(+)+\Pi(-) = 1$, $Z = Z(+)+Z(-)$, and $(\mathbf{u} \cdot \boldsymbol{\sigma})\Pi(\pm) = \pm\Pi(\pm)$ are satisfied. Algebraically, the parts aligned and anti-aligned with the direction $\mathbf{u}$ are isolated by $\Pi(+)$ and $\Pi(-)$. Their role is purely structural: an object is split into two parts adapted to a chosen axis.

For a scalar-valued function $f : \{\pm 1\} \rightarrow \mathbb{C}$,

$$f(\mathbf{u} \cdot \boldsymbol{\sigma}) := f(+)\Pi(+)+f(-)\Pi(-). \quad (67)$$

A compact projector-based formula for roots, inverses, exponentials, logarithms, and related functions is thereby obtained. Once the minimal polynomial of an element of this Pauli algebra is known, polynomial and suitably analytic functions can be reduced within the algebraic framework, with derivative data required at a defective root. The detailed formulas are presented first as projector and eigenvalue identities and then as ladder operations associated with a chosen axis.

About the z axis, the ladder elements are defined by

$$a(\pm) := \frac{1}{2}(\sigma_1 \pm i\sigma_2). \quad (68)$$

The associated commutator, nilpotence, and product identities are

$$\begin{aligned} \text{com}(\sigma_3, a(\pm)) &= \pm 2a(\pm), & a(\pm)^2 &= 0, \\ a(+)a(-) &= \Pi(+), & a(-)a(+) &= \Pi(-). \end{aligned} \quad (69)$$

The nilpotent combinations of σ_1 and σ_2 by which the two projector parts selected by Eq. (65) are exchanged are recorded in Eq. (68). The projector eigenvalue formulas, mixed projector products, and ladder identities built from Eqs. (65), (67), and (68) are collected next.

1.9.1 Functions of $\mathbf{u} \cdot \boldsymbol{\sigma}$

The two-root decomposition of $\mathbf{u} \cdot \boldsymbol{\sigma}$, whose two values are $+1$ and -1 , is given by the projector pair and identities in Eq. (65) together with the functional-calculus rule in Eq. (67). The function is determined by specified scalar values at those roots:

$$\begin{aligned} f(\mathbf{u} \cdot \boldsymbol{\sigma}) &= f(+)\Pi(+)+f(-)\Pi(-) \\ &= \frac{1}{2}(f(+)+f(-)) + \frac{1}{2}(f(+)-f(-))\mathbf{u} \cdot \boldsymbol{\sigma}. \end{aligned}$$

These are the basic projector formulas for a unit-direction element. The square root, logarithm branch data, rational inverse, Cayley transform, and the two exponential families use the branch-sign function defined by

$$\epsilon(\pm) := \text{an independently chosen member of } \{+1, -1\}. \quad (70)$$

The resulting identities are

$$\begin{aligned}
\sqrt{\mathbf{u} \cdot \boldsymbol{\sigma}} &= \epsilon(+)\Pi(+) + \epsilon(-)i\Pi(-), \\
\log(\mathbf{u} \cdot \boldsymbol{\sigma}) &= 2i\pi(n\Pi(+) + (\tfrac{1}{2} + m)\Pi(-)), \\
(z + \mathbf{u} \cdot \boldsymbol{\sigma})^{-1} &= \frac{1}{z+1}\Pi(+) + \frac{1}{z-1}\Pi(-) \\
&= \frac{z - \mathbf{u} \cdot \boldsymbol{\sigma}}{z^2 - 1}, \\
\frac{i - \mathbf{u} \cdot \boldsymbol{\sigma}}{i + \mathbf{u} \cdot \boldsymbol{\sigma}} &= i\mathbf{u} \cdot \boldsymbol{\sigma}, \\
\exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}) &= \cosh(\theta) + \sinh(\theta) \mathbf{u} \cdot \boldsymbol{\sigma}, \\
\exp(i\theta \mathbf{u} \cdot \boldsymbol{\sigma}) &= \cos(\theta) + i\sin(\theta) \mathbf{u} \cdot \boldsymbol{\sigma},
\end{aligned}$$

where m and n are integers indexing independent logarithm branches, and the inverse formula requires $z \neq \pm 1$. The two values from Eq. (70) are independent; the conventional scalar branches $\sqrt{+1} = 1$ and $\sqrt{-1} = i$ are obtained when $\epsilon(+)=\epsilon(-)=+1$. These identities are obtained by applying the chosen scalar value separately on the two projector ranges. The corresponding domains for real and complex paravectors are stated with the exponential and logarithm formulas.

1.9.2 Functions of X

The scalar, vector, and unit-direction data are specified by

$$\begin{aligned}
X &= t + \mathbf{r} \cdot \boldsymbol{\sigma}, \\
t &= \text{scalar}(X) \in \mathbb{R}, \\
\mathbf{r} &= \text{vector}(X) \in \mathbb{R}^3, \\
\mathbf{u} &:= \mathbf{r}/\|\mathbf{r}\| \quad \text{when } \mathbf{r} \neq 0.
\end{aligned}$$

If $\mathbf{r} \neq 0$, its decomposition is $\mathbf{r} = \|\mathbf{r}\|\mathbf{u}$, with $\mathbf{u}^2 = 1$. The two eigenvalue values are defined by

$$\lambda(X, \pm) := t \pm \|\mathbf{r}\|. \quad (71)$$

With the eigenvalue function fixed by Eq. (71), the minimal polynomial of X is

$$\begin{aligned}
X^2 - 2tX + (t^2 - \mathbf{r}^2) &= 0, \\
0 &= (X - \lambda(X, -))(X - \lambda(X, +)) \\
&= X^2 - \lambda(X, -)X \\
&\quad - X\lambda(X, +) + \lambda(X, -)\lambda(X, +).
\end{aligned}$$

With

$$a := \lambda(X, +), \quad b := \lambda(X, -),$$

the projector function from Eq. (65) gives the corresponding decomposition:

$$\begin{aligned}
\Pi(\pm) &= \tfrac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \\
X &= a\Pi(+) + b\Pi(-), \\
\Pi(\pm)X &= \lambda(X, \pm)\Pi(\pm).
\end{aligned}$$

If f is defined at the two eigenvalues, then

$$\begin{aligned}
f(X) &= f(a)\Pi(+) + f(b)\Pi(-) \\
&= \tfrac{1}{2}(f(a) + f(b)) + \tfrac{1}{2}(f(a) - f(b))\mathbf{u} \cdot \boldsymbol{\sigma}.
\end{aligned}$$

The scalar and signed axial coefficients of $f(X)$ are defined by

$$\begin{aligned}
t' &= \tfrac{1}{2}(f(a) + f(b)), \\
\rho' &:= \tfrac{1}{2}(f(a) - f(b)), \\
f(X) &= t' + \rho'\mathbf{u} \cdot \boldsymbol{\sigma}.
\end{aligned}$$

For complex-valued f , both t' and ρ' may be complex. Even when they are real, ρ' is signed; the magnitude of the vector part is $|\rho'|$, not ρ' itself. The projector summary obtained from Eqs. (71) and (65) may therefore be recorded as

$$\begin{aligned}
\lambda(X, \pm) &= t \pm \|\mathbf{r}\|, \\
\mathbf{u} &= \mathbf{r}/\|\mathbf{r}\|, \\
\Pi(\pm) &= \tfrac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \\
\Pi(\pm)X &= \lambda(X, \pm)\Pi(\pm), \\
X &= \lambda(X, +)\Pi(+) + \lambda(X, -)\Pi(-), \\
f(X) &= f(\lambda(X, +))\Pi(+) + f(\lambda(X, -))\Pi(-).
\end{aligned}$$

The standard examples are

$$\begin{aligned}
\exp(X) &= \exp(t) \left(\cosh(\|\mathbf{r}\|) + \sinh(\|\mathbf{r}\|) \mathbf{u} \cdot \boldsymbol{\sigma} \right), \\
\sqrt{X} &= \epsilon(+)\sqrt{a}\Pi(+) + \epsilon(-)\sqrt{b}\Pi(-), \\
&\quad \epsilon(+), \epsilon(-) \in \{+1, -1\}, \\
\log(X) &= \log(a)\Pi(+) + \log(b)\Pi(-), \\
X^{-1} &= \frac{\Pi(+)}{a} + \frac{\Pi(-)}{b} \\
&= \frac{\text{adj}(X)}{\det(X)} = \frac{X^*}{t^2 - \mathbf{r}^2}, \\
X^n &= a^n\Pi(+) + b^n\Pi(-) \\
&= \tfrac{1}{2}(a^n + b^n) + \tfrac{1}{2}(a^n - b^n)\mathbf{u} \cdot \boldsymbol{\sigma}.
\end{aligned}$$

The scalar square roots and logarithms must be taken on specified branches. A square root that is itself a real paravector exists when $a, b \geq 0$, equivalently $t \geq \|\mathbf{r}\|$; otherwise the displayed roots are complex. The displayed logarithm is defined only when $a, b \neq 0$; it is the real principal logarithm when $a, b > 0$. The condition $ab \neq 0$ is required for the inverse. The power formula is valid for $n \geq 0$, and also for negative integers when $ab \neq 0$. For two paravectors diagonalized along possibly different axes, the unit axis of X' is denoted by $\mathbf{u}'$. The second projector pair is the full specialization of Eq. (65),

$$\Pi'(\pm) := \tfrac{1}{2}(1 \pm \mathbf{u}' \cdot \boldsymbol{\sigma}). \quad (72)$$

Here $(\mathbf{u}')^2 = 1$. With eigenvalues a' and b' assigned respectively to $\Pi'(+)$ and $\Pi'(-)$,

$$\begin{aligned}
f(X)g(X') &= (f(a)\Pi(+) + f(b)\Pi(-)) \\
&\quad \times (g(a')\Pi'(+) + g(b')\Pi'(-)),
\end{aligned}$$

so

$$\begin{aligned}
f(X)g(X') &= f(a)g(a')\Pi(+)\Pi'(+) \\
&\quad + f(a)g(b')\Pi(+)\Pi'(-) \\
&\quad + f(b)g(a')\Pi(-)\Pi'(+) \\
&\quad + f(b)g(b')\Pi(-)\Pi'(-).
\end{aligned}$$

The mixed projector products are

$$\begin{aligned}
\Pi(+)\Pi'(+)&= \frac{1}{4}(1 + \mathbf{u} \cdot \mathbf{u}') \\
&\quad + \frac{1}{4}(\mathbf{u}' + \mathbf{u} + i(\mathbf{u} \times \mathbf{u}')) \cdot \boldsymbol{\sigma}, \\
\Pi(+)\Pi'(-)&= \frac{1}{4}(1 - \mathbf{u} \cdot \mathbf{u}') \\
&\quad + \frac{1}{4}(-\mathbf{u}' + \mathbf{u} - i(\mathbf{u} \times \mathbf{u}')) \cdot \boldsymbol{\sigma}, \\
\Pi(-)\Pi'(+)&= \frac{1}{4}(1 - \mathbf{u} \cdot \mathbf{u}') \\
&\quad + \frac{1}{4}(\mathbf{u}' - \mathbf{u} - i(\mathbf{u} \times \mathbf{u}')) \cdot \boldsymbol{\sigma}, \\
\Pi(-)\Pi'(-)&= \frac{1}{4}(1 + \mathbf{u} \cdot \mathbf{u}') \\
&\quad + \frac{1}{4}(-\mathbf{u}' - \mathbf{u} + i(\mathbf{u} \times \mathbf{u}')) \cdot \boldsymbol{\sigma}.
\end{aligned}$$

As a useful example, the Cayley-type transform is itself a projector exponential:

$$\begin{aligned}
\frac{1 - \mathbf{r} \cdot \boldsymbol{\sigma}}{1 + \mathbf{r} \cdot \boldsymbol{\sigma}} &= \exp(-2 \operatorname{arctanh}(\mathbf{r} \cdot \boldsymbol{\sigma})) \\
&= \exp(-2 \operatorname{arctanh}(\|\mathbf{r}\|))\Pi(+) \\
&\quad + \exp(+2 \operatorname{arctanh}(\|\mathbf{r}\|))\Pi(-).
\end{aligned}$$

The rational expression is defined only when $\|\mathbf{r}\| \neq 1$. For a real-valued $\operatorname{arctanh}$ and the displayed real exponentials, the additional condition $\|\mathbf{r}\| < 1$ is required; otherwise a complex branch must be selected.

1.9.3 Minimal Polynomial and Projector Decomposition

The function formulas, projector identities, and exponential identities are determined by the two-root factorization of a complex paravector, as shown by Eqs. (37) and (67):

$$\begin{aligned}
Z &= S + \mathbf{V} \cdot \boldsymbol{\sigma} \in \mathbb{C} + \mathbb{C}^3, \\
Z^2 - 2SZ + (S^2 - \mathbf{V}^2) &= 0.
\end{aligned}$$

When $\mathbf{V}^2 \neq 0$, a value z with $z^2 = \mathbf{V}^2$ is chosen. The two eigenvalue values are defined by

$$\lambda(Z, \pm) := S \pm z. \quad (73)$$

The corresponding projector function is defined by

$$\Pi(Z, \pm) := \frac{1}{2} \left(1 \pm \frac{\mathbf{V} \cdot \boldsymbol{\sigma}}{z} \right). \quad (74)$$

Then

$$\begin{aligned}
(Z - \lambda(Z, -))(Z - \lambda(Z, +)) &= 0, \\
Z &= \lambda(Z, +)\Pi(Z, +) \\
&\quad + \lambda(Z, -)\Pi(Z, -), \\
f(Z) &= f(\lambda(Z, +))\Pi(Z, +) \\
&\quad + f(\lambda(Z, -))\Pi(Z, -)
\end{aligned}$$

for any scalar function defined on the two eigenvalues. The two labels are merely exchanged when the sign of z is changed.

The case $\mathbf{V}^2 = 0$ must be treated separately. If $\mathbf{V} = \mathbf{0}$, then $Z = S$ is scalar. If $\mathbf{V} \neq \mathbf{0}$, the nilpotent element is defined by

$N := \mathbf{V} \cdot \boldsymbol{\sigma}$; then $N^2 = 0$, $Z = S + N$ is not diagonalizable, and for a function analytic at S ,

$$f(Z) = f(S) + f'(S)N.$$

This nilpotent formula is the continuous replacement for the two-projector formula when the two roots coalesce.

1.10 Exponential and Logarithmic Algebra

With the projector calculus fixed, exponential and logarithmic formulas can be developed in one place. Hyperbolic half-angle transform elements are produced by the sigma-number-one exponential, and the circular half-angle elements used later for rotors and for the quaternion correspondence are produced by the sigma-number-two exponential.

1.10.1 Real Exponential and Half-Angle Elements

The sigma-number-one exponential is resolved by the projector calculus. For the real paravector

$$X = a + \mathbf{A} \cdot \boldsymbol{\sigma} \in \mathbb{R} + \mathbb{R}^3,$$

the vector part is decomposed as

$$\begin{aligned}
\mathbf{A} &= \theta \mathbf{u}, \quad \theta = \|\mathbf{A}\|, \\
\mathbf{u} &= \frac{\mathbf{A}}{\|\mathbf{A}\|} \quad (\mathbf{A} \neq 0), \quad \mathbf{u} = (u_1, u_2, u_3), \\
\mathbf{u}^2 &= 1.
\end{aligned}$$

The exponential is given by the functional calculus of Eq. (67):

$$\begin{aligned}
\exp(\pm \mathbf{A} \cdot \boldsymbol{\sigma}) &= \exp(\pm \theta \mathbf{u} \cdot \boldsymbol{\sigma}) \\
&= \exp(\pm \theta)\Pi(+) + \exp(\mp \theta)\Pi(-) \\
&= \cosh(\theta) \pm \sinh(\theta) \mathbf{u} \cdot \boldsymbol{\sigma}. \quad (75)
\end{aligned}$$

An equivalent derivation is obtained from the power series by separating the even and odd powers:

$$\begin{aligned}
\exp(\mathbf{A} \cdot \boldsymbol{\sigma}) &= \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}) \\
&= \sum_{n \geq 0} \frac{\theta^n (\mathbf{u} \cdot \boldsymbol{\sigma})^n}{n!} \\
&= \sum_{n \geq 0} \frac{\theta^{2n}}{(2n)!} + \sum_{n \geq 0} \frac{\theta^{2n+1}}{(2n+1)!} \mathbf{u} \cdot \boldsymbol{\sigma} \\
&= \cosh(\theta) + \sinh(\theta) \mathbf{u} \cdot \boldsymbol{\sigma}.
\end{aligned}$$

With

$$\begin{aligned}
\gamma &= \cosh(\theta), \\
\mathbf{v} &= \tanh(\theta) \mathbf{u},
\end{aligned}$$

the basic hyperbolic parametrization is obtained from Eq. (75):

$$\begin{aligned}
\exp(\mathbf{A} \cdot \boldsymbol{\sigma}) &= \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}), \\
\exp(-\mathbf{A} \cdot \boldsymbol{\sigma}) &= \gamma(1 - \mathbf{v} \cdot \boldsymbol{\sigma}).
\end{aligned}$$

For the real paravector X from Eq. (12), the scalar part commutes with the vector part, so

$$\begin{aligned}\exp(X) &= \exp(a) \exp(\mathbf{A} \cdot \boldsymbol{\sigma}), \\ \exp(X^*) &= \exp(a) \exp(-\mathbf{A} \cdot \boldsymbol{\sigma}) = \exp(X)^*, \\ \exp(-X) &= \exp(-a) \exp(-\mathbf{A} \cdot \boldsymbol{\sigma}) = \exp(X)^{-1}.\end{aligned}$$

The corresponding half-angle element is

$$L(\theta, \mathbf{u}) := \exp\left(\frac{\theta}{2} \mathbf{u} \cdot \boldsymbol{\sigma}\right). \quad (76)$$

Its conjugation properties are

$$L^H = L, \quad L^* = L^{-1}.$$

The action of this element on real events is fixed by the generalized transform law and differs from hyperbolic similarity. The exponential and its inverse are specified by Eq. (76). The two projector weights are exchanged by the conjugate operator:

$$\begin{aligned}\exp(\mathbf{A} \cdot \boldsymbol{\sigma})^* &= \exp(-\theta)\Pi(+) + \exp(\theta)\Pi(-) \\ &= \gamma(1 - \mathbf{v} \cdot \boldsymbol{\sigma}) \\ &= \exp(-\mathbf{A} \cdot \boldsymbol{\sigma}) \\ &= \exp((\mathbf{A} \cdot \boldsymbol{\sigma})^*) \\ &= \exp(\mathbf{A} \cdot \boldsymbol{\sigma})^{-1}.\end{aligned}$$

1.10.2 Real Logarithm of a Paravector

The logarithm is the inverse problem for the same projector formulas. The real paravector Y from Eq. (13) is assumed to satisfy $b > \|\mathbf{B}\|$, so that the two real eigenvalues are positive and the principal logarithm is defined. For $\mathbf{B} \neq 0$, the decomposition $\mathbf{B} = \|\mathbf{B}\|\mathbf{u}$ is used. The eigenvalue function from Eq. (71) and the projector function from Eq. (65) are then specialized to Y :

$$\begin{aligned}\lambda(Y, \pm) &= b \pm \|\mathbf{B}\|, \\ Y &= \lambda(Y, +)\Pi(+) + \lambda(Y, -)\Pi(-),\end{aligned}$$

with the same projector pair $\Pi(\pm) = \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})$. The logarithm is therefore applied separately to the two eigenvalues:

$$\begin{aligned}\log(Y) &= \log(\lambda(Y, +))\Pi(+) \\ &\quad + \log(\lambda(Y, -))\Pi(-) \\ &= \frac{1}{2} \log(\lambda(Y, +)\lambda(Y, -)) \\ &\quad + \frac{1}{2} \log(\lambda(Y, +)/\lambda(Y, -)) \mathbf{u} \cdot \boldsymbol{\sigma}.\end{aligned} \quad (77)$$

When $\mathbf{B} = \mathbf{0}$, the axis is unnecessary and this formula is reduced directly to $\log(Y) = \log(b)$. For $\mathbf{B} \neq 0$, if $Y = \exp(X)$ with X from Eq. (12) and $\mathbf{A} = \theta\mathbf{u}$, then $\lambda(Y, \pm) = \exp(a \pm \theta)$, so the recovered parameters are

$$\begin{aligned}Y &= \exp(X) \\ &= \exp(a + \mathbf{A} \cdot \boldsymbol{\sigma}) \\ &= \exp(a) \exp(\mathbf{A} \cdot \boldsymbol{\sigma}) \\ &= \exp(a) (\cosh(\theta) + \sinh(\theta) \mathbf{u} \cdot \boldsymbol{\sigma}) \\ &= b + \|\mathbf{B}\| \mathbf{u} \cdot \boldsymbol{\sigma},\end{aligned}$$

and the recovered parameters are

$$\begin{aligned}\theta &= \operatorname{arctanh}(\|\mathbf{B}\|/b), \\ \exp(a) &= \sqrt{b^2 - \mathbf{B}^2}, \\ a &= \frac{1}{2} \log(b^2 - \mathbf{B}^2), \\ \mathbf{A} = \theta \mathbf{u} &= \operatorname{arctanh}(\|\mathbf{B}\|/b) \frac{\mathbf{B}}{\|\mathbf{B}\|}.\end{aligned}$$

Thus the principal logarithm of a future-directed time-like real paravector is again a real paravector,

$$\log(Y) = a + \mathbf{A} \cdot \boldsymbol{\sigma}.$$

1.10.3 Circular Exponentials and Half-Angle Elements

The trigonometric form of the sigma-number-two exponential is obtained from the same projector calculation by replacing θ with $i\theta$:

$$\begin{aligned}\exp(\pm i\mathbf{A} \cdot \boldsymbol{\sigma}) &= \exp(\pm i\theta \mathbf{u} \cdot \boldsymbol{\sigma}) \\ &= \exp(\pm i\theta)\Pi(+) + \exp(\mp i\theta)\Pi(-) \\ &= \cos(\theta) \pm i \sin(\theta) \mathbf{u} \cdot \boldsymbol{\sigma}.\end{aligned} \quad (78)$$

In particular, the negative-sign form may be written as

$$\exp(-i\mathbf{A} \cdot \boldsymbol{\sigma}) = \cos(\theta) + \sin(\theta) \mathbf{u} \cdot (-i\boldsymbol{\sigma}).$$

In the quaternion basis $\mathbf{q} = -i\boldsymbol{\sigma}$, the same formula is written as

$$\exp(-i\mathbf{A} \cdot \boldsymbol{\sigma}) = \cos(\theta) + \sin(\theta) \mathbf{u} \cdot \mathbf{q}.$$

For a real angle $\theta \in \mathbb{R}$ and a unit real vector $\mathbf{u} \in \mathbb{R}^3$, the corresponding circular half-angle exponential is

$$R(\theta, \mathbf{u}) := \exp\left(\frac{\theta}{2} \mathbf{u} \cdot i\boldsymbol{\sigma}\right). \quad (79)$$

Its conjugation properties are

$$R^H R = 1, \quad R^* = R.$$

Equation (79) is the circular half-angle element built from Eq. (78). Its transform action is fixed by the general similarity and event-congruence laws, from which Rodrigues's formula is obtained. The equivalent quaternion form is obtained from the same sigma-number-two algebra.

1.10.4 Exponentials of Complex Paravectors

The real paravectors are written as

$$\begin{aligned}X &= a + \mathbf{A} \cdot \boldsymbol{\sigma} \in \mathbb{R} + \mathbb{R}^3, \\ Y &= b + \mathbf{B} \cdot \boldsymbol{\sigma} \in \mathbb{R} + \mathbb{R}^3, \\ Z &= S + \mathbf{V} \cdot \boldsymbol{\sigma} \in \mathbb{C} + \mathbb{C}^3, \\ &= X + iY,\end{aligned}$$

where

$$S = a + ib \in \mathbb{C}, \quad \mathbf{V} = \mathbf{A} + i\mathbf{B} \in \mathbb{C}^3.$$

Because the scalar part S commutes with $\mathbf{V} \cdot \boldsymbol{\sigma}$, the exponential is separated into scalar and directional factors:

$$\begin{aligned}\exp(Z) &= \exp(S + \mathbf{V} \cdot \boldsymbol{\sigma}), \\ &= \exp(S) \exp(\mathbf{V} \cdot \boldsymbol{\sigma}).\end{aligned}$$

$$\begin{aligned}\exp(Z^*) &= \exp(S^*) \exp(-\mathbf{V}^* \cdot \boldsymbol{\sigma}), \\ \exp(Z)^{-1} &= \exp(-S) \exp(-\mathbf{V} \cdot \boldsymbol{\sigma}), \\ \exp(Z^*)^{-1} &= \exp(-S^*) \exp(\mathbf{V}^* \cdot \boldsymbol{\sigma}), \\ (\mathbf{V} \cdot \boldsymbol{\sigma})^2 &= \mathbf{V} \cdot \mathbf{V} = \mathbf{V}^2, \\ &= \mathbf{A}^2 - \mathbf{B}^2 + 2i\mathbf{A} \cdot \mathbf{B}.\end{aligned}$$

The key simplification is that all higher powers are reduced to the two basis elements 1 and $\mathbf{V} \cdot \boldsymbol{\sigma}$:

$$\begin{aligned}(\mathbf{V} \cdot \boldsymbol{\sigma})^{2n} &= (\mathbf{V}^2)^n, \\ z &= \sqrt{\mathbf{V}^2} \quad \text{complex scalar,} \\ \exp(\mathbf{V} \cdot \boldsymbol{\sigma}) &= \sum_{n \geq 0} \frac{(\mathbf{V} \cdot \boldsymbol{\sigma})^n}{n!}, \\ &= \sum_{n \geq 0} \frac{(\mathbf{V}^2)^n}{(2n)!} + \sum_{n \geq 0} \frac{(\mathbf{V}^2)^n}{(2n+1)!} (\mathbf{V} \cdot \boldsymbol{\sigma}), \\ &= \cosh(z) + \sinh(z) \frac{\mathbf{V} \cdot \boldsymbol{\sigma}}{z}.\end{aligned}$$

The final expression is independent of the sign chosen for z . When $\mathbf{V}^2 = 0$, the quotient $\sinh(z)/z$ is understood by its continuous value 1, and the formula is reduced to

$$\exp(Z) = \exp(S)(1 + \mathbf{V} \cdot \boldsymbol{\sigma}).$$

1.11 Generalized Transform Algebra

With the projector and exponential formulas fixed, the remaining purely mathematical transform material can be collected in one place. A generic-to-special order is used: the similarity law is given first, followed by the basis-transport identities and then by the circular and hyperbolic transforms built from the earlier exponential elements:

$$TZ' = ZT, \quad (80)$$

$$Z' = T^{-1}ZT. \quad (81)$$

Here Z is a generic complex paravector and T is a generic invertible algebra element. The equivalence of the primed and unprimed algebra actions is defined by the intertwining requirement in Eq. (80). Its unique solution is the similarity (81), which is used for basis changes, rotations, and electromagnetic fields. This similarity is also separated in the derivation below from the congruence required for a real event paravector.

1.11.1 Intertwining and the General Similarity Transform

Compatibility between two descriptions of the same algebraic action is expressed by the starting relation (80). A primed carrier

is carried into the unprimed description by T . If an algebraic action is represented by Z' before that identification and by Z afterward, then commutation of the action with the change of description is required:

$$T(Z'\psi') = Z(T\psi') \quad \text{for every carrier } \psi'.$$

The relation $TZ' = ZT$ is obtained by removing the arbitrary carrier. Because T is invertible, its unique solution is

$$Z' = T^{-1}ZT.$$

Thus Eq. (81) is not an additional dynamical law: it is the change-of-description law forced by the intertwining requirement. Under the opposite carrier direction, T is replaced everywhere by T^{-1} , and the equivalent convention $Z' = TZT^{-1}$ is obtained. Under the inverse-first convention, the displayed positive sign in Rodrigues's formula is produced by the positive exponent in Eq. (79).

The fully general complex paravectors are written as

$$\begin{aligned}Z &= S + \mathbf{V} \cdot \boldsymbol{\sigma}, & S &\in \mathbb{C}, & \mathbf{V} &\in \mathbb{C}^3, \\ T &= f + \mathbf{g} \cdot \boldsymbol{\sigma}, & f &\in \mathbb{C}, & \mathbf{g} &\in \mathbb{C}^3,\end{aligned}$$

where the dot and cross products are extended complex-bilinearly. The central determinant of T is $\det(T) = f^2 - \mathbf{g}^2$. When $\det(T) \neq 0$, the inverse is given by the general formula (60):

$$T^{-1} = \frac{T^{*H}}{\det(T)} = \frac{f - \mathbf{g} \cdot \boldsymbol{\sigma}}{f^2 - \mathbf{g}^2}.$$

Because the scalar part S is central,

$$Z' = S + \frac{(f - \mathbf{g} \cdot \boldsymbol{\sigma})(\mathbf{V} \cdot \boldsymbol{\sigma})(f + \mathbf{g} \cdot \boldsymbol{\sigma})}{\det(T)}.$$

The first multiplication is

$$(f - \mathbf{g} \cdot \boldsymbol{\sigma})(\mathbf{V} \cdot \boldsymbol{\sigma}) = -\mathbf{g} \cdot \mathbf{V} + (f\mathbf{V} - i\mathbf{g} \times \mathbf{V}) \cdot \boldsymbol{\sigma}.$$

After multiplication by the final factor, the scalar terms are canceled and the remaining vector terms are

$$\begin{aligned}Z' &= S + \mathbf{V}' \cdot \boldsymbol{\sigma}, \\ \mathbf{V}' &= \frac{(f^2 + \mathbf{g}^2)\mathbf{V} - 2(\mathbf{g} \cdot \mathbf{V})\mathbf{g} - 2if(\mathbf{g} \times \mathbf{V})}{f^2 - \mathbf{g}^2}.\end{aligned} \quad (82)$$

This is the similarity law for every complex paravector Z and every invertible complex paravector T . The complex scalar part is fixed. Neither Eq. (81) nor Eq. (82) is changed when T is multiplied by a nonzero central complex scalar.

Products, inverses, and functions are governed by the same intertwining. For a second algebra element W ,

$$\begin{aligned}(ZW)' &= T^{-1}ZWT = T^{-1}ZT T^{-1}WT = Z'W', \\ (Z')^{-1} &= T^{-1}Z^{-1}T \quad \text{when } Z^{-1} \text{ exists,} \\ (Z')^n &= T^{-1}Z^nT, \\ h(Z') &= T^{-1}h(Z)T.\end{aligned} \quad (83)$$

The last line is valid for every convergent power-series function h , and therefore in particular

$$\exp(Z') = T^{-1} \exp(Z) T.$$

The same relation is obeyed by logarithms, roots, and rational functions wherever the same branch and domain are defined.

Commutation of the adjugate with this similarity is also obtained. Since $T^{*\text{H}} = \det(T) T^{-1}$ and $(T^{-1})^{*\text{H}} = T / \det(T)$,

$$\begin{aligned} Z'^{*\text{H}} &= T^{*\text{H}} Z^{*\text{H}} (T^{-1})^{*\text{H}} = T^{-1} Z^{*\text{H}} T, \\ Z' Z'^{*\text{H}} &= T^{-1} (Z Z^{*\text{H}}) T = Z Z^{*\text{H}}. \end{aligned}$$

Thus the determinant form $S^2 - \mathbf{V}^2$, the spectrum, and the inverse framework are preserved by similarity for every invertible T . The positive coefficient norm is a different construction and is not generally preserved by a nonunitary similarity.

Algebra elements whose action is represented by multiplication are transported by the similarity law; this law is not automatically the Lorentz law for every physical representation. A real event paravector is characterized by $X^{\text{H}} = X$. Such paravectors are spanned by Hermitian carrier bilinears $\psi\psi^{\text{H}}$: concretely, in the Pauli realization every Hermitian two-by-two matrix is a real linear combination of bilinears of this form. The carrier identification is $\psi' = T^{-1}\psi$, and therefore

$$\psi' \psi'^{\text{H}} = T^{-1} (\psi \psi^{\text{H}}) (T^{-1})^{\text{H}}.$$

By real linearity, the event law is consequently forced to be the inverse-carried Hermitian congruence. For a normalized Lorentz element $TT^{*\text{H}} = 1$, it is

$$\begin{aligned} X' &= T^{-1} X (T^{-1})^{\text{H}} \\ &= T^{*\text{H}} X T^*. \end{aligned} \quad (84)$$

Distinct roles are assigned to the involutions. In the Pauli realization, $^{\text{H}}$ is Hermitian adjunction, whereas $^{*\text{H}}$ is the adjugate. The latter equals T^{-1} only after the determinant-one normalization $TT^{*\text{H}} = 1$ has been imposed. The real subspace is preserved because $(X')^{\text{H}} = X'$, and the event determinant is preserved because T^{-1} has unit determinant. When X is invertible, the transformed inverse is obtained without a new definition:

$$(X')^{-1} = T^{\text{H}} X^{-1} T, \quad (85)$$

because the identity is obtained by direct multiplication with $X' = T^{-1} X (T^{-1})^{\text{H}}$. The algebraic pattern for the dual derivative paravector is also supplied by this complementary law. The event intertwiner is obtained by multiplication of Eq. (84) on the left by T :

$$TX' = XT^*. \quad (86)$$

The form $TX' = XT$ is obtained for a circular rotor because $R^* = R$, but not for a boost. The similarity and event intertwiners are therefore identical for circular rotors but remain distinct for boosts.

In contrast, the action of a bivector such as the electromagnetic field is represented by left multiplication on a carrier. The

following similarity is then required by the original intertwining argument:

$$F' = T^{-1} F T. \quad (87)$$

For a circular rotor, $R^* = R$ and $R^{-1} = R^{\text{H}}$, so Eq. (84) is reduced to the active spatial rotation

$$X' = R^{\text{H}} X R. \quad (88)$$

For the hyperbolic element in Eq. (76), $L^{\text{H}} = L$ and $L^{-1} = L^* = L^{*\text{H}}$, so the same general event law is reduced to the passive boost

$$X' = L^* X L^*. \quad (89)$$

By contrast, Eq. (87) is reduced to $F' = L^* F L$. The unequal right factors are therefore consequences of the representations being transported, not of an inconsistent choice of inverse. Under the definition of $L(\theta, \mathbf{u})$, the sandwich $L^{\text{H}} X L = L X L$ is the inverse-direction event boost, obtained by replacing θ with $-\theta$. It is physically legitimate, but it is not the passive $+\theta$ convention used in Eq. (89).

The hyperbolic reduction of the general complex formula is obtained directly with

$$f = \cosh(\theta/2), \quad \mathbf{g} = \sinh(\theta/2)\mathbf{u}, \quad \mathbf{u}^2 = 1,$$

Eq. (82) is reduced to

$$\mathbf{V}' = (\mathbf{u} \cdot \mathbf{V})\mathbf{u} + \cosh(\theta)(\mathbf{V} - (\mathbf{u} \cdot \mathbf{V})\mathbf{u}) - i \sinh(\theta) \mathbf{u} \times \mathbf{V}. \quad (90)$$

For $\mathbf{V} = \mathbf{E} + i\mathbf{B}$, its real and imaginary parts are exactly the electric-magnetic boost formulas. For a real $\mathbf{V}$, however, the cross term is generally imaginary; hyperbolic similarity is therefore a bivector or basis transformation rather than an event boost.

The distinction is exposed by the infinitesimal forms. If $T = 1 + K + O(K^2)$, the order functions defined in Eqs. (38) and (39) give

$$\begin{aligned} Z' &= Z + \text{com}(Z, K) + O(K^2), \\ X' &= X - KX - XK^{\text{H}} + O(K^2). \end{aligned}$$

For a circular generator $K^{\text{H}} = -K$, the event law is reduced to the same commutator by which the similarity rotation is generated. For a hyperbolic generator $K^{\text{H}} = K$, an anticommutator is obtained instead: $X' = X - \text{acom}(K, X) + O(K^2)$. Spatial vectors are rotated by the former, while the scalar time component is mixed with the vector component parallel to the boost by the latter. Thus similarity rotations are produced by circular generators, whereas event congruences are produced by hyperbolic generators.

A transform element $T(\lambda)$ is taken to depend on one real parameter λ . All displayed derivatives are ordinary derivatives with respect to that parameter. The similarity generator is defined by

$$\Omega := T^{-1} \frac{dT}{d\lambda}. \quad (91)$$

Differentiation of the inverse and of the transported paravector then gives

$$\begin{aligned}\frac{d(T^{-1})}{d\lambda} &= -T^{-1} \frac{dT}{d\lambda} T^{-1}, \\ \frac{dZ'}{d\lambda} &= T^{-1} \frac{dZ}{d\lambda} T + \text{com}(Z', \Omega), \\ \frac{d^2 Z'}{d\lambda^2} &= T^{-1} \frac{d^2 Z}{d\lambda^2} T \\ &\quad + 2 \text{com}\left(\frac{dZ'}{d\lambda}, \Omega\right), \\ &\quad + \text{com}\left(Z', \frac{d\Omega}{d\lambda}\right), \\ &\quad + \text{com}(\Omega, \text{com}(Z', \Omega)).\end{aligned}$$

They are not space-time derivatives.

1.11.2 Transformed Basis

The parameter-dependent element is written as

$$\begin{aligned}T(\lambda) &= f(\lambda) + \mathbf{g}(\lambda) \cdot \boldsymbol{\sigma}, \quad f \in \mathbb{C}, \quad \mathbf{g} \in \mathbb{C}^3, \\ T^{-1} &= \frac{f - \mathbf{g} \cdot \boldsymbol{\sigma}}{f^2 - \mathbf{g}^2}, \quad f^2 - \mathbf{g}^2 \neq 0.\end{aligned}$$

The transformed basis function is defined by

$$\sigma'_n := T^{-1} \sigma_n T. \quad (92)$$

Its inverse relation is

$$\sigma_n = T \sigma'_n T^{-1}.$$

Both real hyperbolic elements and imaginary-vector rotors are included by allowing complex $\mathbf{g}$. The following relations are obtained immediately by similarity:

$$\begin{aligned}(\sigma'_n)^2 &= 1, \\ \sigma'_n \sigma'_m &= \delta_{nm} + \sum_{k=1}^3 \epsilon_{nmk} i \sigma'_k, \\ \sigma'_1 \sigma'_2 \sigma'_3 &= i.\end{aligned}$$

Thus the same Clifford algebra is satisfied by the transformed generators for every invertible T .

The separate actions of the two basic involutions on the transformed generators are represented by

$$\begin{aligned}\sigma'^{*}_n &= (T^{-1})^* \sigma_n^* T^* = -(T^*)^{-1} \sigma_n T^*, \\ \sigma'^{\text{H}}_n &= T^{\text{H}} \sigma_n^{\text{H}} (T^{-1})^{\text{H}} = T^{\text{H}} \sigma_n (T^{\text{H}})^{-1}.\end{aligned}$$

The scalar part is also preserved by similarity, so $\text{scalar}(\sigma'_n) = \text{scalar}(\sigma_n) = 0$. The individual star and H actions are recorded separately by these formulas and are combined by the adjugate relation.

For every invertible T , commutation of the adjugate with the similarity is obtained from $T^{*\text{H}} = (f^2 - \mathbf{g}^2) T^{-1}$ and $(T^{-1})^{*\text{H}} =$

$T/(f^2 - \mathbf{g}^2)$:

$$\begin{aligned}(\sigma'_n)^{* \text{H}} &= (f^2 - \mathbf{g}^2) T^{-1} (-\sigma_n) \frac{T}{f^2 - \mathbf{g}^2} \\ &= -T^{-1} \sigma_n T = -\sigma'_n.\end{aligned}$$

Consequently a paravector written in the transformed basis,

$$Z = Z_s + \mathbf{Z}_v \cdot \boldsymbol{\sigma}',$$

is assigned the same determinant quadratic form,

$$Z Z^{*\text{H}} = Z_s^2 - \mathbf{Z}_v^2.$$

For differentiation with respect to λ , the generator from Eq. (91) gives

$$\frac{d\sigma'_n}{d\lambda} = \text{com}(\sigma'_n, \Omega), \quad \frac{d\boldsymbol{\sigma}'}{d\lambda} = \text{com}(\boldsymbol{\sigma}', \Omega).$$

The coefficient form is obtained directly from

$$\begin{aligned}\Omega &= \frac{f \frac{df}{d\lambda} - \mathbf{g} \cdot \frac{d\mathbf{g}}{d\lambda}}{f^2 - \mathbf{g}^2} \\ &\quad + \frac{f \frac{d\mathbf{g}}{d\lambda} - \frac{df}{d\lambda} \mathbf{g} - i \mathbf{g} \times \frac{d\mathbf{g}}{d\lambda}}{f^2 - \mathbf{g}^2} \cdot \boldsymbol{\sigma}.\end{aligned}$$

Relative to the primed basis, its vector coefficient is written uniquely as

$$\Omega = \Omega_s + \frac{1}{2}(\mathbf{k} + i\boldsymbol{\omega}) \cdot \boldsymbol{\sigma}', \quad \mathbf{k}, \boldsymbol{\omega} \in \mathbb{R}^3.$$

The same generator carried into the unprimed basis is defined by

$$\Omega' := T \Omega T^{-1}. \quad (93)$$

The definition in Eq. (91) then gives

$$\Omega' = \frac{dT}{d\lambda} T^{-1}.$$

The coefficients are recovered through scalar extraction. For $n \in \{1, 2, 3\}$, the coefficients are given by cyclicity of the algebraic scalar part:

$$k_n + i\omega_n = 2 \text{scalar}(\Omega \sigma'_n) = 2 \text{scalar}(\Omega' \sigma_n).$$

The scalar part is central with respect to the basis, while

$$\begin{aligned}\frac{d\boldsymbol{\sigma}'}{d\lambda} &= (-\boldsymbol{\omega} + i\mathbf{k}) \times \boldsymbol{\sigma}', \\ \frac{dZ}{d\lambda} &= \frac{dZ_s}{d\lambda} + \left(\frac{d\mathbf{Z}_v}{d\lambda} - \mathbf{Z}_v \times \boldsymbol{\omega} + i \mathbf{Z}_v \times \mathbf{k} \right) \cdot \boldsymbol{\sigma}'.\end{aligned}$$

These are algebraic moving-basis identities. No space-time, energy, or proper-time interpretation is implied by similarity alone; a specified Lorentz congruence and frame geometry are required for such an interpretation.

1.11.3 Rotation

Equation (88) is obtained as the rotation case of the general transform law by specializing the transform element to $T = R$. Sigma number two is assigned to the generator, so the exponential is circular rather than hyperbolic. The rotor function defined in Eq. (79) has the expanded form

$$\begin{aligned} R(\theta, \mathbf{u}) &= \exp\left(\frac{1}{2}\theta(\mathbf{u} \cdot i\boldsymbol{\sigma})\right), \\ &= \cos\left(\frac{1}{2}\theta\right) + \sin\left(\frac{1}{2}\theta\right)(\mathbf{u} \cdot i\boldsymbol{\sigma}) \quad \text{rotor.} \\ R^{-1} &= \exp\left(-\frac{1}{2}\theta(\mathbf{u} \cdot i\boldsymbol{\sigma})\right). \end{aligned}$$

$\theta \in \mathbb{R}$, $\mathbf{u} \in \mathbb{R}^3$ is a unit real vector, and R is generated by the sigma-number-two element $\mathbf{u} \cdot i\boldsymbol{\sigma}$. The inverse relation is $R^{-1} = R^H = R^{*\text{H}}$. The orientation sign is fixed by the substitution

$$f = \cos(\theta/2), \quad \mathbf{g} = i \sin(\theta/2)\mathbf{u}$$

into Eq. (82). Then

$$\begin{aligned} f^2 - \mathbf{g}^2 &= 1, \\ f^2 + \mathbf{g}^2 &= \cos(\theta), \\ -2(\mathbf{g} \cdot \mathbf{r})\mathbf{g} &= (1 - \cos(\theta))(\mathbf{u} \cdot \mathbf{r})\mathbf{u}, \\ -2if(\mathbf{g} \times \mathbf{r}) &= \sin(\theta)(\mathbf{u} \times \mathbf{r}). \end{aligned}$$

Therefore the transformed paravector is

$$\begin{aligned} X' &= R^H X R, \\ &= t' + \mathbf{r}' \cdot \boldsymbol{\sigma}. \\ t' &= t, \\ \mathbf{r}' &= \mathbf{r} \cos(\theta) + (\mathbf{u} \times \mathbf{r}) \sin(\theta), \\ &\quad + (\mathbf{u} \cdot \mathbf{r})(1 - \cos(\theta))\mathbf{u}, \end{aligned}$$

Rodrigues's formula.

For a parameter-dependent rotor, the specialization of the general generator is $R^H(dR/d\lambda)$. Its form carried to the other side is $(dR/d\lambda)R^H$. The following identities are obtained by differentiating $R^H R = 1$:

$$\left(R^H \frac{dR}{d\lambda}\right)^H = -R^H \frac{dR}{d\lambda}, \quad \left(\frac{dR}{d\lambda} R^H\right)^H = -\frac{dR}{d\lambda} R^H.$$

The rotation transport law is the specialization of the general one-parameter similarity formula:

$$\begin{aligned} \frac{dX'}{d\lambda} &= R^H \frac{dX}{d\lambda} R + \text{com}\left(X', R^H \frac{dR}{d\lambda}\right) \\ &= R^H \left(\frac{dX}{d\lambda} + \text{com}\left(X, \frac{dR}{d\lambda} R^H\right)\right) R. \end{aligned}$$

1.11.4 Spin Rotors

Each oriented complementary bivector is identified by the pseudoscalar without an index-order ambiguity:

$$i\sigma_1 = \sigma_2\sigma_3, \quad i\sigma_2 = \sigma_3\sigma_1, \quad i\sigma_3 = \sigma_1\sigma_2.$$

For unit $\mathbf{n}, \mathbf{u} \in \mathbb{R}^3$, the rotor function defined in Eq. (79) is evaluated as

$$\begin{aligned} R(\phi, \mathbf{n}) &= \exp\left(\frac{\phi}{2} \mathbf{n} \cdot i\boldsymbol{\sigma}\right) \\ &= \cos(\phi/2) + \sin(\phi/2) \mathbf{n} \cdot i\boldsymbol{\sigma}. \end{aligned}$$

Then

$$R^{-1} = R^H = R^{*\text{H}}, \quad R^H R = 1,$$

and

$$R^H(\mathbf{u} \cdot \boldsymbol{\sigma})R = \mathbf{u}' \cdot \boldsymbol{\sigma}$$

with $\mathbf{u}'$ defined by Rodrigues's formula. The rotor itself is single-valued and 4π -periodic:

$$R(\phi + 2\pi, \mathbf{n}) = -R(\phi, \mathbf{n}), \quad R(\phi + 4\pi, \mathbf{n}) = R(\phi, \mathbf{n}).$$

The induced spatial rotation is two-to-one because the same conjugation is produced by R and $-R$.

In the Pauli matrix realization, the two-by-two representation is transposed and complex-conjugated by H . A general $\psi \in \mathbb{C} + \mathbb{C}^3$ is therefore a full two-by-two matrix, not one selected two-entry column: its two columns are rotated independently by left multiplication. One column is selected by a fixed projector on the right when a single Pauli wave function is intended. Under the active convention associated with this vector action,

$$\psi' = R^H \psi, \quad \psi' \psi'^H = R^H(\psi \psi^H)R.$$

A scalar pairing, normalization, expectation map, projected carrier, and specified operator action are additionally required for a physical spin state. The algebra action is thereby separated from the choice of a single physical spin state.

1.11.5 Spin-Axis Projectors

The left action of a chosen unit axis $\mathbf{u} \in \mathbb{R}^3$ is resolved by the projector family from Eq. (65). For clarity, that definition is restated here as an ordinary equality:

$$\Pi(\pm) = \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \quad \mathbf{u}^2 = 1, \quad \text{from Eq. (65).}$$

In addition to the idempotence, orthogonality, completeness, and axis-eigenvalue identities listed there, the following involution actions are obtained:

$$\begin{aligned} \Pi(\pm)^H &= \Pi(\pm), \\ \Pi(\pm)^* &= \Pi(\mp). \end{aligned}$$

Accordingly,

$$\mathbf{u} \cdot \boldsymbol{\sigma} = (+1)\Pi(+) + (-1)\Pi(-),$$

so the eigenvalues $+1$ and -1 of the axis operator are carried by the two projector parts.

In the Pauli matrix realization, each $\Pi(\pm)$ is of rank one. A general element of $\mathbb{C} + \mathbb{C}^3$, however, is a full 2×2 complex matrix, not one selected two-entry column. The projection formulas are therefore valid column by column. A single Pauli wave-function

column is described after a fixed projector is placed on the right; without that restriction, two independent columns are contained in the full paravector.

If the axis is rotated by the rotor R from Eq. (88), the transformed axis and its second projector pair satisfy

$$\mathbf{u}' \cdot \boldsymbol{\sigma} = R^H(\mathbf{u} \cdot \boldsymbol{\sigma})R, \quad \Pi'(\pm) = \frac{1}{2}(1 \pm \mathbf{u}' \cdot \boldsymbol{\sigma}) = R^H \Pi(\pm) R. \quad (94)$$

Then

$$\mathbf{u}' \cdot \boldsymbol{\sigma} = (+1)\Pi'(+) + (-1)\Pi'(-).$$

For an algebra element $\psi \in \mathbb{C} + \mathbb{C}^3$, specialize the left-projected parts from Eq. (65):

$$\psi(\pm) = \Pi(\pm)\psi.$$

The following identities are obtained:

$$\begin{aligned} (\mathbf{u} \cdot \boldsymbol{\sigma})\psi(\pm) &= \pm\psi(\pm), \\ \psi &= \psi(+) + \psi(-). \end{aligned}$$

Thus each spinor column is split into the two eigenspaces of the chosen axis. The Hermitian operation is represented by

$$\begin{aligned} \psi(\pm)^H &= \psi^H \Pi(\pm)^H \\ &= \psi^H \Pi(\pm), \end{aligned}$$

and therefore

$$\begin{aligned} \psi(\pm)^H \psi(\pm) &= \psi^H \Pi(\pm) \psi, \\ \psi(\pm)^H \psi(\mp) &= \psi^H \Pi(\pm) \Pi(\mp) \psi = 0, \\ \psi(\mp)^H \psi(\pm) &= \psi^H \Pi(\mp) \Pi(\pm) \psi = 0. \end{aligned}$$

For two algebra elements ϕ, ψ , the pointwise algebraic pairing associated with M is $\text{scalar}(\phi^H M \psi)$. For a nonzero ψ , the corresponding normalized diagonal matrix element is

$$\frac{\text{scalar}(\psi^H M \psi)}{\text{scalar}(\psi^H \psi)}.$$

When M is Hermitian for the relevant pairing, this diagonal element is real and is an expectation value. For a general M , it is only a normalized matrix element and may be complex. Scalar extraction is one half of the matrix trace. Consequently, for a full matrix ψ the numerator and denominator are each one half of the sum of their two ordinary column pairings. After one column is selected by a fixed right projector, they are each one half of the corresponding ordinary selected-column quantity. The common factor is canceled in the normalized matrix element. The spatially integrated quantum bracket is defined separately in Eq. (209).

Rotational covariance is made explicit by the same convention. Under the active rotation

$$\psi' = R^H \psi, \quad M' = R^H M R, \quad R R^H = 1,$$

both the numerator and denominator of the normalized matrix element are unchanged:

$$\frac{\text{scalar}(\psi'^H M' \psi')}{\text{scalar}(\psi'^H \psi')} = \frac{\text{scalar}(\psi^H M \psi)}{\text{scalar}(\psi^H \psi)}. \quad (95)$$

Equivalently, if the observable M is held fixed while only the carrier is actively rotated, its value in the original carrier is computed with the rotated-back observable $R M R^H$. This is the active-versus-passive equivalence expressed in Eq. (94).

For an axis observable, the following identity is obtained by cyclic scalar extraction:

$$\text{scalar}(\psi^H(\mathbf{u} \cdot \boldsymbol{\sigma})\psi) = \mathbf{u} \cdot \text{vector}(\psi\psi^H). \quad (96)$$

Consequently its normalized pointwise value is

$$\frac{\mathbf{u} \cdot \text{vector}(\psi\psi^H)}{\text{scalar}(\psi^H \psi)}.$$

In the selected-column matrix realization, twice the numerator is the ordinary column matrix element and twice the denominator is the ordinary column norm, so the common factor is canceled. For a spatial wave field the same statement is valid with both quantities integrated as in Eqs. (209) and (245).

The unnormalized pointwise numerator for the rotated axis is expanded as

$$\begin{aligned} \text{scalar}(\psi^H(\mathbf{u}' \cdot \boldsymbol{\sigma})\psi) &= \text{scalar}(\psi(+)^H(\mathbf{u}' \cdot \boldsymbol{\sigma})\psi(+)) \\ &\quad + \text{scalar}(\psi(+)^H(\mathbf{u}' \cdot \boldsymbol{\sigma})\psi(-)) \\ &\quad + \text{scalar}(\psi(-)^H(\mathbf{u}' \cdot \boldsymbol{\sigma})\psi(+)) \\ &\quad + \text{scalar}(\psi(-)^H(\mathbf{u}' \cdot \boldsymbol{\sigma})\psi(-)). \end{aligned}$$

The algebra governing these same-side and cross-side terms is

$$\begin{aligned} (\mathbf{u}' \cdot \boldsymbol{\sigma})(\mathbf{u} \cdot \boldsymbol{\sigma}) &= \mathbf{u}' \cdot \mathbf{u} + i(\mathbf{u}' \times \mathbf{u}) \cdot \boldsymbol{\sigma}, \\ (\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{u}' \cdot \boldsymbol{\sigma}) &= \mathbf{u}' \cdot \mathbf{u} - i(\mathbf{u}' \times \mathbf{u}) \cdot \boldsymbol{\sigma}, \\ (\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{u}' \cdot \boldsymbol{\sigma})(\mathbf{u} \cdot \boldsymbol{\sigma}) &= (2(\mathbf{u}' \cdot \mathbf{u})\mathbf{u} - \mathbf{u}') \cdot \boldsymbol{\sigma}. \end{aligned}$$

The second axis is decomposed into its par and prp parts by the axis functions from Eqs. (34) and (35):

$$\mathbf{u}' = \text{par}(\mathbf{u}') + \text{prp}(\mathbf{u}'), \quad \text{par}(\mathbf{u}') = (\mathbf{u} \cdot \mathbf{u}')\mathbf{u}.$$

Then the projector products are

$$\Pi(\pm)(\mathbf{u}' \cdot \boldsymbol{\sigma})\Pi(\pm) = \pm(\mathbf{u} \cdot \mathbf{u}')\Pi(\pm),$$

$$\Pi(\pm)(\mathbf{u}' \cdot \boldsymbol{\sigma})\Pi(\mp) = \frac{1}{2}(\text{prp}(\mathbf{u}') \pm i\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma} \Pi(\mp).$$

The diagonal terms are therefore weighted by the axis overlap $\mathbf{u} \cdot \mathbf{u}'$, while the off-diagonal terms are controlled by the prp and commutator parts of the rotated axis.

If θ is the angle between $\mathbf{u}$ and $\mathbf{u}'$, then

$$\mathbf{u}' \cdot \mathbf{u} = \cos(\theta),$$

Because scalar extraction is one half of the matrix trace, the trace-normalized rank-one projector overlap is

$$2 \text{scalar}(\Pi(s)\Pi'(s')) = \frac{1}{2}(1 + ss' \mathbf{u} \cdot \mathbf{u}'), \quad s, s' \in \{+1, -1\}.$$

For a normalized irreducible axis eigenstate this overlap is the transition probability. In particular, the same-side and opposite-side probabilities are

$$\frac{1}{2}(1 + \mathbf{u}' \cdot \mathbf{u}) = \cos^2(\theta/2), \quad \frac{1}{2}(1 - \mathbf{u}' \cdot \mathbf{u}) = \sin^2(\theta/2).$$

If $\mathbf{u}'$ is obtained by rotating $\mathbf{u}$ through angle ϕ about a unit axis $\mathbf{n}$, the overlap is given by Rodrigues's formula:

$$\begin{aligned} \mathbf{u}' \cdot \mathbf{u} &= (\mathbf{u} \cdot \mathbf{u}) \cos(\phi) + (\mathbf{u} \cdot (\mathbf{n} \times \mathbf{u})) \sin(\phi) \\ &\quad + (\mathbf{n} \cdot \mathbf{u})(1 - \cos(\phi))(\mathbf{n} \cdot \mathbf{u}) \\ &= (\mathbf{n} \cdot \mathbf{u})^2 + (1 - (\mathbf{n} \cdot \mathbf{u})^2) \cos(\phi). \end{aligned}$$

In the perpendicular case $\mathbf{n} \cdot \mathbf{u} = 0$, this is reduced to

$$\mathbf{u}' \cdot \mathbf{u} = \cos(\phi),$$

hence

$$\frac{1}{2}(1 + \mathbf{u}' \cdot \mathbf{u}) = \cos^2(\phi/2), \quad \frac{1}{2}(1 - \mathbf{u}' \cdot \mathbf{u}) = \sin^2(\phi/2).$$

These are the familiar spin- $\frac{1}{2}$ transition probabilities. Without normalization and the irreducible-state qualification, only projector overlaps are represented.

1.11.6 Ladder Operations About the z Axis

For raising and lowering about the z axis, the following axis and projector pair are chosen:

$$\mathbf{u} = \mathbf{e}_3 = (0, 0, 1), \quad \Pi(\pm) = \frac{1}{2}(1 \pm \sigma_3).$$

The ladder elements already defined in Eq. (68) are used:

$$a(\pm) = \frac{1}{2}(\sigma_1 \pm i\sigma_2).$$

The commutator relation is satisfied:

$$\text{com}(\sigma_3, a(\pm)) = \pm 2a(\pm),$$

so the σ_3 eigenvalue is raised by $a(+)$ and lowered by $a(-)$.

Their projector products are

$$a(+)a(-) = \Pi(+), \quad a(-)a(+) = \Pi(-),$$

and each ladder operator is nilpotent:

$$a(\pm)^2 = 0.$$

The one-sided projector actions are

$$\begin{aligned} a(+)\Pi(+) &= 0, & a(+)\Pi(-) &= a(+), \\ \Pi(+)a(+) &= a(+), & \Pi(-)a(+) &= 0, \\ a(-)\Pi(-) &= 0, & a(-)\Pi(+) &= a(-), \\ \Pi(-)a(-) &= a(-), & \Pi(+)a(-) &= 0, \end{aligned}$$

by which the raising/lowering interpretation is made explicit. The original sigma generators are recovered from the same ladder/projector data:

$$\begin{aligned} \sigma_1 &= a(+) + a(-), & \sigma_2 &= \frac{a(+) - a(-)}{i}, \\ \sigma_3 &= \Pi(+) - \Pi(-). \end{aligned}$$

1.11.7 Spin-Axis Noncommutativity

For unit directions $\mathbf{u}, \mathbf{u}' \in \mathbb{R}^3$,

$$\begin{aligned} \text{acom}(\mathbf{u} \cdot \boldsymbol{\sigma}, \mathbf{u}' \cdot \boldsymbol{\sigma}) &= 2\mathbf{u} \cdot \mathbf{u}', \\ \text{com}(\mathbf{u} \cdot \boldsymbol{\sigma}, \mathbf{u}' \cdot \boldsymbol{\sigma}) &= 2i(\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}. \end{aligned}$$

The axis overlap is recorded by the anticommutator, while the oriented area spanned by the two axes is recorded by the commutator. Thus exact commutation of the axis operators is obtained when the directions are parallel or antiparallel. These identities are purely algebraic. A normalized state, expectation functional, and state-dependent variances are additionally required for a Robertson or Robertson–Schrödinger uncertainty relation; no such inequality is inferred from the commutator alone [11].

1.12 Quaternion Correspondence

After development of the circular exponential and rotor law, the same sigma-number-two algebra can be restated in Hamilton's quaternion language without introduction of a new multiplication rule. The correspondence is collected here after the rotation discussion.

1.12.1 Quaternion Basis and Involutions

Hamilton's quaternion units are also reproduced by the same sigma-number-two basis used in Eq. (78), so the rotation algebra can be written in quaternion form without leaving the STA multiplication table [1, 3]:

$$\begin{aligned} q_1 &= -\sigma_2\sigma_3, & q_2 &= -\sigma_3\sigma_1, \\ q_3 &= -\sigma_1\sigma_2, & \mathbf{q} &:= (q_1, q_2, q_3) = -i\boldsymbol{\sigma}. \end{aligned} \quad (97)$$

The standard quaternion identities are satisfied by these basis elements:

$$q_1^2 = q_2^2 = q_3^2 = q_1q_2q_3 = -1.$$

Thus the quaternion units are not new generators; the existing sigma-number-two basis is only relabeled.

If $\mathbf{B} = (B_1, B_2, B_3) \in \mathbb{R}^3$, then $\mathbf{B} \cdot \mathbf{q} = -i\mathbf{B} \cdot \boldsymbol{\sigma}$. Since $q_k = -i\sigma_k$, the following relations are obtained immediately from the abstract operator rules:

$$\mathbf{q}^* = \mathbf{q}, \quad \mathbf{q}^H = -\mathbf{q}.$$

Thus the quaternion basis is fixed by the conjugate operator, while its sign is changed by the Hermitian operation.

1.12.2 Quaternion Products

For $a \in \mathbb{R}$ and $\mathbf{B} \in \mathbb{R}^3$, the quaternion is defined by

$$Q := a + \mathbf{B} \cdot \mathbf{q} = a - i\mathbf{B} \cdot \boldsymbol{\sigma}. \quad (98)$$

In complex-paravector notation, the same quaternion may be written as

$$Q = X - iY, \quad X = a + \mathbf{A} \cdot \boldsymbol{\sigma}, \quad Y = b + \mathbf{B} \cdot \boldsymbol{\sigma},$$

with the quaternion specialization

$$\mathbf{A} = 0, \quad b = 0.$$

Under this specialization, $X = a$ is purely scalar and $Y = \mathbf{B} \cdot \boldsymbol{\sigma}$ is purely vector. Because $\mathbf{q}^* = \mathbf{q}$,

$$Q^* = Q.$$

Therefore

$$\begin{aligned} QQ^* &= Q^2 \\ &= (a - i\mathbf{B} \cdot \boldsymbol{\sigma})^2 \\ &= a^2 - 2ia\mathbf{B} \cdot \boldsymbol{\sigma} + (-i)^2(\mathbf{B} \cdot \boldsymbol{\sigma})^2 \\ &= a^2 - \mathbf{B}^2 - 2ia\mathbf{B} \cdot \boldsymbol{\sigma}. \end{aligned} \quad (99)$$

The factor a must be carried by the linear term. This is just the sigma-number-two multiplication law written in quaternion language.

Since $\mathbf{q}^H = -\mathbf{q}$, the Hermitian counterpart of Q is

$$Q^H = a - \mathbf{B} \cdot \mathbf{q} = a + i\mathbf{B} \cdot \boldsymbol{\sigma}.$$

The result of multiplication by the Hermitian counterpart is

$$QQ^H = a^2 + \mathbf{B}^2. \quad (100)$$

Thus the standard positive quaternion norm square is recovered by the Hermitian product, while the ordinary algebra square is recorded by Eq. (99).

1.12.3 Quaternion Form of the Rotor Law

For a unit quaternion Q satisfying $QQ^H = 1$, and for a real vector $\mathbf{r} \in \mathbb{R}^3$, the quaternion rotation law is

$$\mathbf{r}' \cdot \mathbf{q} = Q(\mathbf{r} \cdot \mathbf{q})Q^H. \quad (101)$$

This is the quaternion version of the rotor action $\mathbf{r}' \cdot \boldsymbol{\sigma} = R^H(\mathbf{r} \cdot \boldsymbol{\sigma})R$. With the choice

$$Q = \cos\left(\frac{\theta}{2}\right) + \sin\left(\frac{\theta}{2}\right)\mathbf{u} \cdot \mathbf{q}$$

the relation $Q = R^H$ is obtained because $\mathbf{q} = -i\boldsymbol{\sigma}$. Consequently

$$Q(\mathbf{r} \cdot \mathbf{q})Q^H = R^H(\mathbf{r} \cdot \mathbf{q})R,$$

which has exactly the same orientation as the sigma-vector rotor law. The equivalent quaternion sandwich $Q^H(\mathbf{r} \cdot \mathbf{q})Q$ is required under the alternative choice $Q = R$. Thus quaternion rotations are the same circular half-angle algebra, with the order fixed by the selected orientation convention.

1.13 Matrix and Auxiliary Realizations

The algebra above is complete on its own, but two realizations are useful. In the Pauli representation, one paravector is identified with a two-by-two complex matrix and $Z \text{adj}(Z)$ is identified with its determinant [9]. An auxiliary two-by-two block map whose entries are paravectors is defined after the standard realization; after the Pauli realization it is represented by a four-by-four complex matrix. It is used as a packaging map, not as a second representation or an algebra homomorphism.

1.13.1 Standard Two-by-Two Matrix Realization

The real associative algebra generated by the three sigma elements is commonly called the real Pauli algebra or the three-dimensional real Clifford algebra. The information carried by the even products in the usual flat space-time algebra is retained, and a faithful representation is provided by two-by-two complex matrices. Its eight real coefficients are grouped by the internal pseudoscalar into four complex coefficients: one scalar and one three-vector. Context is provided by these established descriptions, while only the previously defined operations are used in the construction [9].

For compactness, the notation $\{a, b; c, d\}$ is used for the two-by-two matrix with first row a, b and second row c, d , where every entry is a complex scalar. Each matrix operation used below is defined separately:

$$\mathbf{I} := \{1, 0; 0, 1\}. \quad (102)$$

$$\text{tr}(\{a, b; c, d\}) := a + d. \quad (103)$$

$$\text{diag}(A, B) := \{A, 0; 0, B\}. \quad (104)$$

$$\{a, b; c, d\}^T := \{a, c; b, d\}. \quad (105)$$

For a scalar entry $z = x + iy$, the matrix-only conjugate is

$$z^c := x - iy. \quad (106)$$

$$\{a, b; c, d\}^c := \{a^c, b^c; c^c, d^c\}. \quad (107)$$

$$\{a, b; c, d\}^{cT} := \{a^c, c^c; b^c, d^c\}. \quad (108)$$

$$\text{adj}(\{a, b; c, d\}) := \{d, -b; -c, a\}. \quad (109)$$

$$\det(\{a, b; c, d\}) := ad - bc. \quad (110)$$

The sign of i in each scalar entry is changed by the matrix-only superscript c , and nothing else is changed. Entrywise conjugation followed by transpose is denoted by the superscript cT ; the two operations commute. The superscript c is reserved for matrices and their scalar entries. It is not an additional intrinsic operation on the sigma algebra.

The standard realization is

$$\sigma_1 = \{0, 1; 1, 0\}, \quad \sigma_2 = \{0, -i; i, 0\}, \quad \sigma_3 = \{1, 0; 0, -1\}. \quad (111)$$

The entry i is the same central unit defined in Eq. (2); as a scalar matrix entry it obeys $i^2 = -1$. No second imaginary unit is introduced. The distinction in the matrix model lies in the operation being applied. Every displayed i , including the one built into σ_2 , is conjugated by entrywise c . Every abstract sigma generator is additionally flipped by intrinsic star.

The abstract pseudoscalar is represented by the matrix product $\sigma_1\sigma_2\sigma_3 = i\mathbf{I}$. In this realization,

$$\sigma_n^2 = \mathbf{I}, \quad \sigma_m\sigma_n = -\sigma_n\sigma_m \quad (m \neq n), \quad \sigma_1\sigma_2\sigma_3 = i\mathbf{I}.$$

Thus Eq. (111) is a concrete model of the sigma generators and their pseudoscalar.

For the generic complex paravector $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$, its represented matrix is denoted by $\mathbf{M}$:

$$\mathbf{M} := \{S + V_3, V_1 - iV_2; V_1 + iV_2, S - V_3\}. \quad (112)$$

Thus Z is represented by $\mathbf{M}$. Conversely, if $\mathbf{M} = \{a, b; c, d\}$, then

$$\begin{aligned} S &= \frac{a+d}{2}, & V_1 &= \frac{b+c}{2}, \\ V_2 &= \frac{c-b}{2i}, & V_3 &= \frac{a-d}{2}. \end{aligned} \quad (113)$$

The realization in both directions is therefore given by Eqs. (112) and (113).

For a real paravector $X = t + \mathbf{r} \cdot \boldsymbol{\sigma} \in \mathbb{R} + \mathbb{R}^3$, with $\mathbf{r} = (r_1, r_2, r_3) \in \mathbb{R}^3$, the matrix form is

$$\begin{aligned} X &= t + \mathbf{r} \cdot \boldsymbol{\sigma} \\ &\text{is represented by} \\ \{t + r_3, r_1 - ir_2; r_1 + ir_2, t - r_3\}, \\ \det(X) &= t^2 - \mathbf{r}^2. \end{aligned} \quad (114)$$

The real Minkowski quadratic form in Eq. (55) is reproduced by Eq. (114), together with

$$\begin{aligned} \text{tr}(\mathbf{M}) &= 2t, \\ \lambda^2 - \text{tr}(\mathbf{M})\lambda + \det(\mathbf{M}) &= 0, \\ \lambda(X, \pm) &= t \pm \|\mathbf{r}\|. \end{aligned}$$

These eigenvalues are the matrix form of the split into scalar part t and vector magnitude $\|\mathbf{r}\|$.

The sigma-basis action not captured by entrywise c is accounted for by the fixed matrix

$$\mathbf{T} := \{0, 1; -1, 0\}. \quad (115)$$

It satisfies $\mathbf{T}^{-1} = -\mathbf{T}$. The complete intrinsic-to-matrix dictionary is

$$\begin{aligned} Z &\text{ is represented by } \mathbf{M}, \\ Z^* &\text{ is represented by } \mathbf{M}^*, \\ \mathbf{M}^* &= \mathbf{T} \mathbf{M}^c \mathbf{T}^{-1} \\ &= \{d^c, -c^c; -b^c, a^c\}, \\ Z^H &\text{ is represented by } \mathbf{M}^H, \\ \mathbf{M}^H &= \mathbf{M}^{cT} \\ &= \{a^c, c^c; b^c, d^c\}, \\ \text{adj}(Z) &= Z^{*H}, \\ \text{adj}(Z) &\text{ is represented by } \text{adj}(\mathbf{M}), \\ \text{adj}(\mathbf{M}) &= \mathbf{T} \mathbf{M}^T \mathbf{T}^{-1} \\ &= \{d, -b; -c, a\}, \\ \det(Z) &\text{ is represented by } \det(\mathbf{M}) = ad - bc, \\ 0 &\text{ is represented by } \{0, 0; 0, 0\}, \\ 1 &\text{ is represented by } \mathbf{I}, \\ Z^{-1} &\text{ is represented by } \mathbf{M}^{-1}, \\ \mathbf{M}^{-1} &= \frac{\text{adj}(\mathbf{M})}{\det(\mathbf{M})}, \quad \det(\mathbf{M}) \neq 0. \end{aligned} \quad (116)$$

Thus $\mathbf{M}^H = \mathbf{M}^{cT}$ is the ordinary Hermitian adjoint. A matrix is Hermitian precisely when $\mathbf{M}^H = \mathbf{M}$, while the matrix adjugate is always denoted by $\text{adj}(\mathbf{M})$. Crucially, intrinsic star is not matrix c : $\mathbf{M}^* \neq \mathbf{M}^c$ in general.

The star on the left is the intrinsic sigma conjugate transported into the matrix realization; the c on the right is only entrywise complex conjugation. They agree on central complex scalars, but not on a generic paravector. When Z is represented by $\mathbf{M}$, the direct bridge is that Z^H is represented by $\mathbf{M}^{cT}$. Intrinsic star instead requires the basis correction $\mathbf{M}^* = \mathbf{T} \mathbf{M}^c \mathbf{T}^{-1}$.

The extraction functions have the equally direct matrix form

$$\begin{aligned} \text{scalar}(Z) &\text{ is represented by } \frac{1}{2} \text{tr}(\mathbf{M}), \\ \text{vector}(Z) \cdot \boldsymbol{\sigma} &\text{ is represented by } \mathbf{M} - \frac{1}{2} \text{tr}(\mathbf{M}) \mathbf{I}, \\ \text{real}(Z) &\text{ is represented by } \frac{1}{2} (\mathbf{M} + \mathbf{M}^H), \\ \text{img}(Z) &\text{ is represented by } \frac{\mathbf{M} - \mathbf{M}^H}{2i}. \end{aligned} \quad (117)$$

Consequently, the algebraic determinant is exactly the ordinary two-by-two matrix determinant:

$$\mathbf{M} \text{adj}(\mathbf{M}) = \text{adj}(\mathbf{M}) \mathbf{M} = \det(\mathbf{M}) \mathbf{I}. \quad (118)$$

An unambiguous correspondence is obtained among the operation types. Matrix c is an antilinear automorphism and matrix T is a linear anti-automorphism. Their composition cT , represented intrinsically by H , is an antilinear anti-automorphism. The intrinsic star is a different antilinear automorphism, and the combined intrinsic operation $*H$ is the linear anti-automorphism represented by the matrix adjugate. Consequently, Eq. (50) is represented by

$$\mathbf{M}^H = \text{adj}(\mathbf{M})^* = \text{adj}(\mathbf{M}^*),$$

where the star in $\mathbf{M}^*$ remains the induced sigma conjugate, not entrywise c .

The distinction is made concrete by the basis actions:

$$\begin{aligned} \sigma_1^c &= \sigma_1, & \sigma_2^c &= -\sigma_2, & \sigma_3^c &= \sigma_3, \\ \sigma_1^T &= \sigma_1, & \sigma_2^T &= -\sigma_2, & \sigma_3^T &= \sigma_3, \\ \sigma_n^{cT} &= \sigma_n, & \sigma_n^* &= -\sigma_n, & \sigma_n^{*H} &= -\sigma_n. \end{aligned} \quad (119)$$

On these three individual generator matrices, the same sign table is obtained from c and T . Distinct operations are nevertheless represented: $(\mathbf{A}\mathbf{B})^c = \mathbf{A}^c \mathbf{B}^c$, whereas $(\mathbf{A}\mathbf{B})^T = \mathbf{B}^T \mathbf{A}^T$. Only the represented second generator is changed by entrywise c , while all three are changed by intrinsic star. Every σ_n is fixed by their composition cT , exactly matching intrinsic H . Thus c is a basis-dependent matrix-coordinate operation and is distinct from both intrinsic involutions. In particular, it is not intrinsic star. For $n \neq m$,

$$\sigma_n \sigma_m^H = \sigma_m \sigma_n = -\sigma_n \sigma_m,$$

exactly as required by the intrinsic order-reversal rule.

1.13.2 Auxiliary Block Map

A scalar-coefficient complex paravector $Z \in \mathbb{C} + \mathbb{C}^3$ is packaged into the 2×2 block map

$$\mathbf{W}(Z) := \{0, Z; \text{adj}(Z), 0\}. \quad (120)$$

$$\mathbf{W}(Z)^2 = \text{diag}(\det(Z), \det(Z)) = \det(Z) \mathbf{I}. \quad (121)$$

The block map $\mathbf{W}(Z)$ is a 2×2 matrix over the paravector algebra. After each block is replaced by its Pauli matrix, it is a 4×4 complex matrix. It is a useful linear packaging map, not an algebra homomorphism. Equation (121) is obtained because its diagonal block products are reduced to the determinant in Eq. (53); the off-diagonal blocks are zero [12, 14].

If $\det(Z) \neq 0$, the annihilating polynomial $x^2 - \det(Z)$ has distinct roots, so the block matrix is diagonalizable after a square-root branch is chosen. If $\det(Z) = 0$ and $Z \neq 0$, then $\mathbf{W}(Z)^2 = 0$ while $\mathbf{W}(Z) \neq 0$: it is nilpotent and not diagonalizable. For differential or other noncommuting operator-valued blocks, the unreduced statement is $\mathbf{W}(Z)^2 = \text{diag}(ZZ^{*\text{H}}, Z^{*\text{H}}Z)$. A separate operator calculation is required before either ordered product can be called a determinant or both can be reduced to one scalar multiple of the identity.

2 Space-Time Kinematics

On real event paravectors, the Minkowski quadratic form is obtained from the determinant form and four-velocities are represented by normalized timelike paravectors. Standard special-relativistic kinematics is thereby expressed in paravector form [4–6, 12, 13].

2.1 Differential space-time and four-velocity

The differential event is obtained by specializing the real paravector X from Eq. (12):

$$dX := dt + d\mathbf{r} \cdot \boldsymbol{\sigma} \quad (122)$$

The interval is the differential version of Eq. (53):

$$ds^2 = \det(dX) = dX dX^* = dt^2 - d\mathbf{r}^2. \quad (123)$$

Here $t, s, \gamma \in \mathbb{R}$, $\mathbf{r}, \mathbf{v}, \mathbf{P} \in \mathbb{R}^3$, and X, U are real paravectors. Equation (123) is the real-event restriction of the determinant in Eq. (53). For a future-directed timelike worldline, $dt > 0$ and $ds = \sqrt{dt^2 - d\mathbf{r}^2} > 0$.

The coordinate velocity and Lorentz factor are defined by $\mathbf{v} := d\mathbf{r}/dt \in \mathbb{R}^3$ and $\gamma := (1 - \mathbf{v}^2)^{-1/2} \in \mathbb{R}$. The four-velocity is defined by

$$U := \frac{dX}{ds} = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}). \quad (124)$$

Its normalization is

$$\det(U) = 1. \quad (125)$$

The energy and spatial momentum of a particle with real rest mass $m > 0$ are defined by

$$E := m\gamma. \quad (126)$$

$$\mathbf{P} := m\gamma\mathbf{v}. \quad (127)$$

The four-momentum is therefore

$$mU = E + \mathbf{P} \cdot \boldsymbol{\sigma}. \quad (128)$$

from which $E^2 - \mathbf{P}^2 = m^2$ is obtained immediately. The normalized tangent U is conventionally called the four-velocity; only its spatial part $\gamma\mathbf{v}$ is often denoted by “proper velocity.” The four-momentum of a massive particle is obtained by multiplication by $m > 0$. For a massless particle $ds = 0$, so U is undefined and a null momentum paravector must instead be introduced directly.

2.2 Four-acceleration and free motion

The four-acceleration is the real paravector

$$A := \frac{dU}{ds} = \frac{d^2X}{ds^2}. \quad (129)$$

It obeys

$$\text{scalar}(U^*A) = 0. \quad (130)$$

The orthogonality condition in Eq. (129) is the standard four-acceleration orthogonality relation. The invariant proper acceleration is the magnitude $\sqrt{-\det(A)}$, not the paravector A itself.

A free-particle Lagrangian is defined by

$$\mathcal{L} := -\frac{m}{\gamma}, \quad (131)$$

from whose Euler–Lagrange equations inertial motion and the usual nonrelativistic expansion are reproduced:

$$\mathcal{L} = -m + \frac{1}{2}m\mathbf{v}^2 + O(\mathbf{v}^4). \quad (132)$$

Equation (132) is displayed explicitly to exhibit the reduction of the relativistic action to the classical kinetic expression plus an additive rest-energy constant in the low-speed limit.

2.3 Supporting Kinematic Derivations

Equations (123), (124), and (128) are expanded in the supporting derivations, followed by the generalized-coordinate and metric remarks associated with the geometric part of the development.

2.3.1 Space-Time Numbers

A spacetime increment is the real paravector

$$dX = dt + dr_1\sigma_1 + dr_2\sigma_2 + dr_3\sigma_3 = dt + d\mathbf{r} \cdot \boldsymbol{\sigma},$$

with conjugate

$$dX^* = dt - dr_1\sigma_1 - dr_2\sigma_2 - dr_3\sigma_3 = dt - d\mathbf{r} \cdot \boldsymbol{\sigma}.$$

Here $dt \in \mathbb{R}$ is the scalar part and $d\mathbf{r} \in \mathbb{R}^3$ is the spatial vector part. Their recovery formulas are

$$\frac{dX + dX^*}{2} = dt, \quad \frac{dX - dX^*}{2} = d\mathbf{r} \cdot \boldsymbol{\sigma},$$

so

$$\text{scalar}(dX) = dt, \quad \text{vector}(dX) = d\mathbf{r}.$$

The determinant of the increment is

$$\det(dX) = dX dX^* = dX^* dX = dt^2 - d\mathbf{r}^2 = ds^2.$$

The familiar Lorentzian identities are obeyed:

$$\begin{aligned} dt^2 > d\mathbf{r}^2, dX \neq 0 &\text{ implies } \det(dX) > 0, \\ \det(dX + dX') - \det(dX) &= 2(dt dt' - d\mathbf{r} \cdot d\mathbf{r}'), \\ \det(a dX) = a^2 \det(dX), & \quad a \in \mathbb{R}. \end{aligned}$$

The commutator of two spacetime increments is purely spatial:

$$\begin{aligned} \text{com}(dX, dX') &= \text{com}(d\mathbf{r} \cdot \boldsymbol{\sigma}, d\mathbf{r}' \cdot \boldsymbol{\sigma}) \\ &= 2i(d\mathbf{r} \times d\mathbf{r}') \cdot \boldsymbol{\sigma}. \end{aligned}$$

Along a future-directed time-like path, so that $dt > 0$, the following relation is satisfied by the positive arc length:

$$\begin{aligned} ds &= \sqrt{dt^2 - d\mathbf{r}^2} \\ &= \sqrt{1 - d\mathbf{r}^2/dt^2} dt \\ &= \sqrt{1 - \mathbf{v}^2} dt, \end{aligned}$$

where

$$\mathbf{v} := \frac{d\mathbf{r}}{dt} = (v_1, v_2, v_3) \in \mathbb{R}^3, \quad \gamma := \frac{dt}{ds} = \frac{1}{\sqrt{1 - \mathbf{v}^2}}.$$

For a real non-null increment, $dX^H = dX$, so the inverse exists and

$$\begin{aligned} (dX)^{-1} &= \frac{\text{adj}(dX)}{\det(dX)} \\ &= \frac{dX^*}{\det(dX)} \\ &= \frac{dX^*}{dX dX^*} \\ &= \frac{dX^*}{ds^2}. \end{aligned}$$

Null displacements are singled out algebraically because this inverse breaks down exactly when $ds^2 = 0$.

2.3.2 Interval Classes and Proper Time

For a spacetime increment $dX = dt + d\mathbf{r} \cdot \boldsymbol{\sigma}$, the interval is

$$\det(dX) = dX dX^* = dt^2 - d\mathbf{r}^2 = ds^2.$$

The three standard interval classes are obtained from this identity without any additional geometric machinery:

$$\begin{aligned} ds^2 > 0 &\text{ implies a time-like interval,} \\ ds^2 = 0, dX \neq 0 &\text{ implies a null interval,} \\ ds^2 < 0 &\text{ implies a space-like interval.} \end{aligned}$$

If $\det(dX) > 0$, the interval is time-like. Then the arc length s equals the proper time τ measured along the path:

$$\begin{aligned} \tau = s &= \int ds = \int d\tau = \int \sqrt{1 - \mathbf{v}^2} dt \\ &= \int \frac{1}{\gamma} dt. \end{aligned}$$

This is the time-dilation relation in natural units. In SI units the same statement is

$$ds = c d\tau.$$

If $\det(dX) = 0$ with $dX \neq 0$, the interval is null or light-like. Then

$$dt^2 = d\mathbf{r}^2 \quad \text{implies} \quad dt = \pm \|d\mathbf{r}\|,$$

and $(dX)^{-1}$ does not exist.

If $\det(dX) < 0$, the interval is space-like.

In the paravector notation, the interval classification, the proper-time formula, and the null singularity of $(dX)^{-1}$ are all obtained from the single algebraic identity $dX dX^* = dt^2 - d\mathbf{r}^2$.

2.3.3 Four-Velocity

The four-velocity paravector is the normalized tangent to a time-like worldline:

$$\begin{aligned} \frac{dX}{ds} &= \frac{dt}{ds} + \frac{d\mathbf{r}}{ds} \cdot \boldsymbol{\sigma} \\ &= \frac{dt + d\mathbf{r} \cdot \boldsymbol{\sigma}}{\sqrt{dt^2 - d\mathbf{r}^2}} \\ &= \frac{dt/dt + \boldsymbol{\sigma} \cdot (d\mathbf{r}/dt)}{\sqrt{(dt/dt)^2 - (d\mathbf{r}/dt)^2}} \\ &= \frac{1 + \mathbf{v} \cdot \boldsymbol{\sigma}}{\sqrt{1 - \mathbf{v}^2}} \\ &= \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}). \end{aligned}$$

Its scalar and vector parts are therefore

$$\text{scalar}\left(\frac{dX}{ds}\right) = \gamma, \quad \text{vector}\left(\frac{dX}{ds}\right) = \gamma\mathbf{v}.$$

Its determinant is unity:

$$\begin{aligned} \det\left(\frac{dX}{ds}\right) &= \gamma^2(1 + \mathbf{v} \cdot \boldsymbol{\sigma})(1 - \mathbf{v} \cdot \boldsymbol{\sigma}) \\ &= \gamma^2(1 - \mathbf{v}^2) \\ &= 1. \end{aligned}$$

Also,

$$\frac{dX}{dt} = \frac{dX}{ds} \frac{ds}{dt} = \frac{1}{\gamma} \frac{dX}{ds}.$$

The four-velocity paravector is defined by

$$\begin{aligned} U &= \frac{dX}{ds} \\ &= \gamma + \gamma\mathbf{v} \cdot \boldsymbol{\sigma} \\ &= U_s + \mathbf{U}_v \cdot \boldsymbol{\sigma}, \\ U_s &= \gamma, \quad \mathbf{U}_v = \gamma\mathbf{v}. \end{aligned}$$

Since $ds^2 = dX dX^*$ with $ds > 0$,

$$\begin{aligned}\frac{dX^*}{ds} &= (dX)^{-1} ds \\ &= \left(\frac{dX}{ds}\right)^{-1},\end{aligned}$$

so $U^{-1} = U^*$ and

$$\det(U) = UU^* = 1.$$

The following relation is obtained by differentiation of this normalization:

$$\begin{aligned}\frac{d}{ds} \det(U) &= \frac{dU}{ds} U^* + U \left(\frac{dU}{ds}\right)^* \\ &= \left(\frac{dU}{ds} U^*\right) + \left(\frac{dU}{ds} U^*\right)^{*H},\end{aligned}$$

hence

$$\text{scalar}\left(U^* \frac{dU}{ds}\right) = 0.$$

The determinant identity is recovered as

$$\begin{aligned}\det(dX) &= dX dX^* \\ &= \left(\frac{dX}{ds} ds\right) \left(\left(\frac{dX}{ds}\right)^* ds\right) \\ &= UU^* ds^2 \\ &= \det(U) ds^2 \\ &= ds^2.\end{aligned}$$

The proper-acceleration formulas are based on this normalization and orthogonality structure. The four-velocity paravector is resolved into its scalar and spatial components in the following table.

Table 16. Components of dX/ds .

scalar:	γ
vector:	$\gamma \mathbf{v}$

2.3.4 Four-Acceleration

The coordinate acceleration is defined by

$$\mathbf{a} := \frac{d\mathbf{v}}{dt} = \{a_1, a_2, a_3\} \in \mathbb{R}^3.$$

With $\gamma = (1 - \mathbf{v}^2)^{-1/2}$, the following relations hold:

$$\begin{aligned}\frac{d\gamma}{dt} &= \gamma^3(\mathbf{v} \cdot \mathbf{a}), \\ \frac{d\gamma}{ds} &= \gamma^4(\mathbf{v} \cdot \mathbf{a}).\end{aligned}$$

The notation $\partial_{\mathbf{v}} := \partial/\partial \mathbf{v}$ is used for the gradient with respect to velocity. The scalar velocity derivative needed below is

$$\partial_{\mathbf{v}}\left(-\frac{1}{\gamma}\right) = \gamma \mathbf{v}.$$

The following derivatives are obtained by direct differentiation of the vector product $\gamma \mathbf{v}$:

$$\begin{aligned}\frac{d(\gamma \mathbf{v})}{dt} &= \gamma \mathbf{a} + \frac{d\gamma}{dt} \mathbf{v} \\ &= \gamma \mathbf{a} + \gamma^3(\mathbf{v} \cdot \mathbf{a}) \mathbf{v}, \\ \frac{d(\gamma \mathbf{v})}{ds} &= \frac{d(\gamma \mathbf{v})}{dt} \frac{dt}{ds} \\ &= \gamma^2 \mathbf{a} + \gamma^4(\mathbf{v} \cdot \mathbf{a}) \mathbf{v}.\end{aligned}$$

The four-velocity paravector is decomposed as

$$\begin{aligned}U &= U_s + \mathbf{U}_{\mathbf{v}} \cdot \boldsymbol{\sigma}, \\ U_s &= \gamma, \quad \mathbf{U}_{\mathbf{v}} = \gamma \mathbf{v}.\end{aligned}$$

Therefore the four-acceleration is

$$\begin{aligned}A &= \frac{dU}{ds} \\ &= \frac{d\gamma}{ds} + \frac{d(\gamma \mathbf{v})}{ds} \cdot \boldsymbol{\sigma} \\ &= \gamma^4(\mathbf{v} \cdot \mathbf{a}) + \left(\gamma^4(\mathbf{v} \cdot \mathbf{a}) \mathbf{v} + \gamma^2 \mathbf{a}\right) \cdot \boldsymbol{\sigma}.\end{aligned}$$

Equivalently,

$$A = A_s + \mathbf{A}_{\mathbf{v}} \cdot \boldsymbol{\sigma}, \quad A_s = \gamma^4(\mathbf{v} \cdot \mathbf{a}), \quad \mathbf{A}_{\mathbf{v}} = \gamma^4(\mathbf{v} \cdot \mathbf{a}) \mathbf{v} + \gamma^2 \mathbf{a}.$$

For $\mathbf{v} \neq 0$, the unit direction is defined by $\mathbf{u} = \mathbf{v}/\|\mathbf{v}\|$, with $\mathbf{u}^2 = 1$. The following components are then obtained from the axis functions in Eqs. (34) and (35):

$$\begin{aligned}\text{par}(\mathbf{a}) &= (\mathbf{a} \cdot \mathbf{u}) \mathbf{u}, \\ \text{prp}(\mathbf{a}) &= \mathbf{a} - \text{par}(\mathbf{a}).\end{aligned}$$

Then

$$\begin{aligned}\mathbf{a}^2 &= \text{par}(\mathbf{a})^2 + \text{prp}(\mathbf{a})^2, \\ (\mathbf{v} \cdot \mathbf{a})^2 &= \mathbf{v}^2 \text{par}(\mathbf{a})^2,\end{aligned}$$

so the proper-acceleration determinant is

$$\begin{aligned}\det(A) &= -\gamma^4 \left(\mathbf{a}^2 + \gamma^2 (\mathbf{v} \cdot \mathbf{a})^2 \right) \\ &= -\gamma^4 \left(\text{prp}(\mathbf{a})^2 + \gamma^2 \text{par}(\mathbf{a})^2 \right).\end{aligned}$$

This is strictly space-like unless the acceleration vanishes. Its invariant magnitude $\sqrt{-\det(A)}$ is the proper acceleration measured in the instantaneous rest frame.

The products with U are

$$\begin{aligned}UA &= (U_s A_s + \mathbf{U}_{\mathbf{v}} \cdot \mathbf{A}_{\mathbf{v}}) \\ &\quad + (U_s \mathbf{A}_{\mathbf{v}} + A_s \mathbf{U}_{\mathbf{v}} + i(\mathbf{U}_{\mathbf{v}} \times \mathbf{A}_{\mathbf{v}})) \cdot \boldsymbol{\sigma}, \\ U^* A &= (U_s A_s - \mathbf{U}_{\mathbf{v}} \cdot \mathbf{A}_{\mathbf{v}}) \\ &\quad + (U_s \mathbf{A}_{\mathbf{v}} - A_s \mathbf{U}_{\mathbf{v}} - i(\mathbf{U}_{\mathbf{v}} \times \mathbf{A}_{\mathbf{v}})) \cdot \boldsymbol{\sigma}.\end{aligned}$$

Their component forms are

$$\begin{aligned} UA &= 2\gamma^5(\mathbf{v} \cdot \mathbf{a}) \\ &\quad + (\gamma^3\mathbf{a} + 2\gamma^5(\mathbf{v} \cdot \mathbf{a})\mathbf{v}) \cdot \boldsymbol{\sigma} \\ &\quad + i\gamma^3(\mathbf{v} \times \mathbf{a}) \cdot \boldsymbol{\sigma}, \\ U^*A &= \gamma^3\mathbf{a} \cdot \boldsymbol{\sigma} - i\gamma^3(\mathbf{v} \times \mathbf{a}) \cdot \boldsymbol{\sigma}. \end{aligned}$$

In particular,

$$\text{scalar}(U^*A) = 0,$$

which is the orthogonality relation between four-velocity and four-acceleration. The scalar and vector parts of U are recorded in Table 16, while the corresponding scalar and vector parts of UA and U^*A are compared in Tables 17 and 18. The products UA and U^*A are compared in the following two tables, and the orthogonality structure of proper acceleration is thereby made explicit.

Table 17. Components of UA .

real scalar:	$2\gamma^5(\mathbf{v} \cdot \mathbf{a})$
img scalar:	0
real vector:	$\gamma^3\mathbf{a} + 2\gamma^5(\mathbf{v} \cdot \mathbf{a})\mathbf{v}$
img vector:	$\gamma^3(\mathbf{v} \times \mathbf{a})$

Table 18. Components of U^*A .

real scalar:	0
img scalar:	0
real vector:	$\gamma^3\mathbf{a}$
img vector:	$-\gamma^3(\mathbf{v} \times \mathbf{a})$

2.3.5 Four-Momentum

For a massive particle with rest mass $m > 0$, the energy-momentum paravector is produced by multiplication of four-velocity by m :

$$mU = E + \mathbf{P} \cdot \boldsymbol{\sigma}.$$

Its scalar part is the relativistic energy,

$$\begin{aligned} E &:= \text{scalar}(mU) \\ &= m\gamma. \end{aligned}$$

If $\mathbf{v} = 0$, then

$$E = m \quad \text{rest energy} \quad (= mc^2 \text{ in SI units}).$$

The kinetic energy is therefore $E - m$.

Its vector part is the relativistic momentum,

$$\begin{aligned} \mathbf{P} &:= \text{vector}(mU) \\ &= m\gamma\mathbf{v}. \end{aligned}$$

Because $\det(U) = 1$, the mass shell is obtained immediately:

$$\begin{aligned} \det(mU) &= m^2 \\ &= (E + \mathbf{P} \cdot \boldsymbol{\sigma})(E - \mathbf{P} \cdot \boldsymbol{\sigma}) \\ &= E^2 - \mathbf{P}^2. \end{aligned}$$

Hence

$$E^2 = \mathbf{P}^2 + m^2.$$

The null branch $E^2 = \mathbf{P}^2$, appropriate to massless motion, is also admitted by the mass-shell equation. That branch must be introduced directly as a null energy-momentum paravector: Eq. $mU = E + \mathbf{P} \cdot \boldsymbol{\sigma}$ is applicable only to massive time-like motion because $ds = 0$ and $U = dX/ds$ is undefined on a null worldline.

2.3.6 Flat Paravector Coordinate Map

The scalar and vector maps are taken to have the domains

$$\begin{aligned} f &: \mathbb{R} \times \mathbb{R}^3 \rightarrow \mathbb{R}, \\ \mathbf{g} &: \mathbb{R} \times \mathbb{R}^3 \rightarrow \mathbb{R}^3. \end{aligned}$$

A real paravector-valued map into the fixed flat space-time algebra is then defined by

$$X := f(t, \mathbf{r}) + \mathbf{g}(t, \mathbf{r}) \cdot \boldsymbol{\sigma}. \quad (133)$$

The original coordinates are mapped into flat Minkowski paravector components; invariance of the Cartesian metric coefficients under an arbitrary coordinate change is not asserted. Its differential and conjugate are

$$dX = df + d\mathbf{g} \cdot \boldsymbol{\sigma}, \quad dX^* = df - d\mathbf{g} \cdot \boldsymbol{\sigma}.$$

Their product is

$$\begin{aligned} \det(dX) &= dX dX^* \\ &= (df + d\mathbf{g} \cdot \boldsymbol{\sigma})(df - d\mathbf{g} \cdot \boldsymbol{\sigma}) \\ &= df^2 - d\mathbf{g}^2. \end{aligned}$$

Along a future-directed time-like path, the conditions $df > 0$ and $df^2 > d\mathbf{g}^2$ are assumed. Then

$$\begin{aligned} ds &= \sqrt{df^2 - d\mathbf{g}^2} \\ &= \sqrt{1 - \left(\frac{d\mathbf{g}}{df}\right)^2} df. \end{aligned}$$

The corresponding four-velocity is

$$\begin{aligned} U &= \frac{dX}{ds} \\ &= \frac{df + d\mathbf{g} \cdot \boldsymbol{\sigma}}{\sqrt{df^2 - d\mathbf{g}^2}} \\ &= \frac{1 + \left(\frac{d\mathbf{g}}{df}\right) \cdot \boldsymbol{\sigma}}{\sqrt{1 - \left(\frac{d\mathbf{g}}{df}\right)^2}} \\ &= \frac{1 + \mathbf{v} \cdot \boldsymbol{\sigma}}{\sqrt{1 - \mathbf{v}^2}} \\ &= \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}), \end{aligned}$$

where the velocity relative to the scalar component f and its Lorentz factor are

$$\begin{aligned}\mathbf{v} &:= \frac{d\mathbf{g}}{df} \\ &= \frac{d\mathbf{g}/dt}{df/dt} \\ &= \frac{d\mathbf{g}/ds}{df/ds}, \\ \gamma &:= \frac{df}{ds} \\ &= \frac{df}{\sqrt{df^2 - d\mathbf{g}^2}} \\ &= \frac{df/dt}{ds/dt} \\ &= \frac{1}{\sqrt{1 - \mathbf{v}^2}}.\end{aligned}$$

Thus $\mathbf{v}^2 < 1$. For a massive particle, the energy–momentum relation remains unchanged:

$$mU = E + \mathbf{P} \cdot \boldsymbol{\sigma}.$$

The coordinate acceleration relative to f is defined, in direct analogy with $\mathbf{a} = d\mathbf{v}/dt$ above, by

$$\begin{aligned}\mathbf{a} &:= \frac{d\mathbf{v}}{df} \\ &= \frac{d^2\mathbf{g}}{df^2}.\end{aligned}$$

Because $d/ds = \gamma d/df$, the four-acceleration has the same form as before:

$$\frac{dU}{ds} = \gamma^4(\mathbf{v} \cdot \mathbf{a}) + (\gamma^4(\mathbf{v} \cdot \mathbf{a})\mathbf{v} + \gamma^2\mathbf{a}) \cdot \boldsymbol{\sigma}.$$

The commutator of four-velocity and four-acceleration is

$$\begin{aligned}\text{com}\left(U, \frac{dU}{ds}\right) &= \text{com}\left(U, \frac{dU}{df}\right) \frac{df}{ds} \\ &= \gamma \text{com}\left(U, \frac{dU}{df}\right) \\ &= 2\gamma^3(\mathbf{v} \times \mathbf{a}) \cdot (i\boldsymbol{\sigma}).\end{aligned}$$

For a massive free particle in these flat target components,

$$m \frac{dU}{ds} = 0,$$

which is equivalently the straight-worldline equation

$$\frac{d^2X}{ds^2} = 0.$$

After division by the nonzero factor γ , its scalar and vector parts are

$$\begin{aligned}\gamma^3(\mathbf{v} \cdot \mathbf{a}) &= 0, \\ \gamma^3(\mathbf{v} \cdot \mathbf{a})\mathbf{v} + \gamma\mathbf{a} &= 0.\end{aligned}$$

For $\mathbf{v}^2 < 1$, one has $\gamma > 0$. The relation $\mathbf{v} \cdot \mathbf{a} = 0$ is therefore obtained from the scalar part, and $\mathbf{a} = 0$ is then obtained from the vector part. Hence

$$\frac{d^2\mathbf{g}}{df^2} = 0,$$

and along an inertial worldline

$$\mathbf{g} = \mathbf{c}f + \mathbf{c}', \quad \mathbf{v} = \frac{d\mathbf{g}}{df} = \mathbf{c},$$

so inertial motion is linear in the flat target components $(f, \mathbf{g})$ with constant velocity. If the same straight worldline is described in arbitrary curvilinear coordinates, the corresponding connection terms are generally contained in its coordinate equation; no curved-space geodesic equation is asserted.

2.3.7 Lagrangian Mechanics and Conserved Quantities

For a particle trajectory written in terms of time, position, and velocity, the action is taken to be

$$S = \int \mathcal{L}(t, \mathbf{r}, \mathbf{v}) dt.$$

The Euler–Lagrange equation is then

$$\begin{aligned}\left(\partial_{\mathbf{r}} - \frac{d}{dt}\partial_{\mathbf{v}}\right)\mathcal{L} &= 0, \\ \frac{d}{dt}\partial_{\mathbf{v}}\mathcal{L} &= \partial_t\mathcal{L}.\end{aligned}$$

The standard conserved quantities are then obtained immediately. The canonical momentum is

$$\partial_{\mathbf{v}}\mathcal{L},$$

so if $\partial_t\mathcal{L} = 0$, then

$$\frac{d}{dt}\partial_{\mathbf{v}}\mathcal{L} = 0,$$

and the canonical momentum is constant.

The Hamiltonian is defined by

$$\begin{aligned}\mathcal{H} &= \mathbf{v} \cdot (\partial_{\mathbf{v}}\mathcal{L}) - \mathcal{L} \\ &= (\mathbf{v} \cdot \partial_{\mathbf{v}} - 1)\mathcal{L}.\end{aligned}$$

Its time derivative is

$$\begin{aligned}\frac{d\mathcal{H}}{dt} &= \frac{d}{dt}(\partial_{\mathbf{v}}\mathcal{L}) \cdot \mathbf{v} + (\partial_{\mathbf{v}}\mathcal{L}) \cdot \frac{d\mathbf{v}}{dt} - \frac{d\mathcal{L}}{dt} \\ &= (\partial_t\mathcal{L}) \cdot \mathbf{v} + (\partial_{\mathbf{v}}\mathcal{L}) \cdot \frac{d\mathbf{v}}{dt} - \frac{d\mathcal{L}}{dt} \\ &= (\partial_t\mathcal{L}) \cdot \mathbf{v} + (\partial_{\mathbf{v}}\mathcal{L}) \cdot \frac{d\mathbf{v}}{dt} \\ &\quad - \left(\partial_t\mathcal{L} + (\partial_{\mathbf{r}}\mathcal{L}) \cdot \mathbf{v} + (\partial_{\mathbf{v}}\mathcal{L}) \cdot \frac{d\mathbf{v}}{dt}\right) \\ &= -\partial_t\mathcal{L}.\end{aligned}$$

Therefore, if $\partial_t\mathcal{L} = 0$, then

$$\frac{d\mathcal{H}}{dt} = 0,$$

so the Hamiltonian is constant. These identities are applied directly to the free-particle and minimally coupled electromagnetic examples.

2.3.8 Action of a Free Particle

The free-particle action is the mass-weighted proper time along the world line. With $\tau = \int ds$, $U = dX/ds$, and $\gamma = dt/ds$,

$$\begin{aligned}
 S &= \int \mathcal{L} dt = -m\tau \\
 &= -m \int \sqrt{\det(dX)} \\
 &= -m \int ds \\
 &= -m \int \sqrt{\det\left(\frac{dX}{dt}\right)} dt \\
 &= -m \int \frac{ds}{dt} dt \\
 &= -m \int \sqrt{\det\left(\frac{U}{\gamma}\right)} dt \\
 &= -m \int \frac{1}{\gamma} dt \\
 &= -m \int \sqrt{1 - \mathbf{v}^2} dt.
 \end{aligned}$$

Therefore

$$\mathcal{L} = -\frac{m}{\gamma}.$$

The canonical momentum is obtained by differentiation with respect to velocity:

$$\begin{aligned}
 \partial_{\mathbf{v}} \mathcal{L} &= m\gamma \mathbf{v} \\
 &= \mathbf{P}, \\
 (\partial_{\mathbf{v}} \cdot \boldsymbol{\sigma}) \mathcal{L} &= \mathbf{P} \cdot \boldsymbol{\sigma}.
 \end{aligned}$$

The Hamiltonian is

$$\begin{aligned}
 \mathcal{H} &= (\partial_{\mathbf{v}} \mathcal{L}) \cdot \mathbf{v} - \mathcal{L} \\
 &= m\gamma \mathbf{v}^2 + \frac{m}{\gamma} \\
 &= m\gamma \left(\mathbf{v}^2 + \frac{1}{\gamma^2} \right) \\
 &= m\gamma \\
 &= E.
 \end{aligned}$$

Because the free Lagrangian has no explicit time or position dependence,

$$\partial_t \mathcal{L} = 0, \quad \partial_{\mathbf{r}} \mathcal{L} = 0,$$

so the conserved quantities are the relativistic energy $E = m\gamma$ and momentum $\mathbf{P} = m\gamma \mathbf{v}$.

2.3.9 Non-Relativistic Approximation

For the non-relativistic approximation, the free Lagrangian is expanded directly:

$$\begin{aligned}
 \mathcal{L} &= -\frac{m}{\gamma} \\
 &= -m + \frac{1}{2}m\mathbf{v}^2 + O(\mathbf{v}^4).
 \end{aligned}$$

The canonical momentum is then reduced to

$$\begin{aligned}
 \partial_{\mathbf{v}} \mathcal{L} &= \frac{\partial}{\partial \mathbf{v}} \left(\frac{1}{2}m\mathbf{v}^2 + O(\mathbf{v}^4) \right) \\
 &= m\mathbf{v} + O(\mathbf{v}^3) \\
 &\approx m\mathbf{v}.
 \end{aligned}$$

The total energy is the kinetic term plus the rest-mass term:

$$\begin{aligned}
 \mathcal{H} &= (m\mathbf{v} + O(\mathbf{v}^3)) \cdot \mathbf{v} + m - \frac{1}{2}m\mathbf{v}^2 - O(\mathbf{v}^4) \\
 &= m + \frac{1}{2}m\mathbf{v}^2 + O(\mathbf{v}^4) \\
 &\approx m + \frac{1}{2}m\mathbf{v}^2.
 \end{aligned}$$

Equivalently, the low-velocity series is

$$\begin{aligned}
 \mathcal{L} &= -m + \frac{1}{2}m\mathbf{v}^2 + O(\mathbf{v}^4), \\
 \mathbf{P} &= m\mathbf{v} + O(\mathbf{v}^3), \\
 \mathcal{H} &= m + \frac{1}{2}m\mathbf{v}^2 + O(\mathbf{v}^4).
 \end{aligned}$$

The Newtonian kinetic term together with the additive rest-energy constant is thereby shown to be contained in the relativistic action. Classical mechanics is therefore obtained as the approximation, not used as the starting point.

2.3.10 Flat Pullback Metric and Diagonalization

A flat metric in the original coordinate tuple is induced by the map $X = f + \mathbf{g} \cdot \boldsymbol{\sigma}$:

$$\mathbf{x} = (t, r_1, r_2, r_3)^T, \quad d\mathbf{x} = (dt, dr_1, dr_2, dr_3)^T.$$

The transpose operation was defined in Eq. (105). The symmetric pullback matrix $\mathbf{Q}(\mathbf{x})$ is the unique matrix defined by expansion of df and $d\mathbf{g}$ in $d\mathbf{x}$:

$$df^2 - d\mathbf{g}^2 =: d\mathbf{x}^T \mathbf{Q}(\mathbf{x}) d\mathbf{x}. \quad (134)$$

Consequently,

$$ds^2 = d\mathbf{x}^T \mathbf{Q}(\mathbf{x}) d\mathbf{x}.$$

If the map is locally invertible, the signature $(+, -, -, -)$ is preserved by Sylvester's law. Position-dependent entries of $\mathbf{Q}$ can be produced by curvilinear coordinates even though the target geometry remains flat; curvature is not represented by these entries alone. If the map fails to be locally invertible, the pullback may instead be degenerate.

At every point where the map is locally invertible, the ordered eigenvalue function is defined by

$$\lambda(n) := \text{ordered eigenvalue } n \text{ of } \mathbf{Q}, \quad n \in \{1, 2, 3, 4\}. \quad (135)$$

An orthogonal decomposition is then obtained for the real symmetric matrix:

$$\mathbf{Q} = \mathbf{N} \text{diag}(\lambda(1), \lambda(2), \lambda(3), \lambda(4)) \mathbf{N}^T, \quad \mathbf{N}^T \mathbf{N} = \mathbf{I}.$$

The eigenvalues are ordered so that

$$\lambda(1) > 0, \quad \lambda(2) < 0, \quad \lambda(3) < 0, \quad \lambda(4) < 0.$$

The corresponding direction function is defined by

$$\mathbf{u}(n) := \text{column } n \text{ of } \mathbf{N}. \quad (136)$$

The local differential directions are defined by

$$\omega(n) := \sqrt{|\lambda(n)|} \mathbf{u}(n)^T d\mathbf{x}. \quad (137)$$

The interval is consequently

$$ds^2 = \omega(1)^2 - \omega(2)^2 - \omega(3)^2 - \omega(4)^2.$$

The three minus signs cannot be dropped: a Lorentzian interval is not turned into a Euclidean norm by diagonalization. When $\mathbf{Q}$ varies with position, the local differential directions $\omega(n)$ need not be derivatives of globally defined coordinates.

3 Covariant Transform Applications

The generic similarity law and the circular rotor law were established in Section 1.11. The Lorentz action on real event, velocity, and momentum paravectors is separated here from similarity and is applied explicitly.

For $L = \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$, the passive event boost is $X' = L^* X L^*$, the $T = L$ specialization of Eq. (84). The usual Lorentz formulas are supplied by its scalar and vector parts, and $\det(X)$ is preserved. The distinct similarity law (87) is later applied in Section 5 to the electromagnetic field paravector F . The spatial rotor convention used earlier is active; this boost convention is passive.

3.1 Lorentz Boost

The boost element is obtained from the same exponential calculus as the circular rotor, but a space-time paravector is acted on by a congruence rather than by the similarity action used for a spatial rotation. A real event paravector is thereby kept real, while its scalar and parallel-vector parts are allowed to mix. The auxiliary parameters β and γ are introduced first so that the resulting formulas are shortened:

$$\begin{aligned} \theta &= \operatorname{arctanh}(\beta), \\ \beta &= \tanh(\theta), \\ \gamma &= \cosh(\theta), \\ \gamma\beta &= \sinh(\theta), \\ \mathbf{u} &= (u_1, u_2, u_3), \mathbf{u}^2 = 1, \\ L(\theta, \mathbf{u}) &= \exp(\tfrac{1}{2}\theta(\mathbf{u} \cdot \boldsymbol{\sigma})), \\ &= \cosh(\tfrac{1}{2}\theta) + \sinh(\tfrac{1}{2}\theta)(\mathbf{u} \cdot \boldsymbol{\sigma}). \end{aligned}$$

The boost element defined in Eq. (76) is displayed in expanded form for the calculation.

$$L^{-1} = \exp(-\tfrac{1}{2}\theta(\mathbf{u} \cdot \boldsymbol{\sigma})).$$

Here $\theta, \gamma \in \mathbb{R}$, $\beta \in \mathbb{R}$ with $|\beta| < 1$, $\mathbf{u} \in \mathbb{R}^3$ is a unit real vector, and $L \in \mathbb{R} + \mathbb{R}^3$ is the hyperbolic transform element. Its inverse is $L^{-1} = L^* = L^{*\dagger}$. Commutation is obtained for elements along the same axis:

$$\operatorname{com}(L(\theta, \mathbf{u}), L(\theta', \mathbf{u})) = 0.$$

Consequently, collinear boosts are combined by addition of the parameter θ . With $\beta' = \tanh(\theta')$ and $\gamma' = \cosh(\theta')$,

$$\begin{aligned} L(\theta, \mathbf{u})L(\theta', \mathbf{u}) &= L(\theta + \theta', \mathbf{u}), \\ \tanh(\theta + \theta') &= \frac{\beta + \beta'}{1 + \beta\beta'}, \\ \cosh(\theta + \theta') &= \gamma\gamma'(1 + \beta\beta'). \end{aligned}$$

For the passive boost with the signs used below, the real space-time paravector $X = t + \mathbf{r} \cdot \boldsymbol{\sigma}$, with $t \in \mathbb{R}$, $\mathbf{r} \in \mathbb{R}^3$, and $X \in \mathbb{R} + \mathbb{R}^3$, is transformed as

$$\begin{aligned} X &= t + \mathbf{r} \cdot \boldsymbol{\sigma}, \\ X' &= L^* X L^*, \\ &= t' + \mathbf{r}' \cdot \boldsymbol{\sigma}. \end{aligned}$$

$$\begin{aligned} t' &= t \cosh(\theta) - \sinh(\theta)(\mathbf{u} \cdot \mathbf{r}), \\ &= \gamma(t - \beta(\mathbf{u} \cdot \mathbf{r})). \end{aligned}$$

The vector part is made explicit by the axis functions in Eqs. (34) and (35), through which $\mathbf{r}$ is split into components parallel and perpendicular to the chosen direction:

$$\begin{aligned} \operatorname{par}(\mathbf{r}) &= (\mathbf{r} \cdot \mathbf{u})\mathbf{u} \quad \text{parallel to } \mathbf{u}, \\ \operatorname{prp}(\mathbf{r}) &= \mathbf{r} - \operatorname{par}(\mathbf{r}) \quad \text{perpendicular to } \mathbf{u}, \\ \mathbf{r} &= \operatorname{par}(\mathbf{r}) + \operatorname{prp}(\mathbf{r}), \\ \operatorname{par}(\mathbf{r}') &= \gamma(\operatorname{par}(\mathbf{r}) - \beta t\mathbf{u}), \\ \operatorname{prp}(\mathbf{r}') &= \operatorname{prp}(\mathbf{r}), \\ \mathbf{r}' &= \operatorname{par}(\mathbf{r}') + \operatorname{prp}(\mathbf{r}'), \\ &= \operatorname{prp}(\mathbf{r}) + \gamma(\operatorname{par}(\mathbf{r}) - \beta t\mathbf{u}), \\ &= \mathbf{r} + (\gamma(\mathbf{r} \cdot \mathbf{u} - \beta t) - \mathbf{r} \cdot \mathbf{u})\mathbf{u}. \end{aligned}$$

Equivalently, the full transformed paravector is obtained in one line by direct multiplication:

$$L^* X L^* = \gamma(t - \beta \mathbf{u} \cdot \mathbf{r}) + (\operatorname{prp}(\mathbf{r}) + \gamma(\operatorname{par}(\mathbf{r}) - \beta t\mathbf{u})) \cdot \boldsymbol{\sigma}.$$

Thus the component parallel to $\mathbf{u}$ is mixed with the scalar part while the perpendicular component is left unchanged. In the small-parameter limit $\beta \rightarrow 0$,

$$\gamma \rightarrow 1, t' \rightarrow t, \mathbf{r}' \rightarrow \mathbf{r}.$$

The interval invariance is then obtained directly from $L^{-1} = L^*$:

$$\begin{aligned} \det(X') &= X' X'^*, \\ &= (L^* X L^*)(L^* X L^*)^*, \\ &= L^* X L^* L X^* L, \\ &= L^* X X^* L, \\ &= \det(X) L^* L, \\ &= \det(X). \end{aligned}$$

Length contraction is obtained only after the simultaneity condition is specified. A rod is taken to be at rest in the unprimed frame and aligned with $\mathbf{u}$. Its two endpoints are denoted by the function inputs (1) and (2), so

$$\text{prp}(\mathbf{r}(2) - \mathbf{r}(1)) = 0, \quad l := \|\text{par}(\mathbf{r}(2) - \mathbf{r}(1))\|.$$

Thus l is measured in the rod's rest frame. The endpoint events are compared in the primed frame at the same primed time, so $t'(2) - t'(1) = 0$. The following relation is then obtained from the inverse boost:

$$\begin{aligned} \text{par}(\mathbf{r}(2) - \mathbf{r}(1)) &= \gamma \text{par}(\mathbf{r}'(2) - \mathbf{r}'(1)) \\ &\quad + \gamma \beta (t'(2) - t'(1)) \mathbf{u} \\ &= \gamma \text{par}(\mathbf{r}'(2) - \mathbf{r}'(1)). \end{aligned}$$

and therefore the primed length is

$$l' := \|\text{par}(\mathbf{r}'(2) - \mathbf{r}'(1))\| = \frac{l}{\gamma}.$$

The perpendicular length is unchanged.

The four-momentum of a massive particle is acted on by the same congruence:

$$\begin{aligned} (mU)' &= L^*(mU)L^* \\ &= E' + \mathbf{P}' \cdot \boldsymbol{\sigma}. \end{aligned}$$

Consequently,

$$\begin{aligned} E' &= \gamma(E - \beta \mathbf{u} \cdot \mathbf{P}), \\ \text{par}(\mathbf{P}') &= \gamma(\text{par}(\mathbf{P}) - \beta E \mathbf{u}), \\ \text{prp}(\mathbf{P}') &= \text{prp}(\mathbf{P}). \end{aligned}$$

Because the same real-paravector transformation is used as for X , $E^2 - \mathbf{P}^2 = m^2$ is preserved.

3.2 Compositions of Boosts and Rotors

The ordered products and their inverses are listed. The order remains significant even when a simple inverse rule is satisfied separately by each factor.

For one hyperbolic element and one circular element,

$$\begin{aligned} T &= LR, \\ T^{-1} &= T^{*H} = R^{*H} L^{*H} = R^H L^*, \\ T &= RL, \\ T^{-1} &= T^{*H} = L^{*H} R^{*H} = L^* R^H. \end{aligned}$$

For two factors of the same type,

$$\begin{aligned} T &= RR', \\ T^{-1} &= T^{*H} = R'^H R^H, \\ T &= LL', \\ T^{-1} &= T^{*H} = L'^* L^*. \end{aligned}$$

These formulas are simple order identities, but the inverse of each ordered product is fixed by them without ambiguity.

4 Differential Operators

One fixed set of differential conventions is used in the electromagnetic and wave-equation sections. The conventions are collected here so that the same numbered equations can be referenced later without repeated operator definitions.

For spatial derivatives with respect to $\mathbf{r} = (r_1, r_2, r_3)$, the spatial derivative vector is defined by

$$\partial_{\mathbf{r}} := \left\{ \frac{\partial}{\partial r_1}, \frac{\partial}{\partial r_2}, \frac{\partial}{\partial r_3} \right\} = \{\partial_1, \partial_2, \partial_3\}. \quad (138)$$

For shorthand, the coordinate symbol $\mathbf{r}$ is dropped in the last form, so $\partial_i := \partial/\partial r_i$ for $i \in \{1, 2, 3\}$. The corresponding spatial Laplacian is

$$\partial_{\mathbf{r}}^2 := \partial_1^2 + \partial_2^2 + \partial_3^2. \quad (139)$$

The differential paravector and its conjugate are

$$\partial := \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}} = \partial_t + \sigma_1 \partial_1 + \sigma_2 \partial_2 + \sigma_3 \partial_3. \quad (140)$$

$$\partial^* := \partial_t - \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}} = \partial_t - \sigma_1 \partial_1 - \sigma_2 \partial_2 - \sigma_3 \partial_3. \quad (141)$$

Because the coordinate derivatives commute on smooth fields, the scalar d'Alembertian is obtained from their ordered products:

$$\square := \partial \partial^*. \quad (142)$$

The reversed order gives the same scalar operator:

$$\square = \partial^* \partial = \partial_t^2 - \partial_{\mathbf{r}}^2. \quad (143)$$

The ordinary two-term Leibniz rule must be corrected when the left field does not commute with the sigma basis. For smooth paravector fields Z and Z' , the following result is obtained by direct expansion:

$$\partial(ZZ') = (\partial Z)Z' + Z(\partial Z') + \sum_{n=1}^3 \text{com}(\sigma_n, Z) \partial_n Z'. \quad (144)$$

The final term is generated when Z is moved through the left-acting basis element σ_n . It is zero for a scalar Z , or whenever $\text{com}(\sigma_n, Z) = 0$ for every contributing direction, and the ordinary Leibniz rule is then recovered.

4.1 Complex Fields

For a generic complex paravector field, the differential operator and field are written as

$$\begin{aligned} \partial &= \partial_t + \partial_{\mathbf{r}} \cdot \boldsymbol{\sigma}, \\ F &= (a + ib) + (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma}. \end{aligned}$$

The component fields are assumed to have the domains

$$\begin{aligned} a : \mathbb{R} \times \mathbb{R}^3 &\rightarrow \mathbb{R}, & \mathbf{A} : \mathbb{R} \times \mathbb{R}^3 &\rightarrow \mathbb{R}^3, \\ b : \mathbb{R} \times \mathbb{R}^3 &\rightarrow \mathbb{R}, & \mathbf{B} : \mathbb{R} \times \mathbb{R}^3 &\rightarrow \mathbb{R}^3, \end{aligned}$$

so that

$$F : \mathbb{R} \times \mathbb{R}^3 \rightarrow \mathbb{C} + \mathbb{C}^3.$$

The real/imaginary scalar/vector parts of ∂F and $i\partial F$ are then given by: The scalar and vector parts of ∂F and $i\partial F$ for the generic complex paravector field $F = (a + ib) + (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$ are recorded in Tables 19 and 20.

Table 19. Components of ∂F .

real scalar:	$\partial_t a + \partial_{\mathbf{r}} \cdot \mathbf{A}$
img scalar:	$\partial_t b + \partial_{\mathbf{r}} \cdot \mathbf{B}$
real vector:	$\partial_{\mathbf{r}} a + \partial_t \mathbf{A} - \partial_{\mathbf{r}} \times \mathbf{B}$
img vector:	$\partial_{\mathbf{r}} b + \partial_t \mathbf{A} \times \mathbf{A} + \partial_t \mathbf{B}$

Table 20. Components of $i\partial F$.

real scalar:	$-\partial_t b - \partial_{\mathbf{r}} \cdot \mathbf{B}$
img scalar:	$\partial_t a + \partial_{\mathbf{r}} \cdot \mathbf{A}$
real vector:	$-\partial_{\mathbf{r}} b - \partial_{\mathbf{r}} \times \mathbf{A} - \partial_t \mathbf{B}$
img vector:	$\partial_{\mathbf{r}} a + \partial_t \mathbf{A} - \partial_{\mathbf{r}} \times \mathbf{B}$

5 Electromagnetic Fields

Electromagnetic fields, gauge structure, and the Lorentz force can be expressed in compact paravector forms. In physical content, Maxwell theory is recovered in rationalized Heaviside–Lorentz units [2]; the packaging is drawn from later STA presentations [12–14].

5.1 Fields, potentials, and Maxwell equations

The differential conventions are fixed once in Section 4; in particular, the spatial derivative vector, the shorthand ∂_i , the paravector derivative, its conjugate, and the d'Alembertian are defined by Eqs. (138)–(142). With those operators fixed, and with $V, \rho, \lambda, S, q \in \mathbb{R}$, $\mathbf{E}, \mathbf{B}, \mathbf{A}, \mathbf{J} \in \mathbb{R}^3$, the electromagnetic field vector is defined by

$$\mathbf{F} := \mathbf{E} + i\mathbf{B}. \quad (145)$$

The corresponding pure-vector paravector is defined by

$$F := \mathbf{F} \cdot \boldsymbol{\sigma}. \quad (146)$$

The real potential paravector is defined separately by

$$\Phi := V + \mathbf{A} \cdot \boldsymbol{\sigma}. \quad (147)$$

Here Φ is a real paravector, $\mathbf{F}$ is a complex three-vector, and F is a pure-vector complex paravector. Electric and magnetic fields are packaged into F by Eq. (146), and scalar and vector potentials are packaged into Φ .

The scalar part generated by the potential is defined by

$$S := \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}. \quad (148)$$

The relations in Eq. (150) are then encoded by

$$\partial \Phi = S + F^*. \quad (149)$$

with

$$\mathbf{E} = -\partial_t \mathbf{A} - \partial_{\mathbf{r}} V, \quad \mathbf{B} = \partial_{\mathbf{r}} \times \mathbf{A}. \quad (150)$$

The compact Maxwell equation is

$$\partial F = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}. \quad (151)$$

By splitting the real/imaginary scalar/vector parts of (151), the following equations are reproduced:

$$\partial_{\mathbf{r}} \cdot \mathbf{E} = \rho, \quad \partial_{\mathbf{r}} \cdot \mathbf{B} = 0, \quad (152)$$

$$\partial_{\mathbf{r}} \times \mathbf{B} = \mathbf{J} + \partial_t \mathbf{E}, \quad \partial_{\mathbf{r}} \times \mathbf{E} = -\partial_t \mathbf{B}. \quad (153)$$

Agreement with standard space-time-algebra treatments is obtained in Eqs. (152) and (153), subject to the sign conventions fixed by Eqs. (140), (141), (142), and (146). Maxwell's field equations are recovered component by component in the present signature and field convention [2, 12, 13, 15]. Separate primitive divergence and curl operators are avoided at this stage in Eq. (151). They are produced automatically when the scalar and vector parts of the single paravector equation are separated.

5.2 Boost transformation of electromagnetic fields

The field paravector is transformed by the same boost element L from Eq. (89):

$$F' = L^* F L. \quad (154)$$

Equation (154) is deliberately different from the event congruence $X' = L^* X L^*$. The event and field laws were derived from different intertwining requirements in Section 1.11; the usual mixing of $\text{par}(\mathbf{E})$, $\text{prp}(\mathbf{E})$, $\text{par}(\mathbf{B})$, and $\text{prp}(\mathbf{B})$ is obtained from the resulting similarity.

5.3 Wave equations, gauge, and Lorentz force

The equivalent pair is obtained by applying ∂^* to Eq. (151) and ∂ to its star-conjugate equation:

$$\square F = \partial^*(\rho - \mathbf{J} \cdot \boldsymbol{\sigma}), \quad (155)$$

$$\square F^* = \partial(\rho + \mathbf{J} \cdot \boldsymbol{\sigma}). \quad (156)$$

Charge conservation is obtained from the scalar part of the first relation:

$$\partial_t \rho + \partial_{\mathbf{r}} \cdot \mathbf{J} = 0. \quad (157)$$

Equation (157) is the continuity equation written in the present notation. For a real scalar gauge function $\lambda \in \mathbb{R}$, a function input is used so that confusion with a Lorentz-frame prime is avoided:

$$\Phi(\lambda) := \Phi + \partial^* \lambda. \quad (158)$$

The scalar and vector potential functions contained in this paravector are defined by

$$V(\lambda) := V + \partial_t \lambda. \quad (159)$$

$$\mathbf{A}(\lambda) := \mathbf{A} - \partial_{\mathbf{r}} \lambda. \quad (160)$$

The corresponding field function is defined by

$$F(\lambda) := F. \quad (161)$$

The potential Φ is retained even when the field is physically primary because the redundancy is carried by Φ , whereas the gauge-invariant derivative content is carried by F .

For a particle of real mass $m > 0$, real charge q , and four-velocity U , the four-force formula is

$$m \frac{dU}{ds} = q \operatorname{real}(FU) = q\gamma(\mathbf{v} \cdot \mathbf{E} + (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \boldsymbol{\sigma}), \quad (162)$$

in which both relativistic power and the Lorentz three-force are packaged. Since $d/ds = \gamma d/dt$, the equations $dE/dt = q\mathbf{v} \cdot \mathbf{E}$ and $d\mathbf{P}/dt = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$ are obtained by division by γ .

5.4 Lagrangian and energy-momentum remarks

The electromagnetic particle Lagrangian is defined by

$$\mathcal{L} := -\frac{m}{\gamma} - q(V - \mathbf{v} \cdot \mathbf{A}), \quad (163)$$

and $d\mathbf{P}/dt = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$ is recovered from its Euler–Lagrange equations. The free action from Eq. (131) is supplemented by the potential term, and the field-strength form of the force law is recovered after variation.

The frame-dependent field energy-density and flux paravector is defined by

$$T := \frac{1}{2} F F^H. \quad (164)$$

Its scalar and vector parts are

$$T = \frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2) + (\mathbf{E} \times \mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (165)$$

The energy density and Poynting vector are packaged by Eq. (164) into one frame-dependent paravector. The precise caveat and the corresponding covariant stress-energy map are recorded in the supporting subsection.

5.5 Supporting Electromagnetic Derivations

The field decomposition, potential reconstruction, source formulas, gauge transformations, force-law expansions, and variational identities underlying Eqs. (138)–(164) are written out in the supporting derivations.

5.5.1 Electricity and Magnetism

The compact field definitions in Eqs. (146) and (151) are expanded into explicit scalar, vector, and bivector parts. Rationalized Heaviside–Lorentz units are used, with $c = 1$. Here $\rho, S \in \mathbb{R}$, $\mathbf{E}, \mathbf{B}, \mathbf{J} \in \mathbb{R}^3$, $\mathbf{F} = \mathbf{E} + i\mathbf{B} \in \mathbb{C}^3$, and $F = \mathbf{F} \cdot \boldsymbol{\sigma}$ is a pure-vector complex paravector, so $\operatorname{scalar}(F) = 0$. The spatial derivative conventions are

$$\begin{aligned} \partial_{\mathbf{r}} &= \{\partial_1, \partial_2, \partial_3\}, \\ \partial_i &= \partial/\partial r_i, \\ \partial_{\mathbf{r}}^2 &= \partial_1^2 + \partial_2^2 + \partial_3^2, \end{aligned}$$

and the differential paravector is

$$\begin{aligned} \partial &= \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}} \\ &= \partial_t + \sigma_1 \partial_1 + \sigma_2 \partial_2 + \sigma_3 \partial_3, \\ \partial^* &= \partial_t - \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}. \end{aligned}$$

These conventions are restatements of Eqs. (138), (140), and (141); no new definitions are introduced. The generic real/imaginary scalar/vector decomposition of ∂F and $i\partial F$ has already been recorded in Subsection 4.1. With that operator bookkeeping fixed, the electric and magnetic fields can be combined without introducing any new multiplication rule.

The electric and magnetic fields are collected into the complex three-vector $\mathbf{F}$ and the field paravector F :

$$\begin{aligned} \mathbf{E} &= \{E_1, E_2, E_3\} && \text{real vector electric field,} \\ \mathbf{B} &= \{B_1, B_2, B_3\} && \text{real vector magnetic field,} \\ \mathbf{F} &= \mathbf{E} + i\mathbf{B} && \text{electromagnetic field,} \\ F &= \mathbf{F} \cdot \boldsymbol{\sigma} = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, && i = \sigma_1 \sigma_2 \sigma_3. \end{aligned}$$

$$\begin{aligned} F^* &= (-\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \\ &= -\mathbf{F}^* \cdot \boldsymbol{\sigma}. \end{aligned}$$

The two simplest quadratic field combinations are given by

$$\begin{aligned} \frac{1}{2} F^2 &= \frac{1}{2}(\mathbf{E}^2 - \mathbf{B}^2) + i\mathbf{E} \cdot \mathbf{B}, \\ \frac{1}{2} F F^H &= \frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2) + (\mathbf{E} \times \mathbf{B}) \cdot \boldsymbol{\sigma}. \end{aligned}$$

Thus the proper-Lorentz invariants are contained in $\frac{1}{2} F^2$, while $\frac{1}{2} F F^H$ is separated into the energy density and the Poynting vector. Under parity, the sign of $\mathbf{E} \cdot \mathbf{B}$ is changed, and this quantity is therefore a pseudoscalar rather than a scalar invariant of the full Lorentz group.

The real source paravector is defined by

$$D := \rho + \mathbf{J} \cdot \boldsymbol{\sigma}. \quad (166)$$

Thus $D \in \mathbb{R} + \mathbb{R}^3$. Its sigma conjugate is

$$D^* = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}.$$

Here $\rho \in \mathbb{R}$ is the scalar charge density and $\mathbf{J} \in \mathbb{R}^3$ is the current-density vector. Maxwell's equations are then written as

$$\begin{aligned} \partial F &= D^* && \text{Maxwell's equations,} \\ &= \rho - \mathbf{J} \cdot \boldsymbol{\sigma}. \end{aligned}$$

$$\partial^* F^* = D.$$

In this form, the four classical Maxwell equations are reproduced in one step by the scalar, sigma-number-one, and sigma-number-two parts of ∂F . The location of each component is shown by expansion of the product.

The following expression is obtained by expansion of ∂F into scalar, pseudoscalar, vector, and bivector parts:

$$\begin{aligned} \partial F &= (\partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}})((\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}), \\ &= (\partial_{\mathbf{r}} \cdot \mathbf{E}), \\ &\quad + (\partial_{\mathbf{r}} \cdot \mathbf{B})i, \\ &\quad + (\partial_t \mathbf{E} - \partial_{\mathbf{r}} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \\ &\quad + (\partial_t \mathbf{B} + \partial_{\mathbf{r}} \times \mathbf{E}) \cdot (i\boldsymbol{\sigma}). \end{aligned}$$

The first two lines are the scalar and pseudoscalar parts. The last two lines are the sigma-number-one and sigma-number-two vector parts. By comparison with $D^* = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}$, the magnetic-divergence term and Faraday term are immediately forced to vanish, while the electric source terms are fixed.

Accordingly, the component equations of $\partial F = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}$ are

$$\begin{aligned}\partial_{\mathbf{r}} \cdot \mathbf{E} &= \rho, \\ \partial_{\mathbf{r}} \cdot \mathbf{B} &= 0, \\ \partial_t \mathbf{E} - \partial_{\mathbf{r}} \times \mathbf{B} &= -\mathbf{J}, \\ \partial_t \mathbf{B} + \partial_{\mathbf{r}} \times \mathbf{E} &= 0.\end{aligned}$$

The following relations are then obtained by separation of the vector components:

$$\begin{aligned}\partial_{\mathbf{r}} \times \mathbf{B} &= \mathbf{J} + \partial_t \mathbf{E}, \\ \partial_{\mathbf{r}} \times \mathbf{E} &= -\partial_t \mathbf{B}.\end{aligned}$$

5.5.2 Lorentz Transform of Fields

The boost law is specialized to the electromagnetic field paravector

$$F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}.$$

For a constant inertial boost $L = \exp(\frac{1}{2}\theta \mathbf{u} \cdot \boldsymbol{\sigma})$ with $\mathbf{u}^2 = 1$, $\beta = \tanh(\theta)$, and $\gamma = \cosh(\theta)$, the field is transformed by the general similarity (87), even though event and current paravectors are transformed by the congruence (84). In the inverse-first convention the left factor is $L^{-1} = L^*$, so the field is transformed as

$$F' = L^{-1}FL = L^*FL.$$

The final factor is L , rather than L^* , because the field is assigned the similarity $T^{-1}(\cdot)T$. With a double- L^* sandwich, the event congruence would instead be applied to F , and its pure-vector form would not be preserved. For example, if $F = a \mathbf{u} \cdot \boldsymbol{\sigma}$ for a real scalar amplitude a , then

$$L^*FL = F, \quad L^*FL^* = \gamma a \mathbf{u} \cdot \boldsymbol{\sigma} - \gamma \beta a,$$

and an inadmissible scalar part is contained in the second expression. By application of the general complex-vector result (90) to $\mathbf{V} = \mathbf{E} + i\mathbf{B}$, the field and its par and prp parts are obtained:

$$\begin{aligned}F' &= L^*FL, \\ \text{par}(\mathbf{E}') &= \text{par}(\mathbf{E}), \\ \text{par}(\mathbf{B}') &= \text{par}(\mathbf{B}), \\ \text{prp}(\mathbf{E}') &= \gamma(\text{prp}(\mathbf{E}) + \beta \mathbf{u} \times \mathbf{B}), \\ \text{prp}(\mathbf{B}') &= \gamma(\text{prp}(\mathbf{B}) - \beta \mathbf{u} \times \mathbf{E}).\end{aligned}$$

Equivalently, the transformed fields can be written in the compact form

$$\begin{aligned}\mathbf{E}' &= \text{par}(\mathbf{E}) + \gamma(\text{prp}(\mathbf{E}) + \beta \mathbf{u} \times \mathbf{B}), \\ \mathbf{B}' &= \text{par}(\mathbf{B}) + \gamma(\text{prp}(\mathbf{B}) - \beta \mathbf{u} \times \mathbf{E}).\end{aligned}$$

The parallel components are unchanged, while the perpendicular components are mixed by the standard electric-magnetic cross

terms. Because $LL^* = 1$ and F^2 is central, the field invariants are directly preserved:

$$F'^2 = L^*F(LL^*)FL = L^*F^2L = F^2.$$

Thus the real part of F^2 is a scalar and its imaginary part is a pseudoscalar under parity; both are preserved by proper Lorentz transformations.

For any constant normalized Lorentz element T , the potential and source are event-type paravectors, while the complementary action is carried by the derivative:

$$\Phi' = T^{*H}\Phi T^*, \quad D' = T^{*H}DT^*, \quad \partial' = T^H\partial T.$$

For the boost $T = L$, these expressions are reduced to

$$\Phi' = L^*\Phi L^*, \quad D' = L^*DL^*, \quad \partial' = L\partial L.$$

Consequently,

$$\begin{aligned}\partial' F' &= (T^H\partial T)(T^{*H}FT) \\ &= T^H(\partial F)T \\ &= T^H D^* T \\ &= D'^*.\end{aligned}$$

Thus covariance of Maxwell's equation is established. In components, the current transformation is

$$\begin{aligned}\rho' &= \gamma(\rho - \beta \mathbf{u} \cdot \mathbf{J}), \\ \text{par}(\mathbf{J}') &= \gamma(\text{par}(\mathbf{J}) - \beta \rho \mathbf{u}), \\ \text{prp}(\mathbf{J}') &= \text{prp}(\mathbf{J}).\end{aligned}$$

5.5.3 Wave Operator and d'Alembertian

The operator conventions were fixed in Eqs. (140)–(142). They are used only in the derivation of the sourced wave consequences and in the distinction between the scalar wave operator $\square$ and the full square ∂^2 :

$$\begin{aligned}\partial &= \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}, \\ \partial^* &= \partial_t - \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}, \\ \square &= \partial_t^2 - \partial_{\mathbf{r}}^2 \\ &= \partial\partial^* = \partial^*\partial.\end{aligned}$$

These are the previously defined identities (140)–(142), restated for the calculation. The following relation is also obtained from the sigma algebra:

$$\partial^2 = \partial_t^2 + \partial_{\mathbf{r}}^2 + 2\partial_t \partial_{\mathbf{r}} \cdot \boldsymbol{\sigma},$$

so $\partial^2 \neq \square$. With the conventions of Eqs. (140)–(142), the following result is obtained by applying ∂^* to Maxwell's equation:

$$\begin{aligned}\square F &= \partial^* \partial F \\ &= \partial^*(\rho - \mathbf{J} \cdot \boldsymbol{\sigma}) \\ &= (\partial_t \rho + \partial_{\mathbf{r}} \cdot \mathbf{J}) \\ &\quad + (-\partial_t \mathbf{J} - \partial_{\mathbf{r}} \rho + i\partial_{\mathbf{r}} \times \mathbf{J}) \cdot \boldsymbol{\sigma}.\end{aligned}$$

The companion relation is obtained by applying ∂ to the star-conjugate Maxwell equation $\partial^* F^* = D$:

$$\square F^* = \partial D = \partial(\rho + \mathbf{J} \cdot \boldsymbol{\sigma}),$$

which is Eq. (156) in the compact development. Because F is pure vector, the scalar part of the left-hand side vanishes. The scalar and vector parts are therefore given by

$$\begin{aligned}\partial_t \rho + \partial_{\mathbf{r}} \cdot \mathbf{J} &= 0, \\ \square \mathbf{E} &= -\partial_t \mathbf{J} - \partial_{\mathbf{r}} \rho, \\ \square \mathbf{B} &= \partial_{\mathbf{r}} \times \mathbf{J}.\end{aligned}$$

In vacuum, both source terms vanish and the ordinary wave equation is satisfied by each field component.

In vacuum, where $\rho = 0$ and $\mathbf{J} = 0$, the following reduction is obtained:

$$\begin{aligned}\square \mathbf{E} = 0 &\quad \text{implies} \quad \partial_t^2 \mathbf{E} = \partial_{\mathbf{r}}^2 \mathbf{E}, \\ \square \mathbf{B} = 0 &\quad \text{implies} \quad \partial_t^2 \mathbf{B} = \partial_{\mathbf{r}}^2 \mathbf{B}.\end{aligned}$$

The scalar and vector parts of Maxwell's equation and its sourced wave consequence are recorded in Tables 21 and 22.

Table 21. Components of $\partial F = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}$.

real scalar:	$\partial_{\mathbf{r}} \cdot \mathbf{E} = \rho$
img scalar:	$\partial_{\mathbf{r}} \cdot \mathbf{B} = 0$
real vector:	$\partial_t \mathbf{E} - \partial_{\mathbf{r}} \times \mathbf{B} = -\mathbf{J}$
img vector:	$\partial_t \mathbf{B} + \partial_{\mathbf{r}} \times \mathbf{E} = 0$

Table 22. Components of $\square F = \partial^*(\rho - \mathbf{J} \cdot \boldsymbol{\sigma})$.

real scalar:	$\partial_t \rho + \partial_{\mathbf{r}} \cdot \mathbf{J} = 0$
img scalar:	$0 = 0$
real vector:	$\square \mathbf{E} = -\partial_t \mathbf{J} - \partial_{\mathbf{r}} \rho$
img vector:	$\square \mathbf{B} = \partial_{\mathbf{r}} \times \mathbf{J}$

5.5.4 Potential Fields

Gauge freedom, source reconstruction, and the electric/magnetic field split are collected in Eqs. (149) and (150). Here $V, S, \rho, \lambda \in \mathbb{R}$, $\mathbf{A}, \mathbf{E}, \mathbf{B}, \mathbf{J} \in \mathbb{R}^3$, $\Phi \in \mathbb{R} + \mathbb{R}^3$, $\mathbf{F} \in \mathbb{C}^3$, and F is the pure-vector complex paravector defined previously:

$$\begin{aligned}F^* &= (-\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \\ \Phi &= V + \mathbf{A} \cdot \boldsymbol{\sigma}, \\ \partial \Phi &= \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}, \\ &\quad + (\partial_t \mathbf{A} + \partial_{\mathbf{r}} V) \cdot \boldsymbol{\sigma}, \\ &\quad + i(\partial_{\mathbf{r}} \times \mathbf{A}) \cdot \boldsymbol{\sigma}.\end{aligned}$$

The potential gradient is separated by this expansion into one real scalar part and one pure-vector field part. The scalar piece is identified with the Lorenz-gauge scalar S , and the remaining vector piece is identified with the electromagnetic field:

$$\begin{aligned}S &= \text{scalar}(\partial \Phi), \\ &= \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}.\end{aligned}$$

The definition of S in Eq. (149) is thereby recovered. The component identities extracted from $\partial \Phi$ are

$$\begin{aligned}\mathbf{E} &= -\partial_t \mathbf{A} - \partial_{\mathbf{r}} V, \\ \mathbf{B} &= \partial_{\mathbf{r}} \times \mathbf{A}.\end{aligned}$$

The identity is explicit:

$$\begin{aligned}\partial \Phi &= S + F^*, \\ F^* &= \partial \Phi - S, \\ \text{scalar}(F^*) &= \text{scalar}(\partial \Phi - S) = 0, \\ \text{vector}(F^*) &= \text{vector}(\partial \Phi).\end{aligned}$$

The field is therefore the non-scalar part of $\partial \Phi$. Under the displayed gauge change, only the scalar part of $\partial \Phi$ is changed; the field paravector remains unchanged. The component content of the potential identity $\partial \Phi = S + F^*$ is recorded in Table 23.

Table 23. Components of $\partial \Phi = S + F^*$.

real scalar:	$\partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A} = S$
img scalar:	$0 = 0$
real vector:	$\partial_t \mathbf{A} + \partial_{\mathbf{r}} V = -\mathbf{E}$
img vector:	$\partial_{\mathbf{r}} \times \mathbf{A} = \mathbf{B}$

The differential product $\partial \Phi$ is evaluated first; the involutions are then applied to the resulting field. No right-acting differential operator is implied. Equivalently, without introducing S , the relation can be written as

$$F^* = (\partial \Phi - (\partial \Phi)^{*H})/2.$$

5.5.5 Density from Potential

The derivation of the charge density and current density directly from the potential field is collected in one place. From the previously defined potential $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$, scalar part $S = \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}$, and identity $F^* = \partial \Phi - S$, the following relation is obtained:

$$\begin{aligned}F &= \partial^* \Phi^* - S, \\ \partial F &= \partial \partial^* \Phi^* - \partial S, \\ &= \square \Phi^* - \partial S, \\ &= (\square V - \partial_t S) + (-\square \mathbf{A} - \partial_{\mathbf{r}} S) \cdot \boldsymbol{\sigma}, \\ &= \rho - \mathbf{J} \cdot \boldsymbol{\sigma}.\end{aligned}$$

$$\begin{aligned}\rho &= \square V - \partial_t S, \\ \mathbf{J} &= \square \mathbf{A} + \partial_{\mathbf{r}} S.\end{aligned}$$

These are the component source-reconstruction formulas contained in the compact Maxwell equation. Charge density is determined from the potential data by the scalar equation, and current density is determined from the same data by the vector equation.

In Lorenz gauge, where $S = 0$, these relations reduce to

$$\begin{aligned}\partial_t V &= -\partial_{\mathbf{r}} \cdot \mathbf{A}, \\ \rho &= \square V, \\ \mathbf{J} &= \square \mathbf{A}.\end{aligned}$$

5.5.6 Gauge Transform of the Potential

$$\begin{aligned}\text{scalar}(\partial\Phi) &= S, \\ S &= \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}.\end{aligned}$$

A real scalar gauge function λ is introduced. A gauge change is distinguished from the boost primes used above by function inputs: The potential and its components defined in Eqs. (158), (159), and (160) are given by

$$\begin{aligned}\Phi(\lambda) &= \Phi + \partial^* \lambda, \\ V(\lambda) &= V + \partial_t \lambda, \\ \mathbf{A}(\lambda) &= \mathbf{A} - \partial_{\mathbf{r}} \lambda.\end{aligned}$$

The transformed potential gradient is defined separately by

$$(\partial\Phi)(\lambda) := \partial\Phi(\lambda). \quad (167)$$

By direct differentiation, the following relation is obtained:

$$(\partial\Phi)(\lambda) = \partial\Phi + \square\lambda.$$

Since $\square\lambda$ is again a scalar field, only the scalar part is changed. Its transformed value is defined by

$$S(\lambda) := S + \square\lambda. \quad (168)$$

The field invariance already defined in Eq. (161) is

$$F(\lambda) = F.$$

The gauge split is thereby made explicit. The residual condition $\square\lambda = 0$ is satisfied by transformations that preserve Lorenz gauge.

5.5.7 Four-Force

The relativistic work law, the Lorentz-force law, the component split of FU/γ , and the four-force equation are collected in one place.

$$\begin{aligned}mU &= m \frac{dX}{ds}, \\ &= m\gamma + m\gamma \mathbf{v} \cdot \boldsymbol{\sigma}. \\ \frac{d}{ds} &= \gamma \frac{d}{dt}.\end{aligned}$$

The scalar part of $d(mU)/dt$ is the power equation and the vector part is the three-force equation. The usual split is made visible without abandonment of the relativistic form by writing both pieces in one paravector equation.

The relativistic work law is given by the time component,

$$\frac{d(m\gamma)}{dt} = q\mathbf{v} \cdot \mathbf{E}.$$

Here $\mathbf{E}$ denotes the electric field, not the energy operator E , and no work term is contributed by the magnetic field in the scalar

component. The relativistic Lorentz force law is given by the spatial component,

$$\begin{aligned}\mathbf{P} &= m\gamma \mathbf{v}, \\ \frac{d\mathbf{P}}{dt} &= q\mathbf{E} + q\mathbf{v} \times \mathbf{B}.\end{aligned}$$

Together these are assembled into

$$\frac{d(mU)}{dt} = q\mathbf{v} \cdot \mathbf{E} + q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \boldsymbol{\sigma}.$$

The paravector content is exposed explicitly by the expansion

$$\begin{aligned}FU/\gamma &= ((\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma})(1 + \mathbf{v} \cdot \boldsymbol{\sigma}), \\ &= \mathbf{E} \cdot \boldsymbol{\sigma} + i\mathbf{B} \cdot \boldsymbol{\sigma} + (\mathbf{E} \cdot \boldsymbol{\sigma})(\mathbf{v} \cdot \boldsymbol{\sigma}), \\ &\quad + i(\mathbf{B} \cdot \boldsymbol{\sigma})(\mathbf{v} \cdot \boldsymbol{\sigma}), \\ &= \mathbf{E} \cdot \boldsymbol{\sigma} + i\mathbf{B} \cdot \boldsymbol{\sigma} + \mathbf{E} \cdot \mathbf{v}, \\ &\quad + i(\mathbf{E} \times \mathbf{v}) \cdot \boldsymbol{\sigma} + i\mathbf{B} \cdot \mathbf{v} - (\mathbf{B} \times \mathbf{v}) \cdot \boldsymbol{\sigma}.\end{aligned}$$

The product FU is naturally separated into real and imaginary parts. The physical Lorentz-force content is reproduced by the real part, while electric and magnetic roles are exchanged by the imaginary part, which is therefore not used in the force law itself. The component form of the product FU/γ used in the Lorentz-force law is given in Table 24.

Table 24. Components of FU/γ .

real scalar:	$\mathbf{E} \cdot \mathbf{v}$
img scalar:	$\mathbf{B} \cdot \mathbf{v}$
real vector:	$\mathbf{E} + \mathbf{v} \times \mathbf{B}$
img vector:	$\mathbf{B} + \mathbf{E} \times \mathbf{v}$

$$\begin{aligned}\text{real}(FU) &= \frac{1}{2}(FU + UF^H), \\ &= \gamma(\mathbf{v} \cdot \mathbf{E} + (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \boldsymbol{\sigma}).\end{aligned}$$

The physical four-force is therefore

$$\begin{aligned}m \frac{dU}{ds} &= q \text{real}(FU), \\ &= \frac{1}{2}q(FU + UF^H). \\ m \frac{d^2 X}{ds^2} &= q \text{real}\left(F \frac{dX}{ds}\right) \quad \text{four-force}.\end{aligned}$$

5.5.8 Lagrangian Derivation of the Lorentz Force

The minimally coupled action, the Euler–Lagrange derivation of the Lorentz force, the Hamiltonian, and the corresponding power identity are collected in one place. From the Lagrangian of Eq. (163), together with the potential $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$, four-velocity $U = dX/ds$, and $\gamma = dt/ds$, the action is obtained in the form

$$\begin{aligned}\text{action} &= -m \int ds - q \int \text{scalar}(\Phi^* U) ds \\ &= \int \left[-\frac{m}{\gamma} - q \text{scalar}((V - \mathbf{A} \cdot \boldsymbol{\sigma})(1 + \mathbf{v} \cdot \boldsymbol{\sigma})) \right] dt \\ &= \int \left(-\frac{m}{\gamma} - qV + q\mathbf{v} \cdot \mathbf{A} \right) dt.\end{aligned}$$

$$\mathcal{L} = -\frac{m}{\gamma} - q(V - \mathbf{v} \cdot \mathbf{A}).$$

This is the standard minimal-coupling Lagrangian written in the present paravector notation. The canonical momentum is shifted by the vector potential, exactly as in the ordinary component derivation.

The canonical momentum is

$$\begin{aligned}\partial_{\mathbf{v}}\mathcal{L} &= m\gamma\mathbf{v} + q\mathbf{A}, \\ &= \mathbf{P} + q\mathbf{A}.\end{aligned}$$

$$\begin{aligned}\frac{d}{dt}\partial_{\mathbf{v}}\mathcal{L} &= \frac{d\mathbf{P}}{dt} + q\partial_t\mathbf{A} + q(\mathbf{v} \cdot \partial_{\mathbf{r}})\mathbf{A}, \\ \partial_{\mathbf{r}}\mathcal{L} &= -q\partial_{\mathbf{r}}V + q\partial_{\mathbf{r}}(\mathbf{v} \cdot \mathbf{A}).\end{aligned}$$

The vector-calculus identity

$$(\mathbf{v}(t) \cdot \partial_{\mathbf{r}})\mathbf{A}(t, \mathbf{r}) = \partial_{\mathbf{r}}(\mathbf{v} \cdot \mathbf{A}) - \mathbf{v} \times (\partial_{\mathbf{r}} \times \mathbf{A}).$$

is used to obtain the following result from the Euler–Lagrange equation $(d/dt)\partial_{\mathbf{v}}\mathcal{L} = \partial_{\mathbf{r}}\mathcal{L}$:

$$\begin{aligned}\frac{d\mathbf{P}}{dt} &= -q\partial_{\mathbf{r}}V - q\partial_t\mathbf{A}, \\ &\quad + q(\partial_{\mathbf{r}}(\mathbf{v} \cdot \mathbf{A}) - (\mathbf{v} \cdot \partial_{\mathbf{r}})\mathbf{A}), \\ &= q(-\partial_{\mathbf{r}}V - \partial_t\mathbf{A} + \mathbf{v} \times (\partial_{\mathbf{r}} \times \mathbf{A})), \\ &= q(\mathbf{E} + \mathbf{v} \times \mathbf{B}), \\ &= \frac{m}{\gamma}\text{vector}\left(\frac{dU}{ds}\right).\end{aligned}$$

The Hamiltonian is

$$\begin{aligned}\mathcal{H} &= (\partial_{\mathbf{v}}\mathcal{L}) \cdot \mathbf{v} - \mathcal{L} \\ &= (m\gamma\mathbf{v} + q\mathbf{A}) \cdot \mathbf{v} + \frac{m}{\gamma} + qV - q\mathbf{v} \cdot \mathbf{A} \\ &= m\gamma\left(\mathbf{v}^2 + \frac{1}{\gamma^2}\right) + qV \\ &= m\gamma + qV.\end{aligned}$$

Thus $\mathcal{H}$ is given by the mechanical relativistic energy plus the scalar-potential energy. Along a solution,

$$\begin{aligned}\frac{d\mathcal{H}}{dt} &= -\partial_t\mathcal{L} = q\partial_tV - q\mathbf{v} \cdot \partial_t\mathbf{A}, \\ \frac{d\mathcal{H}}{dt} &= \frac{d(m\gamma)}{dt} + q\partial_tV + q\mathbf{v} \cdot \partial_{\mathbf{r}}V.\end{aligned}$$

The following relation is obtained by equating these expressions:

$$\begin{aligned}\frac{d(m\gamma)}{dt} &= q\mathbf{v} \cdot (-\partial_{\mathbf{r}}V - \partial_t\mathbf{A}) \\ &= q\mathbf{v} \cdot \mathbf{E},\end{aligned}$$

in agreement with the scalar component of the Lorentz-force equation.

5.5.9 Energy-Density and Poynting Paravector

The electromagnetic energy density and Poynting vector are packaged into one paravector by Eq. (164). Its components and the scalar energy balance implied by Maxwell’s equation (151) are recorded by the displayed algebra:

$$\begin{aligned}F &= (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \\ F^{\mathbf{H}} &= (\mathbf{E} - i\mathbf{B}) \cdot \boldsymbol{\sigma}, \\ T &= \frac{1}{2}FF^{\mathbf{H}}, \\ &= \frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2) + (\mathbf{E} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \\ &= T_s + \mathbf{T}_{\mathbf{v}} \cdot \boldsymbol{\sigma}.\end{aligned}$$

$$\partial = \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}.$$

The paravector $T = \frac{1}{2}FF^{\mathbf{H}}$ is the energy density and flux measured relative to the chosen time direction. It is not, by itself, the full electromagnetic stress-energy tensor. The covariant content is represented by the following function of a real paravector $a \in \mathbb{R} + \mathbb{R}^3$:

$$T(a) := \frac{1}{2}FaF^{\mathbf{H}}. \quad (169)$$

The argument-free notation is reserved for its value at the identity:

$$T = T(1) = \frac{1}{2}FF^{\mathbf{H}}.$$

Thus the displayed energy-density and flux paravector is the $a = 1$ value of the full stress-energy function [12, 14]. Indeed, for $a' = L^*aL^*$, $F' = L^*FL$, and $(F')^{\mathbf{H}} = LF^{\mathbf{H}}L^*$,

$$T'(a') := \frac{1}{2}F'a'(F')^{\mathbf{H}}. \quad (170)$$

By substitution of the transformed field and argument, the covariance law is obtained:

$$T'(a') = L^*\left(\frac{1}{2}FaF^{\mathbf{H}}\right)L^* = L^*T(a)L^*. \quad (171)$$

so the required event-paravector covariance is satisfied by the function.

The scalar energy balance is obtained immediately from Maxwell’s equations:

$$\begin{aligned}\partial_t T_s + \partial_{\mathbf{r}} \cdot \mathbf{T}_{\mathbf{v}} &= \mathbf{E} \cdot \partial_t \mathbf{E} + \mathbf{B} \cdot \partial_t \mathbf{B} + \partial_{\mathbf{r}} \cdot (\mathbf{E} \times \mathbf{B}) \\ &= -\mathbf{J} \cdot \mathbf{E}.\end{aligned}$$

This is Poynting’s theorem in the present units: the field-energy loss equals the work done on charge. The two quadratic field products and their component content are collected below. The proper-Lorentz invariants and the frame-dependent energy-density/Poynting split are displayed without introduction of a separate field equation for F^2 . The two proper-Lorentz invariants are listed in Table 25. The frame-dependent energy-density/Poynting paravector $T = \frac{1}{2}FF^{\mathbf{H}}$ is given in Table 26.

Table 25. Components of $\frac{1}{2}F^2$.

real scalar:	$\frac{1}{2}(\mathbf{E}^2 - \mathbf{B}^2)$
img scalar:	$\mathbf{E} \cdot \mathbf{B}$
real vector:	$\mathbf{0}$
img vector:	$\mathbf{0}$

Table 26. Components of $\frac{1}{2}FF^H$.

scalar:	$\frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2)$	energy density
vector:	$\mathbf{E} \times \mathbf{B}$	Poynting vector

The differentiated and source-coupled combinations used in the energy discussion are restored in the next five tables. They are algebraic identities and consequences of Maxwell's equation $\partial F = D^*$, not additional field equations. In particular, the derivative is a left-acting paravector, so the noncommutative rule (144) must be used for products. For the energy-density/Poynting paravector, the following relation is obtained:

$$\partial T = \frac{1}{2}(\partial F)F^H + \frac{1}{2}F(\partial F^H) + \frac{1}{2} \sum_{n=1}^3 \text{com}(\sigma_n, F) \partial_n F^H.$$

The direct component split of the left-hand side given in the following table is therefore a derived identity.

Table 27. Components of ∂T .

real scalar:	$\partial_t T_s + \partial_r \cdot \mathbf{T}_v$
img scalar:	0
real vector:	$\partial_r T_s + \partial_t \mathbf{T}_v$
img vector:	$\partial_r \times \mathbf{T}_v$

Because F^2 is a complex scalar, its derivative can also be expanded directly:

Table 28. Components of $\partial(F^2)$.

real scalar:	$\partial_t(\mathbf{E}^2 - \mathbf{B}^2)$
img scalar:	$\partial_t(2\mathbf{E} \cdot \mathbf{B})$
real vector:	$\partial_r(\mathbf{E}^2 - \mathbf{B}^2)$
img vector:	$\partial_r(2\mathbf{E} \cdot \mathbf{B})$

The two ordered source products are given by Maxwell's equation:

Table 29. Components of FD^* .

real scalar:	$-\mathbf{E} \cdot \mathbf{J}$
img scalar:	$-\mathbf{B} \cdot \mathbf{J}$
real vector:	$\rho\mathbf{E} + \mathbf{B} \times \mathbf{J}$
img vector:	$\rho\mathbf{B} - \mathbf{E} \times \mathbf{J}$

Table 30. Components of D^*F .

real scalar:	$-\mathbf{E} \cdot \mathbf{J}$
img scalar:	$-\mathbf{B} \cdot \mathbf{J}$
real vector:	$\rho\mathbf{E} - \mathbf{B} \times \mathbf{J}$
img vector:	$\rho\mathbf{B} + \mathbf{E} \times \mathbf{J}$

Their symmetric sum is

Table 31. Components of $D^*F + FD^*$.

real scalar:	$-2\mathbf{E} \cdot \mathbf{J}$
img scalar:	$-2\mathbf{B} \cdot \mathbf{J}$
real vector:	$2\rho\mathbf{E}$
img vector:	$2\rho\mathbf{B}$

Only part of the derivative of F^2 is contained in the last table. The complete identity is

$$\partial(F^2) = D^*F + FD^* + \sum_{n=1}^3 \text{com}(\sigma_n, F) \partial_n F.$$

With $F = \mathbf{F} \cdot \boldsymbol{\sigma}$, the correction term is

$$\begin{aligned} \sum_{n=1}^3 \text{com}(\sigma_n, F) \partial_n F &= -2i\mathbf{F} \cdot (\partial_r \times \mathbf{F}) \\ &\quad - 2 \left((\partial_r \cdot \mathbf{F})\mathbf{F} - \frac{1}{2}\partial_r(F^2) \right) \cdot \boldsymbol{\sigma}. \end{aligned}$$

It is precisely the correction that would be lost by applying an ordinary two-term Leibniz rule to noncommuting paravector factors.

6 Relativistic Wave Equations and Spin Dynamics

The standard relativistic wave equations are obtained from the same ordered product calculus. Three logically different carriers are separated: a complex scalar for Klein–Gordon theory, a fixed projected column space for a Dirac field, and the surviving two-entry field in the Pauli limit. By keeping these distinctions explicit, algebraic degrees of freedom are prevented from being mistaken for independent physical wave-function components [7–10]. Every spatial integral in this section is evaluated at fixed time. Unless a region is named, the integration is extended over all real values of the three spatial components; the measure is written directly as $dr_1 dr_2 dr_3$.

6.1 Klein–Gordon field and frequency splitting

The form to be represented is fixed by the classical mass-shell relation, while the replacement of classical energy and momentum by differential operators is supplied as the standard quantization input; it is not obtained from the determinant identity by algebra alone. For a complex scalar field $\psi \in \mathbb{C}$, the free Klein–Gordon equation is

$$(\partial\partial^* + m^2)\psi = (\square + m^2)\psi = 0. \quad (172)$$

On a suitable spatial domain, the positive frequency operator is defined by

$$\Omega := \sqrt{m^2 - \partial_{\mathbf{r}}^2}. \quad (173)$$

Here Ω is defined mode by mode: a spatial Fourier mode with wave vector $\mathbf{k}$ is multiplied by the positive real number $\sqrt{m^2 + \mathbf{k}^2}$. When Ω^{-1} exists on the chosen data, the two frequency functions are defined by

$$\psi(\pm) := \frac{1}{2}(\psi \pm i\Omega^{-1}\partial_t\psi). \quad (174)$$

They satisfy

$$\begin{aligned} (\partial_t + i\Omega)\psi(+) &= 0, & (\partial_t - i\Omega)\psi(-) &= 0, \\ \psi &= \psi(+) + \psi(-). \end{aligned} \quad (175)$$

Thus

$$\psi(\pm, t) = \exp(\mp i\Omega t)\psi(\pm, 0).$$

A massless zero mode is treated separately. The scalar $|\psi|^2$ is not the relativistic Klein–Gordon probability density; derivatives are contained in the conserved current, which is not positive on the full positive/negative-frequency solution space. Frequency is labeled in this subsection by the sign in $\psi(\pm)$; that input is unrelated to the plus/minus input of the spin-axis projectors in Eq. (65).

6.2 Minimal coupling and ordered factorization

Under the standard minimal-coupling rule, the free operators are replaced by the following covariant operators. With the real potential $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$, the energy operator is defined by

$$E := i\partial_t - qV. \quad (176)$$

The momentum operator is defined by

$$\mathbf{P} := -i\partial_{\mathbf{r}} - q\mathbf{A}. \quad (177)$$

The minimally coupled scalar equation is

$$(-E^2 + \mathbf{P}^2 + m^2)\psi = 0, \quad (178)$$

or, after applying the Leibniz rule,

$$\left(\square + iqS - q^2 \det(\Phi) + m^2 + 2iq(V\partial_t + \mathbf{A} \cdot \partial_{\mathbf{r}}) \right) \psi = 0. \quad (179)$$

Gauge covariance of the scalar equation is also obtained from the same phase law used for the Dirac field. Under Eq. (158), the transformed scalar field is defined by

$$\psi(\lambda) := \exp(-iq\lambda)\psi. \quad (180)$$

The gauge-dependent energy and momentum operators are defined by

$$E(\lambda) := i\partial_t - qV(\lambda). \quad (181)$$

$$\mathbf{P}(\lambda) := -i\partial_{\mathbf{r}} - q\mathbf{A}(\lambda). \quad (182)$$

The following relations are obtained by direct application of the product rule:

$$E(\lambda)\psi(\lambda) = \exp(-iq\lambda)E\psi, \quad \mathbf{P}(\lambda)\psi(\lambda) = \exp(-iq\lambda)\mathbf{P}\psi, \quad (183)$$

so the same form of Eq. (178) is retained in every gauge. For smooth fields,

$$\begin{aligned} \text{com}(E, P_m) &= iqE_m, \\ \text{com}(P_m, P_n) &= iq(\partial_m A_n - \partial_n A_m). \end{aligned} \quad (184)$$

Consequently, the written order must be retained:

$$(E + \boldsymbol{\sigma} \cdot \mathbf{P})(E - \boldsymbol{\sigma} \cdot \mathbf{P}) = E^2 - \mathbf{P}^2 - iqF, \quad (185)$$

$$(E - \boldsymbol{\sigma} \cdot \mathbf{P})(E + \boldsymbol{\sigma} \cdot \mathbf{P}) = E^2 - \mathbf{P}^2 - iqF^*. \quad (186)$$

Here

$$\begin{aligned} F^* &= (-\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \\ -iqF &= (-iq\mathbf{E} + q\mathbf{B}) \cdot \boldsymbol{\sigma}, \\ -iqF^* &= (iq\mathbf{E} + q\mathbf{B}) \cdot \boldsymbol{\sigma}. \end{aligned}$$

These are ordered star- H products of differential operators, not applications of the scalar-coefficient determinant formula. In a background field the full products are not scalar factorizations of the Klein–Gordon operator: the displayed F or F^* term is retained. The scalar equation is recovered by taking the scalar part or by subtracting that residual term explicitly. The same residuals are converted into the established spin–field terms when the Dirac system is squared; they are not additional scalar-wave components.

6.3 Klein–Gordon current and pairing

For two scalar solutions ϕ and ψ of Eq. (178) in the same real potential, the mixed density function is defined by

$$\rho(\phi, \psi) := \frac{\phi^* E\psi + (E\phi)^* \psi}{2m}. \quad (187)$$

The associated spatial-current function is defined by

$$\mathbf{j}(\phi, \psi) := \frac{\phi^* \mathbf{P}\psi + (\mathbf{P}\phi)^* \psi}{2m}. \quad (188)$$

Star here is ordinary complex conjugation because both fields are scalars. Gauge invariance of this mixed current and its pairing is established immediately by Eq. (183) when the same phase is acquired by both solutions. The local identity is obtained from the product rules:

$$\begin{aligned} \partial_t \rho(\phi, \psi) + \partial_{\mathbf{r}} \cdot \mathbf{j}(\phi, \psi) &= \frac{i}{2m} ((E^2 - \mathbf{P}^2)\phi)^* \psi \\ &\quad - \frac{i}{2m} \phi^* (E^2 - \mathbf{P}^2)\psi \\ &= \frac{i}{2m} (m^2 \phi^* \psi - \phi^* m^2 \psi) = 0. \end{aligned} \quad (189)$$

The conserved Klein–Gordon current is obtained in the equal-field case; its electric current is obtained by multiplication by q . The mixed solution pairing is defined by its spatial integral:

$$\langle \phi | \psi \rangle := \iiint \rho(\phi, \psi) dr_1 dr_2 dr_3. \quad (190)$$

For free positive-frequency solutions, the pairing is

$$\langle \phi | \psi \rangle = \frac{1}{m} \iiint \phi^* \Omega \psi dr_1 dr_2 dr_3,$$

which is positive for equal nonzero fields. The local density can nevertheless be negative at some points in a positive-frequency superposition. Therefore a positive local position probability is not supplied by $\rho(\psi, \psi)$ in the scalar theory; on the full positive- and negative-frequency solution space, even the integrated pairing is indefinite.

6.4 Projected carrier for one Dirac field

In the Pauli realization, a paravector is represented by an arbitrary two-by-two complex matrix. Its two columns are acted upon independently by left multiplication. Therefore two degenerate Dirac fields, rather than one, are described by two unrestricted paravector blocks. A single Dirac field is obtained by selecting one matrix column. The fixed-column function is defined intrinsically by

$$\Pi(\text{column}) := \frac{1}{2}(1 + \sigma_3). \quad (191)$$

One Dirac field is selected by the constraint

$$\Psi_n = \Psi_n \Pi(\text{column}), \quad n \in \{1, 2\}. \quad (192)$$

Every operator below is applied from the left, so the restriction is preserved: $(Z\Psi_n)\Pi(\text{column}) = Z(\Psi_n\Pi(\text{column})) = Z\Psi_n$. Selection of one matrix column by this fixed right factor is denoted by the input column. No measurement of spin along the third axis is represented by it. In the matrix realization, each Ψ_n is therefore represented by one ordinary two-entry complex column, and the usual four complex entries are contained in the block column $\Psi = \{\Psi_1; \Psi_2\}$. The apparent scalar and three-vector coefficients are correspondingly constrained. If $\psi_v := \{\psi_1, \psi_2, \psi_3\}$ and $\psi = \psi_s + \psi_v \cdot \sigma$, then

$$\psi = \psi \Pi(\text{column}) \quad \text{if and only if} \quad \psi_3 = \psi_s, \quad \psi_2 = -i\psi_1, \\ \mathbf{M} = \{2\psi_s, 0; 2\psi_1, 0\}.$$

Thus exactly the two complex amplitudes in the selected column are contained in a projected block, not four independent paravector amplitudes.

For any projected block ψ , the positive density function is defined by

$$\rho(\psi) := 2 \text{ scalar}(\psi^H \psi). \quad (193)$$

The factor 2 converts scalar extraction into the ordinary norm of the selected two-entry column.

Using the block map $\mathbf{W}$ from Eq. (120), the free and coupled Dirac equations are

$$(\mathbf{W}(i\partial) - m\mathbf{I})\Psi = 0, \quad (194)$$

$$(\mathbf{W}(i\partial - q\Phi^*) - m\mathbf{I})\Psi = 0. \quad (195)$$

The coupled system is equivalently

$$(E - \sigma \cdot \mathbf{P})\Psi_2 = m\Psi_1, \quad (E + \sigma \cdot \mathbf{P})\Psi_1 = m\Psi_2. \quad (196)$$

In the matrix realization, the selected columns are represented directly as

$$\Psi_1 \quad \text{is represented by} \quad \begin{Bmatrix} a \\ b \end{Bmatrix}, \\ \Psi_2 \quad \text{is represented by} \quad \begin{Bmatrix} c \\ d \end{Bmatrix}.$$

Equation (196) is then exactly

$$\begin{aligned} (E - P_3)c - (P_1 - iP_2)d &= ma, \\ -(P_1 + iP_2)c + (E + P_3)d &= mb, \\ (E + P_3)a + (P_1 - iP_2)b &= mc, \\ (P_1 + iP_2)a + (E - P_3)b &= md. \end{aligned} \quad (197)$$

These are the four conventional first-order Dirac component equations in the two-block form. The free system is obtained by setting $q = 0$. For a constant determinant-one transform T , the two blocks are fixed differently by covariance:

$$\Psi'_1(X') = T^H \Psi_1(X), \quad (198)$$

$$\Psi'_2(X') = T^{-1} \Psi_2(X). \quad (199)$$

As shown by the supporting derivation of Eqs. (198) and (199), the first block is therefore acted upon by L under a boost, whereas the second is acted upon by L^* ; both are acted upon by R^H under a rotation. This is the wave-function law, distinct from both event congruence and field similarity. The mass-conjugate block operator is defined by

$$\mathbf{D}' := \mathbf{W}(i\partial - q\Phi^*) + m\mathbf{I}. \quad (200)$$

By multiplication by this operator and use of the two ordered factorizations, the squared system is obtained:

$$\begin{cases} (-E^2 + \mathbf{P}^2 + m^2 + iqF^*)\Psi_1 = 0, \\ (-E^2 + \mathbf{P}^2 + m^2 + iqF)\Psi_2 = 0. \end{cases} \quad (201)$$

With $iqF^* = (-iq\mathbf{E} - q\mathbf{B}) \cdot \sigma$ and $iqF = (iq\mathbf{E} - q\mathbf{B}) \cdot \sigma$, the same equations are written as

$$\begin{cases} (-E^2 + \mathbf{P}^2 + m^2 - q\mathbf{B} \cdot \sigma - iq\mathbf{E} \cdot \sigma)\Psi_1 = 0, \\ (-E^2 + \mathbf{P}^2 + m^2 - q\mathbf{B} \cdot \sigma + iq\mathbf{E} \cdot \sigma)\Psi_2 = 0. \end{cases} \quad (202)$$

Thus the expected Klein–Gordon operator, the same magnetic spin coupling in both blocks, and the opposite electric spin couplings of the two blocks are obtained component by component by squaring the first-order system.

Under the gauge law (158),

$$\Psi(\lambda) := \exp(-iq\lambda)\Psi. \quad (203)$$

The form of Eq. (195) is unchanged.

A positive time-slice norm is supplied by the Hermitian product, but its vector part is not a probability current. With the fixed-column convention above, the Dirac density function is defined by

$$\rho(\Psi) := 2 \text{ scalar}(\Psi_1^H \Psi_1 + \Psi_2^H \Psi_2). \quad (204)$$

The spatial-current function is defined by

$$\mathbf{j}(\Psi) := 2 \text{ vector}(\Psi_2 \Psi_2^H - \Psi_1 \Psi_1^H). \quad (205)$$

The full current paravector is defined by

$$J(\Psi) := \rho(\Psi) + \mathbf{j}(\Psi) \cdot \boldsymbol{\sigma}. \quad (206)$$

The two definitions satisfy the continuity equation

$$\partial_t \rho(\Psi) + \partial_{\mathbf{r}} \cdot \mathbf{j}(\Psi) = 0. \quad (207)$$

The normalization condition is the spatial integral

$$\iiint \rho(\Psi) dr_1 dr_2 dr_3 = 1,$$

not a pointwise equation. By Eqs. (116) and (242), H is the ordinary conjugate transpose and $\rho(\Psi)$ is the sum of the two conventional column norm squares. The familiar Dirac spatial current is supplied by the displayed opposite-order bilinears, rather than by the vector part of a single Hermitian norm product. The real current paravector from Eq. (206) obeys the event congruence, as shown in the supporting derivation. The component balances are obtained directly from the two time-evolution equations:

$$\begin{aligned} \partial_t(|a|^2 + |b|^2) &= \partial_{\mathbf{r}} \cdot \left(\{a^* \quad b^*\} \boldsymbol{\sigma} \begin{Bmatrix} a \\ b \end{Bmatrix} \right) \\ &\quad + im(c^*a + d^*b - a^*c - b^*d), \\ \partial_t(|c|^2 + |d|^2) &= -\partial_{\mathbf{r}} \cdot \left(\{c^* \quad d^*\} \boldsymbol{\sigma} \begin{Bmatrix} c \\ d \end{Bmatrix} \right) \\ &\quad + im(a^*c + b^*d - c^*a - d^*b). \end{aligned} \quad (208)$$

By addition of the lines, the mass exchange is canceled and the following expressions are obtained exactly:

$$\begin{aligned} \rho(\Psi) &= |a|^2 + |b|^2 + |c|^2 + |d|^2, \\ \mathbf{j}(\Psi) &= \{c^* \quad d^*\} \boldsymbol{\sigma} \begin{Bmatrix} c \\ d \end{Bmatrix} - \{a^* \quad b^*\} \boldsymbol{\sigma} \begin{Bmatrix} a \\ b \end{Bmatrix}, \end{aligned}$$

and hence the continuity equation in Eq. (207). Neither block norm is conserved separately: norm is exchanged between them by the mass terms.

6.5 Probabilities and measurement expectations

For two projected one-block fields, the bracket is defined by

$$\langle \phi | \psi \rangle := 2 \iiint \text{scalar}(\phi^H \psi) dr_1 dr_2 dr_3. \quad (209)$$

For two projected two-block fields, the complete Dirac pairing is defined by

$$\begin{aligned} &\langle \Psi(\text{left}) | \Psi(\text{right}) \rangle \\ &:= 2 \iiint \sum_{n=1}^2 \text{scalar}(\Psi_n(\text{left})^H \Psi_n(\text{right})) dr_1 dr_2 dr_3. \end{aligned} \quad (210)$$

The two inputs are identified by the words left and right; no new kind of field is introduced. When both inputs are the same, the shorter notation $\langle \Psi | \Psi \rangle$ is used. The conversion of intrinsic scalar extraction into the ordinary selected-column product is supplied by the factor 2. For a specified block operator $\mathcal{O}$, the operator pairing is defined by

$$\begin{aligned} &\langle \Psi(\text{left}) | \mathcal{O} \Psi(\text{right}) \rangle \\ &:= 2 \iiint \sum_{n=1}^2 \text{scalar}(\Psi_n(\text{left})^H (\mathcal{O} \Psi(\text{right}))_n) dr_1 dr_2 dr_3. \end{aligned} \quad (211)$$

Its normalized matrix element is

$$\frac{\langle \Psi | \mathcal{O} \Psi \rangle}{\langle \Psi | \Psi \rangle}. \quad (212)$$

It is a real expectation when $\mathcal{O}$ is Hermitian under this spatial pairing. For multiplication operators, Hermitian diagonal entries and $\mathcal{O}_{21} = \mathcal{O}_{12}^H$ are required for the diagonal and off-diagonal entries, respectively; for derivatives, the stated integration-by-parts boundary conditions and a common operator domain are required. If a Hermitian operator is also idempotent, the probability of the outcome selected by that projector is given by the same ratio.

For a normalized Dirac field, the probability of finding the system in a specified spatial region at the chosen time is defined by

$$p(\text{region}) := \iiint \rho(\Psi) dr_1 dr_2 dr_3, \quad (213)$$

where the integration is restricted to the specified region. The density is nonnegative, but it is the time component of the covariant current, not a Lorentz scalar. The normalization is kept time-independent by conservation when the boundary flux vanishes.

For a projected Pauli block ψ , the factor 2 is canceled from normalized ratios. The probability function supplied by the axis projectors is therefore defined by

$$p(\pm) := \frac{\langle \psi(\pm) | \psi(\pm) \rangle}{\langle \psi | \psi \rangle}. \quad (214)$$

By Eqs. (65) and (66), this can equally be written with $\psi(\pm) = \Pi(\pm)\psi$, and

$$p(+) + p(-) = 1. \quad (215)$$

The expectation of $\mathbf{u} \cdot \boldsymbol{\sigma}$ is $p(+)-p(-)$; the physical spin component along $\mathbf{u}$ is one half of this value in the natural units used here. The corresponding free two-block energy and momentum-axis probability is defined after those fields have been constructed below.

The Klein–Gordon carrier is instead paired by Eq. (190). On a chosen positive-frequency solution sector, the normalized matrix element of an operator $\mathcal{O}$ is

$$\frac{\langle \psi | \mathcal{O} \psi \rangle}{\langle \psi | \psi \rangle}$$

only when that sector is preserved and the operator is Hermitian under the Klein–Gordon pairing. The density $\rho(\psi, \psi)$ is not thereby converted into a pointwise position probability. Likewise, nonscalar channels of a generic paravector sandwich are algebraic bilinears, not additional measured quantities unless they are identified by a field equation and a measurement rule.

6.6 Pauli limit

A single block by itself is not the nonrelativistic large component. First, the slowly varying blocks are defined by removal of the rest phase:

$$\Psi_n(\text{slow}) := \exp(imt) \Psi_n. \quad (216)$$

The large component is then defined by

$$\phi(\text{large}) := \frac{\Psi_1(\text{slow}) + \Psi_2(\text{slow})}{\sqrt{2}}. \quad (217)$$

The small component is defined independently by

$$\phi(\text{small}) := \frac{\Psi_2(\text{slow}) - \Psi_1(\text{slow})}{\sqrt{2}}. \quad (218)$$

Equation (196) is then transformed exactly into

$$\begin{aligned} E\phi(\text{large}) &= (\mathbf{P} \cdot \boldsymbol{\sigma})\phi(\text{small}), \\ (E + 2m)\phi(\text{small}) &= (\mathbf{P} \cdot \boldsymbol{\sigma})\phi(\text{large}). \end{aligned} \quad (219)$$

At energies small compared with m ,

$$\phi(\text{small}) = \frac{\mathbf{P} \cdot \boldsymbol{\sigma}}{2m} \phi(\text{large}) + O(m^{-2}).$$

By use of

$$(\mathbf{P} \cdot \boldsymbol{\sigma})^2 = \mathbf{P}^2 - q\mathbf{B} \cdot \boldsymbol{\sigma},$$

the following result is obtained:

$$E\phi(\text{large}) = \frac{\mathbf{P}^2 - q\mathbf{B} \cdot \boldsymbol{\sigma}}{2m} \phi(\text{large}) + O(m^{-2}). \quad (220)$$

Equivalently,

$$\begin{aligned} i\partial_t \phi(\text{large}) &= \left(\frac{(-i\partial_{\mathbf{r}} - q\mathbf{A})^2}{2m} + qV \right) \phi(\text{large}) \\ &\quad - \frac{q}{2m} (\mathbf{B} \cdot \boldsymbol{\sigma}) \phi(\text{large}) + O(m^{-2}). \end{aligned} \quad (221)$$

The spin–orbit, Darwin, and time-dependent corrections are generated at the next order by electric-field commutators; no term

of that order is retained selectively here. The selected two-entry column representing the large component is defined by

$$\phi(\text{large}) := \text{selected column representing } \phi(\text{large}). \quad (222)$$

The corresponding small-component column is defined by

$$\phi(\text{small}) := \text{selected column representing } \phi(\text{small}). \quad (223)$$

The following expressions are obtained by the exact change of basis:

$$\begin{aligned} \rho(\Psi) &= \phi(\text{large})^{cT} \phi(\text{large}) + \phi(\text{small})^{cT} \phi(\text{small}), \\ \mathbf{j}(\Psi) &= \phi(\text{large})^{cT} \boldsymbol{\sigma} \phi(\text{small}) \\ &\quad + \phi(\text{small})^{cT} \boldsymbol{\sigma} \phi(\text{large}). \end{aligned}$$

The projected-block density from Eq. (193) is represented here by

$$\begin{aligned} \rho(\phi(\text{large})) &= \phi(\text{large})^{cT} \phi(\text{large}) \\ &= 2 \text{ scalar}(\phi(\text{large})^H \phi(\text{large})). \end{aligned} \quad (224)$$

The Pauli spatial-current function is defined by

$$\begin{aligned} \mathbf{j}(\phi(\text{large})) &:= \frac{\phi(\text{large})^{cT} \mathbf{P} \phi(\text{large})}{2m} \\ &\quad + \frac{(\mathbf{P} \phi(\text{large}))^{cT} \phi(\text{large})}{2m} \\ &\quad + \frac{1}{2m} \partial_{\mathbf{r}} \times (\phi(\text{large})^{cT} \boldsymbol{\sigma} \phi(\text{large})). \end{aligned} \quad (225)$$

Because $\phi(\text{small}) = O(m^{-1})$, the original Dirac density and current are reduced to $\rho(\Psi) = \rho(\phi(\text{large})) + O(m^{-2})$ and $\mathbf{j}(\Psi) = \mathbf{j}(\phi(\text{large})) + O(m^{-2})$. The first numerator is read component by component in the three entries of $\mathbf{P}$. The standard convection current plus the divergence-free magnetization current is thereby obtained. The equation $\partial_t \rho(\phi(\text{large})) + \partial_{\mathbf{r}} \cdot \mathbf{j}(\phi(\text{large})) = 0$ is obeyed through the retained order.

If $\mathbf{B} \neq 0$ is uniform and constant, the magnetic-field axis is defined by

$$\mathbf{u} := \frac{\mathbf{B}}{\|\mathbf{B}\|}. \quad (226)$$

Equation (65) is then specialized as $\Pi(\pm) = \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})$. The two spin-axis parts are defined by

$$\phi(\text{large}, \pm) := \Pi(\pm) \phi(\text{large}). \quad (227)$$

They reconstruct the large component:

$$\phi(\text{large}) = \phi(\text{large}, +) + \phi(\text{large}, -). \quad (228)$$

Then

$$\begin{aligned} i\partial_t \phi(\text{large}, \pm) &= \left(\frac{\mathbf{P}^2}{2m} + qV \right) \phi(\text{large}, \pm) \\ &\quad \mp \frac{q\|\mathbf{B}\|}{2m} \phi(\text{large}, \pm). \end{aligned} \quad (229)$$

For a varying axis, the two projected fields are coupled by derivatives of $\Pi(\pm)$ and its commutator with the time-evolution operator on the right of Eq. (221), so this decoupling is lost.

6.7 Free energy and momentum-axis split

For a free momentum mode, the positive energy function is defined by

$$E(\mathbf{p}) := \sqrt{m^2 + \mathbf{p}^2}. \quad (230)$$

The block Hamiltonian is defined by

$$\mathcal{H} := \begin{pmatrix} -\boldsymbol{\sigma} \cdot \mathbf{p} & m \\ m & \boldsymbol{\sigma} \cdot \mathbf{p} \end{pmatrix}. \quad (231)$$

Equation (196) then gives

$$\begin{aligned} i\partial_t \Psi &= \mathcal{H} \Psi, \\ \mathcal{H} \Psi &= \begin{Bmatrix} -(\boldsymbol{\sigma} \cdot \mathbf{p}) \Psi_1 + m \Psi_2 \\ m \Psi_1 + (\boldsymbol{\sigma} \cdot \mathbf{p}) \Psi_2 \end{Bmatrix}, \\ \mathcal{H}^2 &= E(\mathbf{p})^2 \mathbf{I}. \end{aligned} \quad (232)$$

The block matrix $\mathcal{H}$ is called the Hamiltonian because $i\partial_t \Psi$ is supplied by exactly this operator. It is not a new field: the two free Dirac equations are given by its two displayed rows. The calligraphic symbol $\mathcal{H}$ is distinct from the superscript H , which is reserved for the Hermitian operation.

The energy sign e is taken from $\{+1, -1\}$. The corresponding part of the field is defined directly, without naming another operator:

$$\Psi(e) := \frac{1}{2} \left(\Psi + e \frac{\mathcal{H} \Psi}{E(\mathbf{p})} \right). \quad (233)$$

Its two blocks are displayed explicitly as

$$\Psi(e) = \frac{1}{2} \begin{Bmatrix} \Psi_1 + \frac{e}{E(\mathbf{p})} (-(\boldsymbol{\sigma} \cdot \mathbf{p}) \Psi_1 + m \Psi_2) \\ \Psi_2 + \frac{e}{E(\mathbf{p})} (m \Psi_1 + (\boldsymbol{\sigma} \cdot \mathbf{p}) \Psi_2) \end{Bmatrix}. \quad (234)$$

The eigenvalue and reconstruction identities are

$$\mathcal{H} \Psi(e) = e E(\mathbf{p}) \Psi(e), \quad \Psi = \Psi(+) + \Psi(-). \quad (235)$$

Thus the two choices of e are the positive- and negative-energy parts of the same free field. The two blocks displayed in $\Psi(e)$ are denoted below by $\Psi_1(e)$ and $\Psi_2(e)$.

For $\mathbf{p} \neq 0$, the momentum direction is defined by

$$\mathbf{u} := \frac{\mathbf{p}}{\|\mathbf{p}\|}. \quad (236)$$

For $s \in \{+1, -1\}$, the existing axis projector from Eq. (65) is then

$$\Pi(s) = \frac{1}{2} (1 + s \mathbf{u} \cdot \boldsymbol{\sigma}) = \frac{1}{2} \left(1 + s \frac{\mathbf{p} \cdot \boldsymbol{\sigma}}{\|\mathbf{p}\|} \right). \quad (237)$$

The simultaneous energy and momentum-axis part is defined by

$$\Psi(e, s) := \begin{Bmatrix} \Pi(s) \Psi_1(e) \\ \Pi(s) \Psi_2(e) \end{Bmatrix}. \quad (238)$$

It satisfies

$$\begin{aligned} (\mathbf{p} \cdot \boldsymbol{\sigma}) \Pi(s) &= s \|\mathbf{p}\| \Pi(s), \\ \Psi &= \sum_{e \in \{+1, -1\}} \sum_{s \in \{+1, -1\}} \Psi(e, s). \end{aligned} \quad (239)$$

Here s is the eigenvalue of $\mathbf{u} \cdot \boldsymbol{\sigma}$. In the natural units used here, the physical spin component along momentum is $s/2$; its sign is commonly called helicity. The Hamiltonian is Hermitian, and the two energy parts are orthogonal. The already established $\Pi(s)$ projectors are Hermitian, idempotent, and complementary. Because the momentum axis is used, commutation of their left action with $\mathcal{H}$ is obtained, so the four displayed fields are simultaneous energy and momentum-axis parts. At rest any spin axis may be chosen. Negative-frequency classical solutions are not identified with antiparticles until the field is quantized.

The joint probability function is now defined without referring to a field before its construction:

$$p(e, s) := \frac{\langle \Psi(e, s) | \Psi(e, s) \rangle}{\langle \Psi | \Psi \rangle}. \quad (240)$$

Completeness gives

$$\sum_{e \in \{+1, -1\}} \sum_{s \in \{+1, -1\}} p(e, s) = 1. \quad (241)$$

For a normalizable free field this split is applied to each momentum mode before the spatial pairing is evaluated. For a single plane mode, the usual box or delta-function normalization is required instead.

6.8 Supporting Wave-Equation Derivations

The ordered operator expansions, scalar frequency evolution, projected-carrier interpretation, conserved pairings, gauge covariance, squared Dirac system, and controlled Pauli approximation are given in detail in the supporting derivations.

6.8.1 Relativistic Quantum-Mechanical Setting

One fixed operator language is used throughout the wave-equation section. With the differential conventions from Section 4,

$$\begin{aligned} \partial &= \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}, \\ \partial^* &= \partial_t - \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}, \\ (i\partial)^* &= -i\partial^*. \end{aligned}$$

The d'Alembertian is

$$\begin{aligned} \square &= \partial_t^2 - \partial_{\mathbf{r}}^2 \\ &= \partial \partial^* = \partial^* \partial. \end{aligned}$$

All fields are acted on by the differential operators to their left. In an ordered product, the action of the rightmost factor is applied first, and the written order is retained whenever potentials and derivatives do not commute.

6.8.2 Algebra Blocks and the Physical Spinor Carrier

Throughout the spinful part of the wave-equation section, a wave field is not a space-time event but an algebra-valued amplitude field. A generic algebra block may be written

$$\psi : \mathbb{R} \times \mathbb{R}^3 \rightarrow \mathbb{C} + \mathbb{C}^3, \quad \psi(t, \mathbf{r}) = \psi_s(t, \mathbf{r}) + \psi_v(t, \mathbf{r}) \cdot \boldsymbol{\sigma}.$$

The location at which the state is evaluated in space-time is specified by the argument $(t, \mathbf{r})$, while the value $\psi(t, \mathbf{r})$ is contained in the internal complex-paravector algebra. The real paravector

$$X = t + \mathbf{r} \cdot \boldsymbol{\sigma}$$

is used to describe an event or interval variable, whereas ψ is a quantum state amplitude defined *over* such events.

At each fixed point, four complex coefficients are contained in an unrestricted complex paravector. In the Pauli matrix realization it is an arbitrary 2×2 complex matrix, and its two columns are acted on independently by left multiplication. Two independent two-entry columns are therefore contained, not one physical wave-function block. One column is selected without adding component indices by the fixed selector $\Pi(\text{column})$ from Eq. (191), and a physical block is restricted by

$$\psi = \psi \Pi(\text{column}).$$

The column is selected by the fixed factor on the right. Every displayed operator is applied on the left, so the restriction is preserved by associativity: $(Z\psi)\Pi(\text{column}) = Z(\psi\Pi(\text{column})) = Z\psi$. The factor $\Pi(\text{column})$ is only a column selector; it is not a measurement of spin along the third axis.

The scalar and vector coefficients of an algebra block are amplitude channels, not separately measurable particles or classical subfields. A bare component count also does not cause physical spin to be assigned: the transformation law, field equation, and carrier restriction must be specified together.

The field $\psi \in \mathbb{C}$ is used in the spin-0 Klein–Gordon subsections. A two-block field is used in the Dirac construction:

$$\Psi = \begin{Bmatrix} \Psi_1 \\ \Psi_2 \end{Bmatrix},$$

with $\Psi_n = \Psi_n \Pi(\text{column})$. Two complex amplitudes are then carried by each block, and the four complex amplitudes of one Dirac field are carried by the pair. If the fixed-column restriction is omitted, the same equations remain algebraically valid but are applicable to two independent Dirac fields, one in each matrix column.

6.8.3 Complex-Paravector Bilinears and Bra-Ket Extraction

The bra-ket notation is made precise in the paravector setting by the decompositions

$$\psi = \psi_s + \psi_v \cdot \boldsymbol{\sigma}, \quad Z = Z_s + \mathbf{Z}_v \cdot \boldsymbol{\sigma}, \quad X = X_s + \mathbf{X}_v \cdot \boldsymbol{\sigma},$$

with $\psi_s, Z_s \in \mathbb{C}$, $\psi_v, \mathbf{Z}_v \in \mathbb{C}^3$, $X_s \in \mathbb{R}$, $\mathbf{X}_v \in \mathbb{R}^3$, and with fields acted on by operator-valued components to their left. The three involutions on the state are

$$\begin{aligned} \psi^* &= \psi_s^* - \psi_v^* \cdot \boldsymbol{\sigma}, \\ \psi^H &= \psi_s^* + \psi_v^* \cdot \boldsymbol{\sigma}, \\ \psi^{*H} &= \psi_s - \psi_v \cdot \boldsymbol{\sigma}. \end{aligned}$$

The reason for the H pairing is made explicit by Eq. (116). If ϕ and ψ are realized by the fixed right projector as the two-entry

columns $\{a; b\}$ and $\{c; d\}$, then

$$2 \text{ scalar}(\phi^H \psi) = \text{tr}(\phi^H \psi) = \begin{Bmatrix} a^* & b^* \end{Bmatrix} \begin{Bmatrix} c \\ d \end{Bmatrix} = a^* c + b^* d. \quad (242)$$

Thus ϕ^H is precisely the familiar row bra in the selected-column realization, and $2 \text{ scalar}(\psi^H \psi)$ is its nonnegative ordinary column norm. The factor 2 is required because intrinsic scalar extraction is one half of the matrix trace. By summation of this relation over the two projected Dirac blocks, the density is obtained; the spatial current is instead obtained from the opposite-order block bilinears fixed by the Dirac equation. The basic component products are collected below.

The direct action of a complex paravector operator on a complex paravector state is recorded in Table 32.

Table 32. Components of $\mathcal{O}\psi$.

scalar:	$\text{scalar}(\mathcal{O})\psi_s + \text{vector}(\mathcal{O}) \cdot \psi_v$
vector:	$\text{scalar}(\mathcal{O})\psi_v + \text{vector}(\mathcal{O})\psi_s + i(\text{vector}(\mathcal{O}) \times \psi_v)$

The three basic bilinear sandwiches with a generic complex paravector $Z = Z_s + \mathbf{Z}_v \cdot \boldsymbol{\sigma}$ are given in Tables 33–35.

Table 33. Components of $\psi^* Z \psi$.

scalar:	$Z_s(\psi_s ^2 - \psi_v^* \cdot \psi_v) + \psi_s^*(\mathbf{Z}_v \cdot \psi_v) - \psi_s(\mathbf{Z}_v \cdot \psi_v^*) - i\psi_v^* \cdot (\mathbf{Z}_v \times \psi_v)$
vector:	$Z_s(\psi_s^* \psi_v - \psi_s \psi_v^* - i\psi_v^* \times \psi_v) + (\psi_s ^2 + \psi_v^* \cdot \psi_v)\mathbf{Z}_v - (\mathbf{Z}_v \cdot \psi_v^*)\psi_v - (\mathbf{Z}_v \cdot \psi_v)\psi_v^* + i(\psi_s^* \mathbf{Z}_v \times \psi_v + \psi_s \mathbf{Z}_v \times \psi_v^*)$

Table 34. Components of $\psi^H Z \psi$.

scalar:	$Z_s(\psi_s ^2 + \psi_v^* \cdot \psi_v) + \psi_s^*(\mathbf{Z}_v \cdot \psi_v) + \psi_s(\mathbf{Z}_v \cdot \psi_v^*) + i\psi_v^* \cdot (\mathbf{Z}_v \times \psi_v)$
vector:	$Z_s(\psi_s^* \psi_v + \psi_s \psi_v^* + i\psi_v^* \times \psi_v) + (\psi_s ^2 - \psi_v^* \cdot \psi_v)\mathbf{Z}_v + (\mathbf{Z}_v \cdot \psi_v^*)\psi_v + (\mathbf{Z}_v \cdot \psi_v)\psi_v^* + i(\psi_s^* \mathbf{Z}_v \times \psi_v - \psi_s \mathbf{Z}_v \times \psi_v^*)$

Table 35. Components of $\psi^{*H} Z \psi$.

scalar:	$Z_s(\psi_s^2 - \psi_v \cdot \psi_v)$
vector:	$(\psi_s^2 + \psi_v \cdot \psi_v)\mathbf{Z}_v - 2(\mathbf{Z}_v \cdot \psi_v)\psi_v + 2i\psi_s \mathbf{Z}_v \times \psi_v$

Tables 36–38 are obtained by specialization to a real paravector $X = X_s + \mathbf{X}_v \cdot \boldsymbol{\sigma}$, with $X_s \in \mathbb{R}$ and $\mathbf{X}_v \in \mathbb{R}^3$.

Table 36. Components of $\psi^* X \psi$.

scalar:	$X_s(\psi_s ^2 - \psi_v^* \cdot \psi_v)$ $+ \psi_s^*(\mathbf{X}_v \cdot \psi_v) - \psi_s(\mathbf{X}_v \cdot \psi_v^*)$ $- i \psi_v^* \cdot (\mathbf{X}_v \times \psi_v)$
vector:	$X_s(\psi_s^* \psi_v - \psi_s \psi_v^* - i \psi_v^* \times \psi_v)$ $+ (\psi_s ^2 + \psi_v^* \cdot \psi_v) \mathbf{X}_v$ $- (\mathbf{X}_v \cdot \psi_v^*) \psi_v - (\mathbf{X}_v \cdot \psi_v) \psi_v^*$ $+ i(\psi_s^* \mathbf{X}_v \times \psi_v + \psi_s \mathbf{X}_v \times \psi_v^*)$

Table 37. Components of $\psi^H X \psi$.

scalar:	$X_s(\psi_s ^2 + \psi_v^* \cdot \psi_v)$ $+ \psi_s^*(\mathbf{X}_v \cdot \psi_v) + \psi_s(\mathbf{X}_v \cdot \psi_v^*)$ $+ i \psi_v^* \cdot (\mathbf{X}_v \times \psi_v)$
vector:	$X_s(\psi_s^* \psi_v + \psi_s \psi_v^* + i \psi_v^* \times \psi_v)$ $+ (\psi_s ^2 - \psi_v^* \cdot \psi_v) \mathbf{X}_v$ $+ (\mathbf{X}_v \cdot \psi_v^*) \psi_v + (\mathbf{X}_v \cdot \psi_v) \psi_v^*$ $+ i(\psi_s^* \mathbf{X}_v \times \psi_v - \psi_s \mathbf{X}_v \times \psi_v^*)$

Table 38. Components of $\psi^{*H} X \psi$.

scalar:	$X_s(\psi_s^2 - \psi_v \cdot \psi_v)$
vector:	$(\psi_s^2 + \psi_v \cdot \psi_v) \mathbf{X}_v$ $- 2(\mathbf{X}_v \cdot \psi_v) \psi_v + 2i \psi_s \mathbf{X}_v \times \psi_v$

For a scalar function S , the same three sandwiches are reduced to Tables 39–41.

Table 39. Components of $\psi^* S \psi$.

scalar:	$S(\psi_s ^2 - \psi_v^* \cdot \psi_v)$
vector:	$S(\psi_s^* \psi_v - \psi_s \psi_v^* - i \psi_v^* \times \psi_v)$

Table 40. Components of $\psi^H S \psi$.

scalar:	$S(\psi_s ^2 + \psi_v^* \cdot \psi_v)$
vector:	$S(\psi_s^* \psi_v + \psi_s \psi_v^* + i \psi_v^* \times \psi_v)$

Table 41. Components of $\psi^{*H} S \psi$.

scalar:	$S(\psi_s^2 - \psi_v \cdot \psi_v)$
vector:	0

For $S \in \mathbb{R}$, the Hermitian sandwich is

$$\psi^H S \psi = S \psi^H \psi,$$

whose physical selected-column density was defined in Eq. (193). Its local form is

$$\rho(\psi) = 2 \text{scalar}(\psi^H \psi) = 2(|\psi_s|^2 + \psi_v^* \cdot \psi_v) \geq 0.$$

In the matrix realization scalar extraction is one half of the trace, so $\rho(\psi)$ is the conventional selected-column norm. The factor is retained explicitly in the scalar-extraction formula. The time-slice norm for a projected Pauli carrier and, after summation over both blocks, for a Dirac carrier is supplied by this positive scalar. It is not the conserved Klein–Gordon norm of a scalar relativistic field.

The vector part of $\psi^H \psi$, however, is not automatically a spatial probability current. In fact, under the active spinor convention $\psi' = R^H \psi$, it is invariant:

$$\psi'^H \psi' = \psi^H R R^H \psi = \psi^H \psi.$$

Spatial spin or polarization information instead occurs in bilinears with the opposite order, such as $\psi \psi^H$, or in operator sandwiches fixed by the field equation. More generally, four complex, hence eight real, coefficient channels are contained in $\psi^H \mathcal{O} \psi$ before an extraction is chosen. No generic physical interpretation is assigned to its nonscalar channels.

For a nonzero projected block, the scale is defined by

$$N := \rho(\psi). \quad (243)$$

The following useful opposite-order identities are obtained from the fixed-column condition:

$$\begin{aligned} \psi^H \psi &= N \Pi(\text{column}), \\ \left(\frac{\psi \psi^H}{N} \right)^H &= \frac{\psi \psi^H}{N}, \\ \left(\frac{\psi \psi^H}{N} \right)^2 &= \frac{\psi \psi^H}{N}. \end{aligned} \quad (244)$$

Only the selected right column is retained by the first product. In the matrix realization the second product is the ordinary rank-one pure-state density matrix; its determinant is zero and twice the vector part of $\psi \psi^H$ is the selected column's sigma-polarization vector. This is the familiar geometry of a two-entry state, not an additional paravector degree of freedom.

At fixed time, the spatially integrated bracket defined in Eq. (209) is

$$\langle \phi | \psi \rangle = 2 \iiint \text{scalar}(\phi^H \psi) dr_1 dr_2 dr_3.$$

Conversion of intrinsic scalar extraction, which is one half of the matrix trace, into the ordinary inner product of the selected two-entry column in Eq. (242) is provided by the factor 2. It is therefore a normalization convention rather than an additional physical factor. For a projected one-block Pauli carrier, the normalization is its equal-field specialization,

$$\langle \psi | \psi \rangle = \iiint \rho(\psi) dr_1 dr_2 dr_3 = 1.$$

For a complex-paravector operator $\mathcal{O}$,

$$\langle \psi | \mathcal{O} \psi \rangle = 2 \iiint \text{scalar}(\psi^H \mathcal{O} \psi) dr_1 dr_2 dr_3.$$

For a state that is not already normalized, its normalized scalar matrix element is the ratio

$$\frac{\langle \psi | \mathcal{O} \psi \rangle}{\langle \psi | \psi \rangle}. \quad (245)$$

If $\mathcal{O} \psi = \phi_s + \phi_v \cdot \sigma$, then the extracted scalar channel is simply

$$\text{scalar}(\psi^H \mathcal{O} \psi) = \psi_s^* \phi_s + \psi_v^* \cdot \phi_v.$$

When $\mathcal{O}$ is Hermitian under the spatial integral and the boundary terms are benign, this map is a real expectation value. For a general $\mathcal{O}$, it is only a normalized matrix element and may be complex. For the Dirac field the physical norm is the sum of this same selected-column pairing over its two restricted blocks; the associated conserved spatial current is supplied by the field equation. No separately conserved norm is assigned to either massive Dirac block.

By contrast, for an unrestricted algebra-valued paravector field,

$$\text{adj}(\psi) \psi = \psi_s^2 - \psi_v \cdot \psi_v = \det(\psi)$$

is the determinant quadratic form. It is algebraically useful, but it is not positive-definite and so it should not be used as the probabilistic normalization. Moreover, after the fixed-projector restriction to one matrix column, this determinant vanishes identically because the rank of the resulting matrix is at most one. In this framework the single bra-ket bracket $\langle \cdot | \cdot \rangle$ is therefore reserved for the scalar expectation map built from H .

6.8.4 Carrier Interpretation and Pairings

In standard physics, spin is not assigned from component count alone. The physically relevant data are

(transformation law, field equation, constraints).

Thus the spin-zero Klein–Gordon field is carried by a complex scalar only after its scalar transformation law and field equation are imposed. A one-half-unit Dirac block is carried by a selected two-entry column only after the paired first-order equations are imposed. The spin-one electromagnetic field is carried by the pure-vector complex paravector F only with its field transformation law and Maxwell equation. A connection between the present construction and the standard classification is made by these statements in words, without introduction of another symbolic notation.

Bilinear pairings. For two complex-paravector fields ϕ and ψ , the three scalar pairings are

$$\text{scalar}(\phi^H \psi) = \phi_s^* \psi_s + \phi_v^* \cdot \psi_v,$$

$$\text{scalar}(\phi^* H \psi) = \phi_s \psi_s - \phi_v \cdot \psi_v,$$

$$\text{scalar}(\phi^* \psi) = \phi_s^* \psi_s - \phi_v^* \cdot \psi_v.$$

When the two fields agree, the first line is the positive Hermitian pairing; the second is the determinant extension; and the third is its indefinite companion. For $\phi = \psi$, the determinant extension reduces to

$$\text{scalar}(\psi^* H \psi) = \text{adj}(\psi) \psi = \det(\psi).$$

Only the Hermitian pairing is being used here as the quantum bra-ket pairing. The equal-argument determinant is the intrinsic quadratic form used in factorization identities, and the companion pairing is algebraically well-defined but does not carry an independent standard quantum interpretation here.

For the spatial bracket already defined in Eq. (209), an operator $\mathcal{O}$ is Hermitian on the chosen fields and boundary conditions when

$$\langle \phi | \mathcal{O} \psi \rangle = \langle \mathcal{O} \phi | \psi \rangle.$$

For left multiplication by a fixed paravector Z , moving the operator to the first field replaces it by Z^H , because $(Z^H \phi)^H = \phi^H Z$. For a pointwise multiplication operator, H is exactly the matrix Hermitian adjoint from Eq. (116). For a differential operator, integration by parts, its domain, and the boundary terms must also be included before it can be called Hermitian under the spatial bracket. Nonscalar parts of $\psi^H \mathcal{O} \psi$ remain available as algebraic bilinears, but a current, flux, or polarization interpretation must be established separately for the chosen operator and field equation. **Generalized operator products.** The following identities are algebraic. Operator composition on a paravector-valued state is described by them, but no physical spin is assigned by them alone.

For operator-valued components, reduction of the commutator to the pure cross-product law of Eq. (40) no longer occurs. For compactness, the scalar operator is denoted by S and the vector operator by $\mathbf{V}$:

$$\mathcal{O} = S + \mathbf{V} \cdot \boldsymbol{\sigma}, \quad \mathcal{O}' = S' + \mathbf{V}' \cdot \boldsymbol{\sigma},$$

with the state acted on by component operators to its left; those operators commute with the fixed basis elements. Then, with dot and cross products taken componentwise and in the written operator order,

$$\begin{aligned} \text{com}(\mathcal{O}, \mathcal{O}') \psi &= \left(\text{com}(S, S') + \mathbf{V} \cdot \mathbf{V}' - \mathbf{V}' \cdot \mathbf{V} \right) \psi \\ &+ \left(\text{com}(S, \mathbf{V}') + \text{com}(\mathbf{V}, S') \right. \\ &\quad \left. + i(\mathbf{V} \times \mathbf{V}' - \mathbf{V}' \times \mathbf{V}) \right) \cdot \boldsymbol{\sigma} \psi, \end{aligned} \quad (246)$$

where the scalar-vector commutators are understood componentwise. Unlike the purely algebraic paravector commutator, both scalar and vector parts are generally contained in Eq. (246). Only when all component operators mutually commute is the following simpler law obtained:

$$\text{com}(\mathcal{O}, \mathcal{O}') \psi = 2i(\mathbf{V} \times \mathbf{V}') \cdot \boldsymbol{\sigma} \psi.$$

The matching anticommutator is

$$\begin{aligned} \text{acom}(\mathcal{O}, \mathcal{O}') \psi &= (\mathcal{O} \mathcal{O}' + \mathcal{O}' \mathcal{O}) \psi \\ &= \left(\text{acom}(S, S') + \mathbf{V} \cdot \mathbf{V}' + \mathbf{V}' \cdot \mathbf{V} \right) \psi \\ &+ \left(\text{acom}(S, \mathbf{V}') \right. \\ &\quad \left. + \text{acom}(\mathbf{V}, S') \right. \\ &\quad \left. + i(\mathbf{V} \times \mathbf{V}' + \mathbf{V}' \times \mathbf{V}) \right) \cdot \boldsymbol{\sigma} \psi, \end{aligned} \quad (247)$$

so that

$$\mathcal{O} \mathcal{O}' = \frac{1}{2} (\text{acom}(\mathcal{O}, \mathcal{O}') + \text{com}(\mathcal{O}, \mathcal{O}')).$$

This is the natural companion to the operator commutator formula. A covariance or uncertainty identity can be formed from it only after a state space, expectation functional, Hermitian operators,

and common domains have been specified; no such physical statement is obtained from this algebraic identity alone.

Carrier closure. The carrier spaces that are closed under the respective operators are displayed by the same algebra:

$$\begin{aligned}\partial : \mathbb{C} &\rightarrow \mathbb{C} + \mathbb{C}^3, \\ \partial : \{0\} + \mathbb{C}^3 &\rightarrow \mathbb{C} + \mathbb{C}^3, \\ \partial : \mathbb{C} + \mathbb{C}^3 &\rightarrow \mathbb{C} + \mathbb{C}^3, \\ \partial : (\mathbb{C} + \mathbb{C}^3)\Pi(\text{column}) &\rightarrow (\mathbb{C} + \mathbb{C}^3)\Pi(\text{column}).\end{aligned}\quad (248)$$

Indeed,

$$\begin{aligned}\partial\psi_s &= (\partial_t\psi_s) + (\partial_r\psi_s) \cdot \sigma, \\ \partial(\psi_v \cdot \sigma) &= (\partial_r \cdot \psi_v) + (\partial_t\psi_v + i\partial_r \times \psi_v) \cdot \sigma.\end{aligned}$$

The full complex-paravector space is the smallest algebra containing the scalar and vector channels, but the projected fixed-column carrier is smaller. Two complex amplitudes are carried by the carrier $(\mathbb{C} + \mathbb{C}^3)\Pi(\text{column})$, selected by a fixed right factor, and closure is preserved because every differential and potential operator is applied on the left. Closure of the scalar subspace is obtained only after passage to scalar second-order combinations such as $\square + m^2$. A pure-vector specialization is algebraically meaningful, but it is not preserved by generic first-order paravector action; Maxwell equations or separately defined massive-vector constraints are therefore required if a spin-1 field is to be described by it. A real-paravector specialization is not a general quantum carrier because it is not closed under multiplication by the phase $\exp(i\theta)$.

With those algebraic preliminaries fixed, a carrier field is chosen first, followed by the equation and operators under which it is preserved, and then by the bilinears from which physical observables are defined. Bilinears need not take values in the original carrier; scalar, vector, or paravector observables may be obtained.

Scalar carrier. The Klein–Gordon field ψ is a complex scalar. Multiplication by $\Pi(\pm)$ from Eq. (65) would take it out of that carrier, so the axis projectors are not scalar-wave spin projectors. The positive quantity $|\psi|^2$ is not the time component of the relativistic conserved current. For $m > 0$, the mixed current is defined in Eq. (187). The following relations are obtained by direct differentiation:

$$\begin{aligned}2m\partial_t\rho(\phi, \psi) &= i((E^2\phi)^*\psi - \phi^*E^2\psi), \\ 2m\partial_r \cdot \mathbf{j}(\phi, \psi) &= i(\phi^*\mathbf{P}^2\psi - (\mathbf{P}^2\phi)^*\psi).\end{aligned}$$

Equation (189) is established by addition of the lines and application of the two Klein–Gordon equations. For a constant-potential mode

$$\psi = N \exp(-i\omega t + i\mathbf{k} \cdot \mathbf{r}),$$

where N is a constant complex amplitude, the component check is

$$\begin{aligned}E\psi &= (\omega - qV)\psi, & \mathbf{P}\psi &= (\mathbf{k} - q\mathbf{A})\psi, \\ (\omega - qV)^2 &= (\mathbf{k} - q\mathbf{A})^2 + m^2, \\ \rho(\psi, \psi) &= \frac{\omega - qV}{m}|N|^2, & \mathbf{j}(\psi, \psi) &= \frac{\mathbf{k} - q\mathbf{A}}{m}|N|^2.\end{aligned}$$

Every potential and current sign is thereby verified.

The free field is obtained by setting $q = 0$. The integrated pairing in Eq. (190) is positive after restriction to the positive-frequency sector, although its local density need not be pointwise positive for a superposition. That solution sector must be preserved by a scalar measurement operator, which must be Hermitian under this pairing. The split is preserved for the free field, or for a stationary background with a separately established positive-energy subspace; a global frequency split need not be preserved by a general time-dependent electromagnetic background. Accordingly, no local position-probability rule is claimed for the one-particle Klein–Gordon field.

Projected two-block carrier. For two projected blocks, the Hermitian pairing is Eq. (210). For the block operator

$$\mathcal{O} = \begin{pmatrix} \mathcal{O}_{11} & \mathcal{O}_{12} \\ \mathcal{O}_{21} & \mathcal{O}_{22} \end{pmatrix},$$

the operator pairing defined in Eq. (211) is expanded explicitly as

$$\begin{aligned}\langle \Psi(\text{left}) | \mathcal{O} \Psi(\text{right}) \rangle &= 2 \iiint \text{scalar} \left(\Psi_1(\text{left})^H \mathcal{O}_{11} \Psi_1(\text{right}) \right. \\ &\quad + \Psi_1(\text{left})^H \mathcal{O}_{12} \Psi_2(\text{right}) \\ &\quad + \Psi_2(\text{left})^H \mathcal{O}_{21} \Psi_1(\text{right}) \\ &\quad \left. + \Psi_2(\text{left})^H \mathcal{O}_{22} \Psi_2(\text{right}) \right) dr_1 dr_2 dr_3.\end{aligned}$$

Conversion of scalar extraction in the matrix realization to the ordinary two-entry column normalization is provided by the factor 2. Without the fixed right projector, two independent copies of every displayed Dirac and Pauli equation are obtained from the two matrix columns. The conserved density and current in Eq. (207) are supplied by the coupled first-order equation rather than by the vector part of a generic Hermitian product. Equation (212) is the corresponding complete two-block measurement rule.

Pure-vector field carrier. The electromagnetic field F is a pure-vector complex paravector. That pure-vector space is not preserved under left multiplication by the projectors $\Pi(\pm)$ from Eq. (65), so this two-projector split is not the electromagnetic polarization split. A free massless electromagnetic field has two independent transverse polarizations; a massive spin-one vector field would have three. The invariant field square and the observer-dependent energy and flux paravector are the bilinears already derived in the electromagnetic section. A generic Hermitian-product bracket is not substituted for those field observables.

Thus spin is never fixed by component count alone. The carrier, transformation law, field equation, constraints, and observable bilinears must agree. In the present notation, the positive wave-function pairing is supplied by H , the determinant used in factorization is supplied by the star- H composition, and the pure conjugate pairing remains an algebraic companion without an additional physical interpretation.

6.8.5 Spin-0 Wave Mechanics

The scalar relativistic wave operator is motivated by the classical mass-shell identity developed in the kinematic section. Replace-

ment of classical energy and momentum by differential operators is the standard quantization prescription and is an additional physical input, not a consequence of the determinant identity alone. Here $\psi \in \mathbb{C}$ is a complex scalar field, $m \in \mathbb{R}$, and the differential conventions from Section 4 are used for the free-particle operators. The mass-shell identity is

$$\det(mU) = m^2.$$

Its scalar form is

$$-E^2 + \mathbf{P}^2 + m^2 = 0.$$

For a free particle,

$$E = i\partial_t, \quad \mathbf{P} = -i\partial_{\mathbf{r}},$$

and equivalently

$$E + \mathbf{P} \cdot \boldsymbol{\sigma} = i\partial^*.$$

Acting on a scalar field ψ , the operator chain is

$$\begin{aligned} (-E^2 + \mathbf{P}^2 + m^2)\psi &= (-(i\partial_t)^2 - \partial_{\mathbf{r}}^2 + m^2)\psi \\ &= (\partial_t^2 - \partial_{\mathbf{r}}^2 + m^2)\psi \\ &= (\partial\partial^* + m^2)\psi \\ &= (\square + m^2)\psi. \end{aligned}$$

Thus the free spin-0 relativistic wave equation is

$$(\square + m^2)\psi = 0 \quad \text{Klein-Gordon equation.}$$

This is the scalar wave equation recorded in Eq. (172). The dispersion relation $\omega^2 = \mathbf{k}^2 + m^2$ and the later positive/negative-frequency split are consequences of this same operator identity.

6.8.6 Positive and Negative Frequency Sectors

The frequency decomposition is one of the central interpretive steps in the free scalar theory. On a spatial domain where the positive square root is defined, the frequency function from Eq. (173) is given by

$$\begin{aligned} \Omega &= \sqrt{m^2 - \partial_{\mathbf{r}}^2} \\ &= \sqrt{\det(m + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}})}. \end{aligned}$$

Accordingly,

$$\begin{aligned} \square + m^2 &= \partial_t^2 + \Omega^2 \\ &= \partial_t^2 + (m^2 - \partial_{\mathbf{r}}^2) \\ &= (\partial_t - i\Omega)(\partial_t + i\Omega) \\ &= (\partial_t + i\Omega)(\partial_t - i\Omega). \end{aligned}$$

The sector definitions in Eq. (175) are then restated as

$$\begin{aligned} (\partial_t \pm i\Omega)\psi(\pm) &= 0, \\ \psi(\pm, t) &= \exp(\mp i\Omega t)\psi(\pm, 0), \\ \psi &= \psi(+) + \psi(-). \end{aligned}$$

Here the parenthesized sign labels frequency and is not the spin-axis label in the spin-axis action $\Pi(\pm)\phi(\text{large})$ defined by Eq. (65). When Ω^{-1} exists, the two parts of a Klein-Gordon solution at any time can equivalently be extracted as

$$\psi(\pm) = \frac{1}{2} (\psi \pm i\Omega^{-1}\partial_t\psi).$$

For a massless field the spatial zero mode is treated separately. The Klein-Gordon pairing has opposite sign on a pure negative-frequency mode and is not a positive norm on the combined solution space. In a quantized relativistic field the particle-antiparticle interpretation is supplied by the field-operator construction, not by reinterpreting this one-particle density alone [16].

6.8.7 Fourier Meaning of the Frequency Operator

The nonlocal square root Ω is most transparent one spatial Fourier mode at a time. With the standard phase convention, the field and modes are written schematically as

$$\begin{aligned} \psi &= \sum_n c_n f_n, \\ f_n &= \exp(-i\omega_n t + i\mathbf{k}_n \cdot \mathbf{r}). \end{aligned}$$

Then

$$\begin{aligned} \square\psi &= \sum_n (\mathbf{k}_n^2 - \omega_n^2) c_n f_n, \\ (\square + m^2)\psi &= \sum_n (\mathbf{k}_n^2 + m^2 - \omega_n^2) c_n f_n, \\ \Omega\psi &= \sum_n \sqrt{m^2 + \mathbf{k}_n^2} c_n f_n. \end{aligned}$$

For modes satisfying the Klein-Gordon dispersion relation,

$$\mathbf{k}_n^2 + m^2 = \omega_n^2, \quad \omega_n = \pm \sqrt{\mathbf{k}_n^2 + m^2},$$

and each spatial mode is multiplied by its positive energy magnitude by Ω . The only square-root operator needed in the frequency split is thereby defined, and an unnecessary branch is not assigned to $\sqrt{\square}$.

6.8.8 Klein-Gordon with Potential Fields

The charged-particle Lagrangian is taken as the starting point so that the minimally coupled scalar theory is kept in one self-contained place:

$$\mathcal{L} = -\frac{m}{\gamma} - q(V - \mathbf{v} \cdot \mathbf{A}).$$

Here E denotes the energy operator, not the electric field $\mathbf{E}$. The canonical energy and momentum are identified by the classical Lagrangian. After the standard differential replacement, the covariant operators are

$$E = i\partial_t - qV, \quad \mathbf{P} = -i\partial_{\mathbf{r}} - q\mathbf{A}.$$

This differential replacement, together with minimal coupling, is the standard quantization input.

Acting on a smooth test field ψ , the Leibniz rules needed below are

$$\begin{aligned}\partial_t(V\psi) &= (\partial_t V)\psi + V\partial_t\psi, \\ \partial_m(A_n\psi) &= (\partial_m A_n)\psi + A_n\partial_m\psi.\end{aligned}$$

The following expansions are obtained:

$$\begin{aligned}E^2\psi &= E(i\partial_t\psi - qV\psi) \\ &= (-\partial_t^2 - iq\partial_t V - 2iqV\partial_t + q^2V^2)\psi, \\ \mathbf{P}^2\psi &= \mathbf{P} \cdot (-i\partial_{\mathbf{r}}\psi - q\mathbf{A}\psi) \\ &= (-\partial_{\mathbf{r}}^2 + iq\partial_{\mathbf{r}} \cdot \mathbf{A} + 2iq\mathbf{A} \cdot \partial_{\mathbf{r}} + q^2\mathbf{A}^2)\psi.\end{aligned}$$

The required identities are

$$\begin{aligned}\square &= \partial_t^2 - \partial_{\mathbf{r}}^2, \\ \Phi &= V + \mathbf{A} \cdot \boldsymbol{\sigma}, \\ S &= \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}, \\ \det(\partial) &= \square, \\ \det(\Phi) &= V^2 - \mathbf{A}^2.\end{aligned}$$

Here $\det(\partial) = \partial\partial^*$ is valid because the coordinate derivatives commute. The coupled energy and momentum operators below do not commute and therefore remain explicit ordered products rather than determinants. The minimally coupled Klein–Gordon equation is obtained in explicit form by substitution of the operator expansions:

$$\begin{aligned}0 &= (-E^2 + \mathbf{P}^2 + m^2)\psi \\ &= \left(\square + iq(\partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}) + 2iq(V\partial_t + \mathbf{A} \cdot \partial_{\mathbf{r}}) \right. \\ &\quad \left. - q^2(V^2 - \mathbf{A}^2) + m^2\right)\psi \\ &= \left(\square + iqS - q^2\det(\Phi) + m^2 + 2iq(V\partial_t + \mathbf{A} \cdot \partial_{\mathbf{r}})\right)\psi.\end{aligned}$$

For the gauge-shifted potential and components defined in Eqs. (158), (159), and (160),

$$\begin{aligned}V(\lambda) &= V + \partial_t\lambda, \\ \mathbf{A}(\lambda) &= \mathbf{A} - \partial_{\mathbf{r}}\lambda, \\ \psi(\lambda) &= \exp(-iq\lambda)\psi.\end{aligned}$$

The ordinary product rule is obeyed by the central scalar phase. Consequently,

$$\begin{aligned}(i\partial_t - qV(\lambda))\psi(\lambda) &= \exp(-iq\lambda)(i\partial_t - qV)\psi, \\ (-i\partial_{\mathbf{r}} - q\mathbf{A}(\lambda))\psi(\lambda) &= \exp(-iq\lambda)(-i\partial_{\mathbf{r}} - q\mathbf{A})\psi.\end{aligned}$$

Equation (183) and the gauge covariance of the coupled Klein–Gordon equation are established by a second application of either relation. Substitution in Eq. (187) also cancels the phase from both current components.

The electromagnetic fields and commutators are incorporated through

$$\begin{aligned}\mathbf{E} &= -\partial_t\mathbf{A} - \partial_{\mathbf{r}}V, \\ \mathbf{B} &= \partial_{\mathbf{r}} \times \mathbf{A}, \\ F &= (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \quad F^* = (-\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma},\end{aligned}$$

and

$$\begin{aligned}\text{com}(E, P_m)\psi &= (i\partial_t - qV)(-i\partial_m\psi - qA_m\psi) \\ &\quad - (-i\partial_m - qA_m)(i\partial_t\psi - qV\psi) \\ &= -iq((\partial_t A_m)\psi + A_m\partial_t\psi) \\ &\quad - iq((\partial_m V)\psi + V\partial_m\psi) \\ &\quad + iqV\partial_m\psi + iqA_m\partial_t\psi \\ &= -iq(\partial_t A_m + \partial_m V)\psi \\ &= iqE_m\psi, \\ \text{com}(P_m, P_n)\psi &= (-i\partial_m - qA_m)(-i\partial_n\psi - qA_n\psi) \\ &\quad - (-i\partial_n - qA_n)(-i\partial_m\psi - qA_m\psi) \\ &= iq(\partial_m A_n - \partial_n A_m)\psi.\end{aligned}$$

Hence

$$\text{com}(E, \mathbf{P}) = iq\mathbf{E}, \quad \mathbf{P} \times \mathbf{P} = iq\mathbf{B}.$$

The following ordered products are obtained by substitution of those commutators:

$$\begin{aligned}(E + \boldsymbol{\sigma} \cdot \mathbf{P})(E - \boldsymbol{\sigma} \cdot \mathbf{P}) &= E^2 - \mathbf{P}^2 \\ &\quad + \boldsymbol{\sigma} \cdot (-\text{com}(E, \mathbf{P}) - i\mathbf{P} \times \mathbf{P}) \\ &= E^2 - \mathbf{P}^2 - iqF, \\ (E - \boldsymbol{\sigma} \cdot \mathbf{P})(E + \boldsymbol{\sigma} \cdot \mathbf{P}) &= E^2 - \mathbf{P}^2 \\ &\quad + \boldsymbol{\sigma} \cdot (\text{com}(E, \mathbf{P}) - i\mathbf{P} \times \mathbf{P}) \\ &= E^2 - \mathbf{P}^2 - iqF^*.\end{aligned}$$

Since

$$E + \boldsymbol{\sigma} \cdot \mathbf{P} = i\partial^* - q\Phi, \quad E - \boldsymbol{\sigma} \cdot \mathbf{P} = i\partial - q\Phi^* = (E + \boldsymbol{\sigma} \cdot \mathbf{P})^{*\text{H}},$$

the two ordered star- H products are

$$(E \pm \boldsymbol{\sigma} \cdot \mathbf{P})(E \pm \boldsymbol{\sigma} \cdot \mathbf{P})^{*\text{H}} = E^2 - \mathbf{P}^2 + \boldsymbol{\sigma} \cdot (\mp \text{com}(E, \mathbf{P}) - i\mathbf{P} \times \mathbf{P}).$$

Unlike the determinant function defined for commuting complex coefficients, these operator products retain a vector part. Operator ordering therefore matters, and the two first-order products remain distinct. Consequently,

$$\begin{aligned}-(E + \boldsymbol{\sigma} \cdot \mathbf{P})(E + \boldsymbol{\sigma} \cdot \mathbf{P})^{*\text{H}} &= (\partial^* + iq\Phi)(\partial + iq\Phi^*), \\ -(E + \boldsymbol{\sigma} \cdot \mathbf{P})^{*\text{H}}(E + \boldsymbol{\sigma} \cdot \mathbf{P}) &= (\partial + iq\Phi^*)(\partial^* + iq\Phi).\end{aligned}$$

The same factorizations can be rewritten directly in derivative form:

$$\begin{aligned}(E + \boldsymbol{\sigma} \cdot \mathbf{P})(E - \boldsymbol{\sigma} \cdot \mathbf{P})\psi &= -(\partial^* + iq\Phi)(\partial + iq\Phi^*)\psi \\ &= -\square\psi - iq\partial^*(\Phi^*\psi) \\ &\quad - iq\Phi\partial\psi \\ &\quad + q^2\det(\Phi)\psi \\ &= -\square\psi - iqS\psi + q^2\det(\Phi)\psi \\ &\quad - 2iq(V\partial_t + \mathbf{A} \cdot \partial_{\mathbf{r}})\psi \\ &\quad - iqF\psi,\end{aligned}$$

$$\begin{aligned}
(E - \boldsymbol{\sigma} \cdot \mathbf{P})(E + \boldsymbol{\sigma} \cdot \mathbf{P})\psi &= -(\partial + iq\Phi^*)(\partial^* + iq\Phi)\psi \\
&= -\square\psi - iq\partial(\Phi\psi) \\
&\quad - iq\Phi^*\partial^*\psi \\
&\quad + q^2\det(\Phi)\psi \\
&= -\square\psi - iqS\psi + q^2\det(\Phi)\psi \\
&\quad - 2iq(V\partial_t + \mathbf{A} \cdot \partial_{\mathbf{r}})\psi \\
&\quad - iqF^*\psi.
\end{aligned}$$

Equivalently,

$$\begin{aligned}
\partial^*(\Phi^*\psi) + \Phi\partial\psi &= S\psi + 2(V\partial_t + \mathbf{A} \cdot \partial_{\mathbf{r}})\psi + F\psi, \\
\partial(\Phi\psi) + \Phi^*\partial^*\psi &= S\psi + 2(V\partial_t + \mathbf{A} \cdot \partial_{\mathbf{r}})\psi + F^*\psi,
\end{aligned}$$

and their difference isolates the electric contribution:

$$\begin{aligned}
(\partial^*(\Phi^*\psi) + \Phi\partial\psi) - (\partial(\Phi\psi) + \Phi^*\partial^*\psi) \\
&= (\partial\Phi^* - \partial\Phi)\psi \\
&= (F - F^*)\psi \\
&= 2(\mathbf{E} \cdot \boldsymbol{\sigma})\psi.
\end{aligned}$$

From the definitions in Eqs. (176) and (177), (140), (141), and (146), the scalar Klein–Gordon operator can therefore be recovered from either ordered first-order product:

$$\begin{aligned}
0 &= (-E^2 + \mathbf{P}^2 + m^2)\psi \\
&= \left(\square + iqS - q^2\det(\Phi) + m^2 + 2iq(V\partial_t + \mathbf{A} \cdot \partial_{\mathbf{r}})\right)\psi \\
&= (\partial^* + iq\Phi)(\partial + iq\Phi^*)\psi - iqF\psi + m^2\psi \\
&= (\partial + iq\Phi^*)(\partial^* + iq\Phi)\psi - iqF^*\psi + m^2\psi.
\end{aligned}$$

The following identities are satisfied by the ordered first-order products:

$$\begin{aligned}
(E \pm \boldsymbol{\sigma} \cdot \mathbf{P})(E \mp \boldsymbol{\sigma} \cdot \mathbf{P}) &= E^2 - (\boldsymbol{\sigma} \cdot \mathbf{P})^2 \mp \text{com}(E, \boldsymbol{\sigma} \cdot \mathbf{P}) \\
&= E^2 - \mathbf{P}^2 - iq(\pm\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}.
\end{aligned}$$

The following equation is therefore recovered by taking the scalar part:

$$0 = \text{scalar}((\partial + iq\Phi^*)(\partial^* + iq\Phi))\psi + m^2\psi.$$

The Klein–Gordon operator itself is reproduced only by the scalar part; the full ordered products retain the non-scalar field terms $-iqF$ and $-iqF^*$ that distinguish the two factorizations. Explicitly,

$$-iqF = (-iq\mathbf{E} + q\mathbf{B}) \cdot \boldsymbol{\sigma}, \quad -iqF^* = (iq\mathbf{E} + q\mathbf{B}) \cdot \boldsymbol{\sigma}.$$

These residuals are not new components of the scalar equation. They are the standard spin–field terms retained by the two ordered Dirac squares below.

6.8.9 Spin- $\frac{1}{2}$ Wave Mechanics

Before the electromagnetic field is turned on, a self-contained first-order version of the relativistic mass shell is obtained from

the same auxiliary 2×2 map in Eq. (120). The two-block field is considered in the form

$$\Psi = \begin{Bmatrix} \Psi_1 \\ \Psi_2 \end{Bmatrix}.$$

For one Dirac field, the following restriction is satisfied by both blocks: $\Psi_n = \Psi_n \Pi(\text{column})$. If they are left unrestricted, every formula is applied independently to the two matrix columns, and two identical Dirac fields are therefore described. From Eqs. (120) and (121), complex linearity of the auxiliary block map and block diagonality of its square are obtained. Therefore

$$\begin{aligned}
\mathbf{W}(i\partial) &= i \begin{bmatrix} 0 & \partial \\ \partial^* & 0 \end{bmatrix} = i\mathbf{W}(\partial), \\
\mathbf{W}(i\partial)^2 &= -\text{diag}(\partial\partial^*, \partial^*\partial) \\
&= -\square\mathbf{I}.
\end{aligned}$$

The free operator in Eq. (194) is $\mathbf{W}(i\partial) - m\mathbf{I}$. Its factorization partner is obtained by reversing only the sign of the mass. Their product in either order is

$$\begin{aligned}
(\mathbf{W}(i\partial) + m\mathbf{I})(\mathbf{W}(i\partial) - m\mathbf{I}) &= \mathbf{W}(i\partial)^2 - m^2\mathbf{I} \\
&= -(\square + m^2)\mathbf{I}, \\
(\mathbf{W}(i\partial) - m\mathbf{I})(\mathbf{W}(i\partial) + m\mathbf{I}) &= -(\square + m^2)\mathbf{I}.
\end{aligned}$$

Thus the free Dirac equation

$$(\mathbf{W}(i\partial) - m\mathbf{I})\Psi = 0$$

requires the Klein–Gordon operator to be satisfied by each block. No determinant of algebra-valued blocks is needed: the relevant minimal polynomial is supplied directly by Eq. (121).

The equivalence with the standard two-coupled-column form is explicit in the Pauli matrix realization. For a plane wave

$$\exp(-iEt + i\mathbf{p} \cdot \mathbf{r})$$

the selected columns are represented by $\{a; b\}$ for Ψ_1 and $\{c; d\}$ for Ψ_2 . The two equations are

$$(E - \mathbf{p} \cdot \boldsymbol{\sigma})\{c; d\} = m\{a; b\}, \quad (E + \mathbf{p} \cdot \boldsymbol{\sigma})\{a; b\} = m\{c; d\}.$$

The relation $E^2 = \mathbf{p}^2 + m^2$ is obtained from their product. These are the standard two first-order equations for the four complex entries of one Dirac field. Two independent copies of this system are obtained by retaining both matrix columns. With a potential present, the operator equations in Eq. (197) are represented by the same four entries; hence component-by-component agreement with the compact paravector equation is obtained before squaring.

The two coupled equations are obtained by writing out the free equation:

$$\begin{aligned}
i\partial\Psi_2 - m\Psi_1 &= 0, \\
i\partial^*\Psi_1 - m\Psi_2 &= 0.
\end{aligned}$$

Their transformation law is determined by covariance rather than component count. For a constant determinant-one transform T ,

with X' given by Eq. (84), the transformed blocks from Eqs. (198) and (199) obey

$$\begin{aligned}\Psi'_1(X') &= T^H \Psi_1(X), \\ \Psi'_2(X') &= T^{-1} \Psi_2(X).\end{aligned}$$

The corresponding derivative and potential laws are

$$\partial' = T^H \partial T. \quad (249)$$

$$\partial'^* = T^{-1} \partial^* T^*. \quad (250)$$

$$\Phi' = T^{-1} \Phi T^*. \quad (251)$$

$$\Phi'^* = T^H \Phi^* T. \quad (252)$$

For a spatial rotor, the same left factor R^H is assigned to both blocks. For a boost, L is assigned to the first block and L^* to the second. Since $\det(T) = 1$, the identities $TT^{-1} = 1$ and $T^*T^H = 1$ imply the following explicit substitutions:

$$\begin{aligned}(i\partial' - q\Phi'^*)\Psi'_2 &= T^H(i\partial - q\Phi^*)\Psi_2 = m\Psi'_1, \\ (i\partial'^* - q\Phi')\Psi'_1 &= T^{-1}(i\partial^* - q\Phi)\Psi_1 = m\Psi'_2.\end{aligned}$$

Thus both coupled equations are covariant. Because all transform factors are applied from the left, the fixed right projector is preserved.

The real current paravector defined in Eq. (206) is expanded as

$$J(\Psi) = 2 \left(\Psi_2 \Psi_2^H + (\Psi_1 \Psi_1^H)^* \right) = \rho(\Psi) + \mathbf{j}(\Psi) \cdot \sigma.$$

The following transformation is obtained from Eqs. (198) and (199) in each product:

$$J(\Psi')(X') = T^{-1} J(\Psi)(X) (T^{-1})^H,$$

so the current is transformed under the same congruence law as a real event paravector. For the selected columns used in Eq. (197), the component form is

$$\begin{aligned}\rho(\Psi) &= |a|^2 + |b|^2 + |c|^2 + |d|^2, \\ \mathbf{j}(\Psi) &= \begin{Bmatrix} 2 \operatorname{real}(c^*d - a^*b) \\ 2 \operatorname{img}(c^*d - a^*b) \\ |c|^2 - |d|^2 - |a|^2 + |b|^2 \end{Bmatrix}.\end{aligned}$$

These are the conventional Dirac density and current components. The two uncanceled block balances are Eq. (208); by their addition, $\partial_t \rho(\Psi) + \partial_r \cdot \mathbf{j}(\Psi) = 0$ is established. In particular,

$$\mathbf{u} \cdot \mathbf{j}(\Psi) = 2 \operatorname{scalar}(\Psi_2^H (\mathbf{u} \cdot \sigma) \Psi_2 - \Psi_1^H (\mathbf{u} \cdot \sigma) \Psi_1),$$

so this is the local matrix-element density for the ordinary two-block velocity operator. Its spatial integral divided by $\langle \Psi | \Psi \rangle$ is the corresponding velocity expectation. The constant-time density is a frame component of this covariant current, not a Lorentz scalar.

For $m > 0$, the following result is obtained by solving the first equation for the upper component and substituting into the second:

$$\begin{aligned}\Psi_1 &= \frac{i\partial \Psi_2}{m}, \\ 0 &= i\partial^* \Psi_1 - m\Psi_2 \\ &= -\frac{\partial^* \partial \Psi_2}{m} - m\Psi_2 \\ &= -\frac{(\square + m^2)\Psi_2}{m}.\end{aligned}$$

Hence the lower component is a Klein–Gordon solution, and the upper component is obtained from it by one first-order derivative.

Conversely, a scalar Klein–Gordon solution ψ and constant polarization elements $a, b \in (\mathbb{C} + \mathbb{C}^3)\Pi(\text{column})$ are taken. The mass-conjugate operator is applied to the projected seed:

$$\Psi := \frac{1}{m} (\mathbf{W}(i\partial) + m\mathbf{I}) \begin{Bmatrix} \psi a \\ \psi b \end{Bmatrix}. \quad (253)$$

Because the constants occur on the right, only ψ is acted on by the derivatives, and the two blocks are

$$\begin{aligned}\Psi_1 &= \psi a + \frac{i}{m} (\partial \psi) b, \\ \Psi_2 &= \frac{i}{m} (\partial^* \psi) a + \psi b.\end{aligned}$$

The following result is obtained from the factorization:

$$(\mathbf{W}(i\partial) - m\mathbf{I}) \Psi = -\frac{1}{m} (\square + m^2) \psi \begin{Bmatrix} a \\ b \end{Bmatrix} = 0.$$

The spin polarization is carried by the constants a and b ; it is not carried by a bare scalar solution alone. For example, the following field is obtained with $a = \Pi(\text{column})$ and $b = 0$:

$$\Psi = \begin{Bmatrix} \psi \Pi(\text{column}) \\ \frac{i}{m} (\partial^* \psi) \Pi(\text{column}) \end{Bmatrix}.$$

This is the first-order lift from the Klein–Gordon mass shell into the chosen projected Dirac carrier. The two-seed presentation is symmetric but redundant: on the Klein–Gordon mass shell, $\mathbf{W}(i\partial) + m\mathbf{I}$ has a kernel, so the same Dirac field can be produced by different seed pairs. For any fixed nonzero massive momentum mode, the two spin polarizations are already spanned by one projected seed block. Four new polarization degrees of freedom are therefore not introduced by the symbols a and b .

6.8.10 Dirac with Potential Field

The two-block field is again considered:

$$\Psi = \begin{Bmatrix} \Psi_1 \\ \Psi_2 \end{Bmatrix}.$$

The one-field interpretation continues to impose

$$\Psi_n = \Psi_n \Pi(\text{column}).$$

All potential and derivative factors are applied from the left, and the fixed-column carrier is therefore preserved. The auxiliary block function already defined in Eq. (120) is used. The Dirac operator with potential is

$$\mathbf{D} := \mathbf{W}(i\partial - q\Phi^*) - m\mathbf{I}. \quad (254)$$

By linearity of the block function, this definition is equivalently

$$\mathbf{D} = i\mathbf{W}(\partial) - q\mathbf{W}(\Phi^*) - m\mathbf{I}.$$

Its field equation is Eq. (195). Because

$$\begin{aligned} \mathbf{W}(\partial)^2 &= \square\mathbf{I}, \\ \mathbf{W}(\partial)\mathbf{W}(\Phi^*) &= \begin{bmatrix} 0 & \partial \\ \partial^* & 0 \end{bmatrix} \begin{bmatrix} 0 & \Phi^* \\ \Phi & 0 \end{bmatrix} = \text{diag}(\partial\Phi, \partial^*\Phi^*), \\ \mathbf{W}(\Phi^*)\mathbf{W}(\partial) &= \begin{bmatrix} 0 & \Phi^* \\ \Phi & 0 \end{bmatrix} \begin{bmatrix} 0 & \partial \\ \partial^* & 0 \end{bmatrix} = \text{diag}(\Phi^*\partial^*, \Phi\partial), \\ \mathbf{W}(\Phi^*)^2 &= \begin{bmatrix} 0 & \Phi^* \\ \Phi & 0 \end{bmatrix}^2 = \text{diag}(\Phi^*\Phi, \Phi\Phi^*) \\ &= \det(\Phi)\mathbf{I}, \end{aligned}$$

the following result is obtained:

$$\begin{aligned} \mathbf{W}(i\partial - q\Phi^*)^2 &= -\mathbf{W}(\partial)^2 \\ &\quad - iq(\mathbf{W}(\partial)\mathbf{W}(\Phi^*) + \mathbf{W}(\Phi^*)\mathbf{W}(\partial)) \\ &\quad + q^2\mathbf{W}(\Phi^*)^2 \\ &= -\square\mathbf{I} - iq \text{diag}(\partial\Phi + \Phi^*\partial^*, \partial^*\Phi^* + \Phi\partial) \\ &\quad + q^2 \det(\Phi)\mathbf{I} \\ &= -\text{diag}\left((\partial + iq\Phi^*)(\partial^* + iq\Phi), \right. \\ &\quad \left. (\partial^* + iq\Phi)(\partial + iq\Phi^*)\right). \end{aligned}$$

The Dirac equation

$$\mathbf{D}\Psi = 0$$

is therefore reduced to the two coupled equations

$$\begin{aligned} -m\Psi_1 + (i\partial - q\Phi^*)\Psi_2 &= 0, \\ (i\partial^* - q\Phi)\Psi_1 - m\Psi_2 &= 0. \end{aligned}$$

Equivalently, their time evolution is

$$\begin{aligned} i\partial_t\Psi_1 &= (qV - \boldsymbol{\sigma} \cdot \mathbf{P})\Psi_1 + m\Psi_2, \\ i\partial_t\Psi_2 &= (qV + \boldsymbol{\sigma} \cdot \mathbf{P})\Psi_2 + m\Psi_1. \end{aligned}$$

The H operation is applied to these equations, multiplication is performed on the opposite side, and scalar parts are taken. In selected-column form the two separate balance equations are written out in Eq. (208). The real potential terms cancel within each block, while the mass terms are equal and opposite between the blocks. The following relation is obtained by addition of those two equations:

$$\partial_t \text{scalar}(\Psi_1^H\Psi_1 + \Psi_2^H\Psi_2) + \partial_{\mathbf{r}} \cdot \text{vector}(\Psi_2\Psi_2^H - \Psi_1\Psi_1^H) = 0.$$

The stated density and current are thereby verified, and the distinction between the vector part of a single $\psi^H\psi$ and the Dirac current is established. The following result is obtained by solving the first equation for the upper component and substituting into the second:

$$\begin{aligned} \Psi_1 &= \frac{(i\partial - q\Phi^*)\Psi_2}{m}, \\ 0 &= (i\partial^* - q\Phi)\Psi_1 - m\Psi_2 \\ &= -\frac{((\partial^* + iq\Phi)(\partial + iq\Phi^*) + m^2)\Psi_2}{m}. \end{aligned}$$

Thus a Klein–Gordon-type operator with the appropriate ordered field coupling is satisfied by each block.

The mass-conjugate operator defined in Eq. (200) is

$$\mathbf{D}' = \mathbf{W}(i\partial - q\Phi^*) + m\mathbf{I}.$$

The following result is obtained by multiplication by $\mathbf{D}'$:

$$\begin{aligned} 0 &= \mathbf{D}'\mathbf{D}\Psi = (\mathbf{W}(i\partial - q\Phi^*)^2 - m^2\mathbf{I})\Psi \\ &= -\text{diag}\left((\partial + iq\Phi^*)(\partial^* + iq\Phi) + m^2, \right. \\ &\quad \left. (\partial^* + iq\Phi)(\partial + iq\Phi^*) + m^2\right)\Psi. \end{aligned}$$

Equivalently,

$$\begin{aligned} 0 &= ((\partial + iq\Phi^*)(\partial^* + iq\Phi) + m^2)\Psi_1 \\ &= (-E^2 + \mathbf{P}^2 + m^2 + iqF^*)\Psi_1, \\ 0 &= ((\partial^* + iq\Phi)(\partial + iq\Phi^*) + m^2)\Psi_2 \\ &= (-E^2 + \mathbf{P}^2 + m^2 + iqF)\Psi_2. \end{aligned}$$

Since

$$iqF^* = (-iq\mathbf{E} - q\mathbf{B}) \cdot \boldsymbol{\sigma}, \quad iqF = (iq\mathbf{E} - q\mathbf{B}) \cdot \boldsymbol{\sigma},$$

the two equations are exactly the component forms in Eq. (202). Once the square is taken, the coupled Dirac system is separated into two Klein–Gordon-type equations with the same magnetic term and opposite electric terms, as required by the two first-order blocks.

6.8.11 Gauge Transform of the Dirac System

For a real scalar gauge field $\lambda \in \mathbb{R}$, the potential defined in Eq. (158) is

$$\Phi(\lambda) = \Phi + \partial^*\lambda.$$

Because λ is real,

$$\partial\lambda = \Phi(\lambda) - \Phi^*.$$

The field transformation defined in Eq. (203) is equivalently written as

$$\Psi = \exp(iq\lambda)\Psi(\lambda).$$

The phase factor is a central scalar, so every commutator correction in Eq. (144) is zero. The ordinary two-term product rule is therefore obeyed by the derivative of the phase factor:

$$\partial(\exp(iq\lambda)\Psi(\lambda)) = \exp(iq\lambda)(iq\partial\lambda + \partial)\Psi(\lambda).$$

Therefore,

$$\begin{aligned}
0 &= (\mathbf{W}(i\partial - q\Phi^*) - m\mathbf{I})\Psi \\
&= (\mathbf{W}(i\partial - q\Phi^*) - m\mathbf{I}) \exp(iq\lambda)\Psi(\lambda) \\
&= \exp(iq\lambda)(\mathbf{W}(i\partial - q\partial\lambda - q\Phi^*) - m\mathbf{I})\Psi(\lambda) \\
&= \exp(iq\lambda)(\mathbf{W}(i\partial - q\Phi(\lambda) - \Phi^* - q\Phi^*) - m\mathbf{I})\Psi(\lambda) \\
&= \exp(iq\lambda)(\mathbf{W}(i\partial - q\Phi(\lambda)^*) - m\mathbf{I})\Psi(\lambda).
\end{aligned}$$

Hence

$$(\mathbf{W}(i\partial - q\Phi(\lambda)^*) - m\mathbf{I})\Psi(\lambda) = 0.$$

This is the gauge-covariance statement: the phase change of the Dirac field is exactly compensated by the shift of the potential, so the same form of the coupled Dirac equation is retained in the transformed gauge.

6.8.12 Dirac Equation Components

The canonical definitions of ∂ , Φ , and F from Eqs. (140), (141), and (146), together with the generic block $\psi = \psi_s + \psi_v \cdot \sigma$ defined above, are used. The scalar and vector parts of $\partial\psi$, $\partial^*\psi$, $\Phi\psi$, $\Phi^*\psi$, $iqF\psi$, and $iqF^*\psi$ are given explicitly in the supporting tables. They are identities in the full algebra; the fixed-column carrier is selected by imposing $\psi = \psi\Pi(\text{column})$, without alteration of those identities. The actual constrained-column equations are Eq. (197); only the action of each generic algebra factor before application of that constraint is exposed by the tables. The direct component action of ∂ and ∂^* on a complex paravector test field is recorded in Tables 42 and 43.

Table 42. Components of $\partial\psi$.

scalar:	$\partial_t\psi_s + \partial_r \cdot \psi_v$
vector:	$\partial_t\psi_v + \partial_r\psi_s + i(\partial_r \times \psi_v)$

Table 43. Components of $\partial^*\psi$.

scalar:	$\partial_t\psi_s - \partial_r \cdot \psi_v$
vector:	$\partial_t\psi_v - \partial_r\psi_s - i(\partial_r \times \psi_v)$

The component actions of the potential Φ and its conjugate on the same field are then shown in Tables 44 and 45.

Table 44. Components of $\Phi\psi$.

scalar:	$V\psi_s + \mathbf{A} \cdot \psi_v$
vector:	$V\psi_v + \psi_s\mathbf{A} + i(\mathbf{A} \times \psi_v)$

Table 45. Components of $\Phi^*\psi$.

scalar:	$V\psi_s - \mathbf{A} \cdot \psi_v$
vector:	$V\psi_v - \psi_s\mathbf{A} - i(\mathbf{A} \times \psi_v)$

The Dirac-block operator tables are completed by Tables 46 and 47, in which the field-coupling terms $iqF\psi$ and $iqF^*\psi$ are recorded. In the second table, $F^* = -\mathbf{F}^* \cdot \sigma = -(\mathbf{E} - i\mathbf{B}) \cdot \sigma$, so no entrywise matrix conjugation is hidden in the notation.

Table 46. Components of $iqF\psi$.

scalar:	$iq(\mathbf{F} \cdot \psi_v)$
vector:	$iq\psi_s\mathbf{F} - q(\mathbf{F} \times \psi_v)$

Table 47. Components of $iqF^*\psi$.

scalar:	$-iq(\mathbf{F}^* \cdot \psi_v)$
vector:	$-iq\psi_s\mathbf{F}^* + q(\mathbf{F}^* \times \psi_v)$

6.8.13 Low-Energy Spin-0 Limit

For the minimally coupled scalar theory, the covariant operators are

$$E = i\partial_t - qV, \quad \mathbf{P} = -i\partial_r - q\mathbf{A}.$$

The fast rest-mass oscillation is removed by writing

$$\psi = \exp(-imt)\psi(\text{slow}).$$

Then

$$\begin{aligned}
E^2\psi &= \exp(-imt)(E + m)^2\psi(\text{slow}), \\
\mathbf{P}^2\psi &= \exp(-imt)\mathbf{P}^2\psi(\text{slow}).
\end{aligned}$$

The following result is obtained by substitution into the minimally coupled Klein–Gordon equation:

$$\begin{aligned}
0 &= (-E^2 + \mathbf{P}^2 + m^2)\psi \\
&= \exp(-imt)(-(E + m)^2 + \mathbf{P}^2 + m^2)\psi(\text{slow}),
\end{aligned}$$

hence

$$0 = (-E^2 - 2mE + \mathbf{P}^2)\psi(\text{slow}).$$

For the free theory, or more generally when the operators in the iteration commute, the following two branches are admitted by the quadratic equation:

$$E = -m \pm \sqrt{m^2 + \mathbf{P}^2}.$$

The root near zero is selected by the slow positive-energy assumption:

$$\begin{aligned}
E\psi(\text{slow}) &= \left(-m + \sqrt{m^2 + \mathbf{P}^2}\right)\psi(\text{slow}) \\
&= \left(\frac{\mathbf{P}^2}{2m} - \frac{\mathbf{P}^4}{8m^3} + O(\mathbf{P}^6/m^5)\right)\psi(\text{slow}).
\end{aligned}$$

The other root begins at $-2m$ after the rest phase is removed and is incompatible with the assumed slowly varying positive-energy branch. For noncommuting or time-dependent potentials, commutators are also contained in the higher-order terms; the leading low-energy relation remains

$$E\psi(\text{slow}) \approx \frac{\mathbf{P}^2}{2m}\psi(\text{slow}).$$

The Schrödinger equation is obtained by solving for $i\partial_t\psi(\text{slow})$:

$$i\partial_t\psi(\text{slow}) \approx \left(\frac{(-i\partial_r - q\mathbf{A})^2}{2m} + qV\right)\psi(\text{slow}).$$

If $\mathbf{A} = 0$, this reduces to

$$i\partial_t\psi(\text{slow}) \approx \left(-\frac{\partial_{\mathbf{r}}^2}{2m} + qV\right)\psi(\text{slow}).$$

The scalar low-energy limit is therefore already encoded in the relativistic Klein–Gordon theory.

6.8.14 Low-Energy Spin- $\frac{1}{2}$ and Pauli Limit

The fixed-column two-block field from Eq. (195) is used. Its two block equations are

$$\begin{aligned}(i\partial - q\Phi^*)\Psi_2 - m\Psi_1 &= 0, \\ (i\partial^* - q\Phi)\Psi_1 - m\Psi_2 &= 0.\end{aligned}$$

With the covariant operators from Eqs. (176) and (177),

$$E = i\partial_t - qV, \quad \mathbf{P} = -i\partial_{\mathbf{r}} - q\mathbf{A},$$

the system is

$$\begin{aligned}(E - \mathbf{P} \cdot \boldsymbol{\sigma})\Psi_2 - m\Psi_1 &= 0, \\ (E + \mathbf{P} \cdot \boldsymbol{\sigma})\Psi_1 - m\Psi_2 &= 0.\end{aligned}$$

The slow fields defined in Eq. (216) are related by

$$\Psi_n = \exp(-imt)\Psi_n(\text{slow}).$$

Since

$$E\Psi_n = \exp(-imt)(E + m)\Psi_n(\text{slow}),$$

the system is reduced to

$$\begin{aligned}(E + m - \mathbf{P} \cdot \boldsymbol{\sigma})\Psi_2(\text{slow}) - m\Psi_1(\text{slow}) &= 0, \\ (E + m + \mathbf{P} \cdot \boldsymbol{\sigma})\Psi_1(\text{slow}) - m\Psi_2(\text{slow}) &= 0.\end{aligned}$$

Neither original block is small at rest. With the energy-basis combinations from Eq. (217),

$$\begin{aligned}\phi(\text{large}) &= \frac{\Psi_1(\text{slow}) + \Psi_2(\text{slow})}{\sqrt{2}}, \\ \phi(\text{small}) &= \frac{\Psi_2(\text{slow}) - \Psi_1(\text{slow})}{\sqrt{2}}.\end{aligned}$$

The exact system is obtained by addition and subtraction of the reduced equations:

$$\begin{aligned}E\phi(\text{large}) &= (\mathbf{P} \cdot \boldsymbol{\sigma})\phi(\text{small}), \\ (E + 2m)\phi(\text{small}) &= (\mathbf{P} \cdot \boldsymbol{\sigma})\phi(\text{large}).\end{aligned}$$

Thus $\phi(\text{large})$ is the large positive-energy field, and $\phi(\text{small})$ is suppressed by one power of $1/m$. Formally,

$$\begin{aligned}\phi(\text{small}) &= (E + 2m)^{-1}(\mathbf{P} \cdot \boldsymbol{\sigma})\phi(\text{large}), \\ E\phi(\text{large}) &= (\mathbf{P} \cdot \boldsymbol{\sigma})(E + 2m)^{-1}(\mathbf{P} \cdot \boldsymbol{\sigma})\phi(\text{large}) \\ &= \frac{(\mathbf{P} \cdot \boldsymbol{\sigma})^2}{2m}\phi(\text{large}) + O(m^{-2}),\end{aligned}$$

where energies small compared with $2m$ and external fields smooth on the Compton scale are assumed.

The magnetic term is obtained directly from the covariant-momentum commutator:

$$\begin{aligned}\text{com}(P_m, P_n) &= iq(\partial_m A_n - \partial_n A_m), \\ \mathbf{P} \times \mathbf{P} &= iq\mathbf{B}, \\ (\mathbf{P} \cdot \boldsymbol{\sigma})^2 &= \mathbf{P}^2 + i(\mathbf{P} \times \mathbf{P}) \cdot \boldsymbol{\sigma} \\ &= \mathbf{P}^2 - q\mathbf{B} \cdot \boldsymbol{\sigma}.\end{aligned}$$

Consequently,

$$E\phi(\text{large}) = \frac{\mathbf{P}^2 - q\mathbf{B} \cdot \boldsymbol{\sigma}}{2m}\phi(\text{large}) + O(m^{-2}),$$

and the Pauli equation is

$$\begin{aligned}i\partial_t\phi(\text{large}) &= \left(\frac{(-i\partial_{\mathbf{r}} - q\mathbf{A})^2}{2m} + qV\right)\phi(\text{large}) \\ &\quad - \frac{q}{2m}(\mathbf{B} \cdot \boldsymbol{\sigma})\phi(\text{large}) + O(m^{-2}).\end{aligned}$$

The spin–orbit, Darwin, and time-dependent corrections at the next order are generated by electric-field commutators through the noncommutation of E and $\mathbf{P}$. They must be retained consistently rather than discarded from the second-order equation for one original block. The leading spin coupling is therefore a controlled consequence of the Dirac system [9, 17].

The Pauli current in Eq. (225) is obtained from the same controlled reduction. The selected columns of $\phi(\text{large})$ and $\phi(\text{small})$ are denoted by $\phi(\text{large})$ and $\phi(\text{small})$, respectively. Then

$$\begin{aligned}\mathbf{j}(\Psi) &= \phi(\text{large})^{cT}\boldsymbol{\sigma}\phi(\text{small}) \\ &\quad + \phi(\text{small})^{cT}\boldsymbol{\sigma}\phi(\text{large}), \\ \phi(\text{small}) &= \frac{\boldsymbol{\sigma} \cdot \mathbf{P}}{2m}\phi(\text{large}) + O(m^{-2}).\end{aligned}$$

The product identity $\sigma_m\sigma_n = \delta_{mn} + i\epsilon_{mnk}\sigma_k$ is used in the written order, and the result is separated into the following vector expression:

$$\begin{aligned}\mathbf{j}(\phi(\text{large})) &= \frac{1}{2m}\left(\phi(\text{large})^{cT}(\mathbf{P}\phi(\text{large}))\right. \\ &\quad \left.+ (\mathbf{P}\phi(\text{large}))^{cT}\phi(\text{large})\right) \\ &\quad + \frac{\partial_{\mathbf{r}} \times (\phi(\text{large})^{cT}\boldsymbol{\sigma}\phi(\text{large}))}{2m} + O(m^{-2}).\end{aligned}$$

In the first numerator, the three entries are obtained by replacing $\mathbf{P}$ in turn by P_1, P_2, P_3 ; thus this compact line is exactly the three component equations. The first term is the covariant-momentum flow and the second is the magnetization curl. The latter has zero divergence, so agreement of both the current and the continuity equation with the standard Pauli limit is obtained.

6.8.15 Spin Up and Spin Down

Where $\mathbf{B} \neq \mathbf{0}$, its direction is used as the spin axis:

$$\mathbf{u} = \frac{\mathbf{B}}{\|\mathbf{B}\|},$$

The projector pair from Eq. (65) is then used:

$$\Pi(\pm) = \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \quad (\mathbf{u} \cdot \boldsymbol{\sigma})\Pi(\pm) = \pm\Pi(\pm).$$

The Pauli state is decomposed by

$$\begin{aligned} \phi(\text{large}, \pm) &= \Pi(\pm)\phi(\text{large}), \\ (\mathbf{u} \cdot \boldsymbol{\sigma})\phi(\text{large}, \pm) &= \pm\phi(\text{large}, \pm), \\ \phi(\text{large}) &= \phi(\text{large}, +) + \phi(\text{large}, -), \end{aligned}$$

so $\phi(\text{large}, +)$ and $\phi(\text{large}, -)$ are the spin-up and spin-down parts along the field axis. The fixed-column condition $\phi(\text{large}) = \phi(\text{large})\Pi(\text{column})$ is preserved under these left projectors.

At leading order, the Pauli equation is written as

$$\begin{aligned} \mathbf{P} &= -i\partial_{\mathbf{r}} - q\mathbf{A}, \\ i\partial_t\phi(\text{large}) &= \left(\frac{\mathbf{P}^2}{2m} + qV \right) \phi(\text{large}) \\ &\quad - \frac{q}{2m} \|\mathbf{B}\| (\mathbf{u} \cdot \boldsymbol{\sigma}) \phi(\text{large}). \end{aligned}$$

The following result is obtained by applying $\Pi(\pm)$ to the time evolution:

$$\begin{aligned} i\partial_t\phi(\text{large}, \pm) &= \left(\frac{\mathbf{P}^2}{2m} + qV - \frac{q}{2m} \|\mathbf{B}\| \mathbf{u} \cdot \boldsymbol{\sigma} \right) \phi(\text{large}, \pm) \\ &\quad + \left(i\partial_t\Pi(\pm) + \frac{\text{com}(\Pi(\pm), \mathbf{P}^2)}{2m} \right) \phi(\text{large}). \end{aligned}$$

Here qV and the local field-axis term commute pointwise with $\Pi(\pm)$; spatial variation is incorporated through the displayed kinetic commutator. The two local spin sectors are coupled by this term together with $\partial_t\Pi(\pm)$. If $\mathbf{B}$ is uniform and static, then $\Pi(\pm)$ is constant and the coupling term is zero, so the two spin sectors are decoupled. In that case

$$(-\mathbf{B} \cdot \boldsymbol{\sigma})\phi(\text{large}, \pm) = \mp \|\mathbf{B}\| \phi(\text{large}, \pm),$$

and the Zeeman effect is represented by

$$i\partial_t\phi(\text{large}, \pm) = \left(\frac{\mathbf{P}^2 \mp q\|\mathbf{B}\|}{2m} + qV \right) \phi(\text{large}, \pm).$$

A preferred axis is therefore selected by the field direction, and opposite Zeeman shifts are acquired by the two projector parts.

6.8.16 Free Energy and Momentum-Axis Checks

The definitions in Eqs. (231)–(238) are expanded; no additional projector notation is needed. The following result is obtained by direct block multiplication:

$$\begin{aligned} \mathcal{H}^2 &= \begin{pmatrix} (\boldsymbol{\sigma} \cdot \mathbf{p})^2 + m^2 & -m\boldsymbol{\sigma} \cdot \mathbf{p} + m\boldsymbol{\sigma} \cdot \mathbf{p} \\ -m\boldsymbol{\sigma} \cdot \mathbf{p} + m\boldsymbol{\sigma} \cdot \mathbf{p} & (\boldsymbol{\sigma} \cdot \mathbf{p})^2 + m^2 \end{pmatrix} \\ &= (m^2 + \mathbf{p}^2)\mathbf{I} = E(\mathbf{p})^2\mathbf{I}. \end{aligned}$$

Here $(\boldsymbol{\sigma} \cdot \mathbf{p})^2 = \mathbf{p}^2$ is obtained directly from the sigma product. Because m , $\mathbf{p}$, and the sigma matrices are Hermitian, $\mathcal{H}^\dagger = \mathcal{H}$.

The calligraphic $\mathcal{H}$ names this block operator; the superscript H still means the Hermitian operation.

The following relations are obtained from the definition of $\Psi(e)$ and the square:

$$\begin{aligned} \mathcal{H}\Psi(e) &= \frac{1}{2} \left(\mathcal{H}\Psi + e \frac{\mathcal{H}^2\Psi}{E(\mathbf{p})} \right) \\ &= \frac{1}{2} (\mathcal{H}\Psi + eE(\mathbf{p})\Psi) = eE(\mathbf{p})\Psi(e), \\ \Psi(+) + \Psi(-) &= \Psi. \end{aligned}$$

Thus the two displayed fields really are the two energy-sign parts. Since $\mathcal{H}$ is Hermitian, zero Hermitian pairing is obtained between parts with different real eigenvalues.

For $\mathbf{p} \neq \mathbf{0}$, the following relation is obtained from Eq. (238):

$$(\boldsymbol{\sigma} \cdot \mathbf{p})\Pi(s) = s\|\mathbf{p}\|\Pi(s).$$

The projector commutes with $\boldsymbol{\sigma} \cdot \mathbf{p}$, and its left action commutes with the real mass entries of $\mathcal{H}$. Consequently

$$\begin{aligned} \mathcal{H}\Psi(e, s) &= eE(\mathbf{p})\Psi(e, s), \\ (\boldsymbol{\sigma} \cdot \mathbf{p})\Psi_n(e, s) &= s\|\mathbf{p}\|\Psi_n(e, s). \end{aligned}$$

The following especially transparent check is now obtained by writing the Hamiltonian equation one block at a time:

$$\begin{aligned} m\Psi_2(e, s) &= (eE(\mathbf{p}) + s\|\mathbf{p}\|)\Psi_1(e, s), \\ m\Psi_1(e, s) &= (eE(\mathbf{p}) - s\|\mathbf{p}\|)\Psi_2(e, s). \end{aligned} \quad (255)$$

The following relation is recovered by multiplication of these two scalar factors:

$$m^2 = (eE(\mathbf{p}) + s\|\mathbf{p}\|)(eE(\mathbf{p}) - s\|\mathbf{p}\|) = E(\mathbf{p})^2 - \mathbf{p}^2.$$

Within one fixed (e, s) sector, the selected columns are represented locally by $\{a; b\}$ for $\Psi_1(e, s)$ and $\{c; d\}$ for $\Psi_2(e, s)$. The momentum-axis equation is then exactly

$$\begin{aligned} p_3a + (p_1 - ip_2)b &= s\|\mathbf{p}\|a, & (p_1 + ip_2)a - p_3b &= s\|\mathbf{p}\|b, \\ p_3c + (p_1 - ip_2)d &= s\|\mathbf{p}\|c, & (p_1 + ip_2)c - p_3d &= s\|\mathbf{p}\|d. \end{aligned}$$

The following relations are obtained component by component from Eq. (255):

$$\begin{aligned} mc &= (eE(\mathbf{p}) + s\|\mathbf{p}\|)a, & md &= (eE(\mathbf{p}) + s\|\mathbf{p}\|)b, \\ ma &= (eE(\mathbf{p}) - s\|\mathbf{p}\|)c, & mb &= (eE(\mathbf{p}) - s\|\mathbf{p}\|)d. \end{aligned}$$

These are the usual free Dirac energy and momentum-axis spin relations, written entirely with the present paravector conventions. The physical current is also verified directly:

$$\begin{aligned} \frac{\mathbf{u} \cdot \mathbf{j}(\Psi(e, s))}{\rho(\Psi(e, s))} &= s \frac{|c|^2 + |d|^2 - |a|^2 - |b|^2}{|a|^2 + |b|^2 + |c|^2 + |d|^2} = \frac{e\|\mathbf{p}\|}{E(\mathbf{p})}, \\ \frac{\mathbf{j}(\Psi(e, s))}{\rho(\Psi(e, s))} &= \frac{e\mathbf{p}}{E(\mathbf{p})}. \end{aligned}$$

Thus the current divided by the density is the expected relativistic velocity for each energy-sign mode.

Every displayed operation is applied on the left. Therefore the fixed-column condition is preserved directly:

$$\begin{aligned} & (\Pi(s)\Psi_n(e))\Pi(\text{column}) \\ &= \Pi(s)(\Psi_n(e)\Pi(\text{column})) \\ &= \Pi(s)\Psi_n(e). \end{aligned}$$

No second matrix column is introduced. At $\mathbf{p} = 0$ the momentum direction, and hence helicity, is undefined; any separately chosen spin axis may still be used.

7 Conclusions

The algebra used here is the real Pauli algebra, written as one complex scalar plus one complex three-vector. Its economy is supplied by a structural fact rather than by a new physical postulate: the complex unit is supplied by the central pseudoscalar, dot and cross products are combined by the sigma product, and the determinant is obtained by multiplication by the adjugate. On real event paravectors, that determinant is the Minkowski interval.

Within this framework, spatial rotations, Lorentz boosts, four-velocity and four-momentum, Maxwell's equations, the Lorentz force, scalar relativistic waves, the Dirac factorization, and the leading Pauli equation are derived in one notation. The distinctions between object types are essential. Events are transformed by the determinant-preserving congruence derived from the general paravector law, the electromagnetic field is transformed by similarity, and one Dirac field is carried by a fixed projected column space. Once those distinctions are enforced, agreement between the compact notation and the standard component theories is obtained.

Several complementary structures are also exposed by the reformulation. The four Maxwell equations are obtained from the real and imaginary parts of ∂F ; the conjugate spin couplings F and F^* are obtained from the two ordered momentum products; and elementary functions, Pauli spin sectors, and free helicity sectors are organized by the same axis projectors. These are connections among established equations, not evidence for an additional interaction.

In particular, no independent measured components are added by the paravector wave carriers. Each Dirac or Pauli block is reduced by the fixed-column condition to one ordinary two-entry state, its opposite-order bilinear is the familiar rank-one density matrix, and the two squared field terms are the familiar two Dirac spin couplings. Other nonscalar sandwich channels remain algebraic until a physical meaning is assigned by a separately derived observable.

7.1 Scope of the Established Results

The results in this paper are established under the assumptions of flat Minkowski space, smooth fields, and domains on which the displayed differential operators and square roots are defined. The electromagnetic discussion is classical, and the wave-equation discussion is first-quantized. An antiparticle interpretation of a

negative-frequency solution is obtained only after second quantization. Likewise, $FF^H/2$ is the energy-density and flux paravector seen in one frame; the full electromagnetic stress-energy object is the linear function $T(a) = FaF^H/2$ on real paravectors a .

The displayed components are changed by a nonlinear coordinate map or algebraic change of basis in flat space, but curvature is not thereby created. Additional geometric structure—a position-dependent frame, a compatible connection, curvature, and covariant field equations—would be required for a genuine extension to general relativity. Complex space-time, dark-sector, and cosmological claims are outside what has been derived here.

These and other uncompleted directions are recorded in the following chapter without being treated as results.

8 Future Work

The developed results constitute a flat-space reformulation of established physics. Extensions not yet constructed are separated from points requiring greater mathematical precision. A proposed extension is marked **TODO**. A gap requiring resolution before physical claims can be supported is marked **FIXME**. No premise for the established results is taken from this section.

8.1 TODO: Local Frames and Gravitation

One fixed sigma basis is used throughout space and time in the present calculations. A gravitational extension would instead need a position-dependent local basis and a rule for comparing algebra elements defined at neighboring positions. That comparison rule would have to distinguish an ordinary change of description from a physical gravitational field. In particular, a basis written everywhere as a single transform of a fixed basis is locally only a change of description; curvature is not produced by it alone.

A complete construction would therefore require an independently specified connection, a definition of curvature as the failure of successive local transports to commute, and field equations by which that curvature is determined from physical sources. The interval, freely falling motion, Maxwell equation, Dirac system, and energy-momentum source map would then all have to be rebuilt with the same local transport rule. Agreement with the flat formulas in the limit of a constant basis would be a required check, not an additional assumption.

A moving-basis extension could be formed by combining translational motion with rotation-rate and boost-rate cross-position terms, interpreting the result as a velocity, multiplying it by mass to obtain energy and momentum, and constructing a transformed interval. Those physical identifications are not justified by the moving-basis generator derived in Sec. 1.11 alone. In particular, a similarity-transported basis is not the event congruence used for a Lorentz transformation, and a complex velocity-like expression is generally produced by the proposed boost-rate term. For such a construction to be valid, a real event law, an observer and its measured time and distance, and a covariant derivative based on the local connection must first be defined. Real measured velocity, energy, momentum, and interval

must then be derived from those definitions, and the inertial formulas must be recovered when the generator is constant or vanishes.

Space-time curvature is also not yet supplied by a plane-curve analogy. A numerator obtained by differentiating a coordinate quotient is only one piece of a parameter-dependent curve formula. Before it can be used here, the tangent normalization, parameter choice, transport rule, and multidirectional curvature test must all be specified. Curvature cannot be inferred from that numerator alone.

8.2 TODO: Internal Gauge Symmetries

One phase freedom and one potential paravector are used in the electromagnetic coupling. Other interactions involve several coupled internal field components whose transformations need not commute. It remains to be determined whether those transformations can be added while keeping the space-time product and the internal action conceptually separate.

The allowed internal field values, their transformation rule, the corresponding potentials, and the conserved source quantities would need to be defined before any field equation is written. The products applied to space-time paravectors and those applied only to the internal field components would also have to be shown explicitly. Merely replacing the electromagnetic potential by a larger array would not establish a consistent theory.

8.3 TODO: Many-Particle and Quantized Fields

Classical fields or first-quantized single-particle wave mechanics are described by the wave equations in this manuscript. In particular, a negative-frequency solution is not by itself an antiparticle. Field operators, particle creation and removal, charge conjugation, and a vacuum state must be introduced in a fuller theory, and the particle and antiparticle interpretation must be derived from that structure.

The combination of several one-particle carrier spaces, imposition of identical-particle exchange, representation of entangled and reduced states, and extraction of local observables must be specified in the same development. The fixed projected column space used for one Dirac field is a useful starting point, but these many-particle constructions are not supplied by it alone.

The indefinite Klein–Gordon pairing and the absence of a positive local one-particle position density are also not repaired by a change of algebraic packaging. In a quantized-field treatment, the origin of its positive state-space norm, charge operator, particles, and antiparticles must be shown, and the one-particle formulas must be recovered in their proper regime.

8.4 TODO: Media and Constitutive Maps

Vacuum fields and prescribed sources are treated in the electromagnetic section. A constitutive map is required for applications to material media: a locally defined rule by which the electromagnetic field is converted into the polarization, magnetization, current, or other response of the material. Anisotropy, dispersion, conductivity, and dissipation generally make that rule depend on direction, frequency, field history, or more than one input.

The notation-light approach should be tested first on simple linear media and then on direction-dependent and time-dependent responses. The distinction between a field paravector and an operator acting on that field must be preserved by each proposed map, and its energy balance must be derived rather than assumed. Material laws that fit naturally into one paravector product will thereby be distinguished from those requiring a separately defined linear map.

8.5 FIXME: Operator Foundations

Differential, inverse, and square-root operators are used in several factorizations in the wave-mechanics sections. Their algebraic forms are correct on the smooth test fields stated in the derivations, but the allowed fields, boundary conditions, and behavior at infinity must be specified in a complete analytic treatment. The conditions under which integration by parts adds no boundary term, an inverse exists, and an energy operator has real measured values are determined by those choices.

A consistent modewise definition of the square-root frequency operator, including its branch and treatment of zero modes, is also required. Sufficient regularity for both orders of operation to exist is required by commutators containing derivatives. These conditions should eventually be stated once in a common operator-domain section and then cited wherever they are used in a factorization, commutator, or low-energy expansion.

Likewise, a formal expression such as the determinant of a block differential operator minus a spectral parameter is not defined merely because the paravector determinant is defined. A characteristic determinant may be used only after a Fourier-mode, finite-volume, or other finite-matrix reduction has been specified. At the operator level, the ordered factorization and minimal-polynomial identities used in the manuscript are the justified statements; additional analytic construction is required for a spectral determinant.

For scalar relativistic waves, the positive-energy solution space for each permitted stationary background must be constructed by a rigorous treatment, and the measurement operators under which it is preserved must be determined. General time-dependent backgrounds can mix the frequency sectors, and relativistic localization cannot be settled by declaring the local Klein–Gordon density to be a probability. For Dirac and Pauli carriers, the same treatment must specify the common domains on which the displayed H -adjoints, block operators, spectral projectors, and expectation values are genuinely Hermitian.

8.6 FIXME: Uncertainty Relations

A spin uncertainty bound cannot be inferred directly from the axis commutator and a determinant inequality. Such a calculation is not a state-dependent variance inequality and is therefore not established as a result. The normalized projected-wave bracket in Eq. (209) must be used as the starting point, centered Hermitian operators and their variances must be defined, and a common domain on which both operator orders and their adjoints exist must be required. The usual commutator bound, and the stronger form that also retains the centered anticommutator, should then

be proved in the manuscript's $\text{com}()$ and $\text{acom}()$ notation. Axis observables should be checked as a worked special case rather than treated as the proof.

8.7 TODO: Higher-Order Pauli Reduction

The leading kinetic and magnetic spin terms are proved by the present Dirac reduction. In a controlled next-order expansion, the electric spin-orbit term, the contact term from spatially varying electric fields, and the corrections required for time-dependent potentials should be derived with the same sign and charge conventions. The transformed density and current must be expanded to the same order as the Hamiltonian; retaining higher-order energy terms while leaving the measurement map at leading order would be inconsistent. These terms are established consequences of the Dirac equation, but their derivation in this notation remains future work.

8.8 TODO: Classify Companion Components

Paired real and imaginary components and paired scalar and vector components are repeatedly produced by the algebra. Some of these are familiar physical laws: the magnetic-divergence and induction equations, for example, are encoded by the imaginary parts of the Maxwell equation. Other components, such as the imaginary part of the force product, are valid algebraic companions without a separate physical interpretation established here.

Two cases are already identified with established physics: a projected block's opposite-order product is the familiar rank-one two-state density matrix, and the F and F^* residuals in the ordered products are the familiar two Dirac spin couplings. For the remaining nonscalar channels of a generic $\psi^H \mathcal{O} \psi$ sandwich, a separate current, polarization, or measurement derivation is still required before physical meaning is assigned.

The following exact identity may also be studied:

$$\square(F^2) = \partial^* \partial(F^2).$$

Because F^2 is a complex scalar, this equality is an exact use of the defined wave operator, but it is not an independent field equation. Any useful source expansion must be derived from Maxwell's equation with the ordered product rule of Eq. (144). Future work may classify the resulting scalar identity as an invariant balance or a redundant differential consequence; it must not be presented as new dynamics without such a derivation.

Each companion formula should be placed by a systematic classification into one of three categories: an alternate form of an established law, redundant information implied by another displayed equation, or an algebraic channel whose physical meaning is presently unassigned. The involutions, order functions, projectors, and field bilinears should all be included in that classification. No unassigned component should be advertised as a new interaction merely because it is made available by the algebra. Automorphisms and projector, factorization, exponential, and logarithmic constructions that are intrinsic to the algebra should be distinguished from those dependent on the chosen matrix realization.

8.9 TODO: Complex Space-Time and Imaginary Coordinates

A complex paravector is already permitted by the generic algebra: $Z = X + iY$, with X and Y real paravectors. Treating such an element as a physical event, rather than as an algebraic field value, would be a new hypothesis. The following self-contained consistency identity is obtained from the existing determinant and adjugate definitions:

$$\det(Z) = \det(X) - \det(Y) + i(X \text{adj}(Y) + Y \text{adj}(X)).$$

The quantity in parentheses is a scalar, but the determinant of Z is generally complex. No interval interpretation is fixed by this identity. In a proposed complex-event theory, it would have to be stated which, if any, real quantity is measured as an interval and why that choice is preserved by the allowed transformations.

The scalar part of Y could be described as an imaginary-time coordinate, and its vector part could be described as imaginary space. Those descriptions alone carry no physical content. A complex space-time metric would require a specified real measurement map, a nondegenerate interval, and a transformation law that preserves that interval. Taking the real part, magnitude, or positive norm of a complex determinant would produce different geometries and therefore cannot be selected by terminology alone. The ordinary real interval would have to be recovered when the imaginary part vanishes.

Only after that choice is justified could proper time, causal order, velocity, momentum, or force be defined for a complex event. It would also have to be explained why real events and the usual light cone are recovered by ordinary experiments to their observed accuracy. Complex mass, hidden trajectories, or new force channels should not be introduced before these more basic questions are resolved.

Faster-than-light propagation is not implied by complex coordinates. A propagation speed can be assigned only after field equations, the surfaces along which disturbances propagate, observers, and measured time and distance have been defined. If the measured projection remains the real paravector X , the ordinary light cone may remain unchanged. If Y changes the propagation characteristics, the resulting causal cone and the possibility of controllable signaling would have to be derived explicitly.

Mixed terms in $\det(Z)$ show an algebraic relationship between the real and imaginary parts, but they do not establish a weak physical interaction. Such an interaction would require a dynamical equation and a source law. Only then could empirical parameters be identified. Possible parameters could include a real-imaginary mixing strength, a mass or length scale suppressing the imaginary sector, and a relative propagation or decay scale. None of these constants is predicted by the algebra alone; each would have to be connected to a definite observable and constrained experimentally.

The algebraic complexification cannot by itself be falsified because it is a mathematical extension. A proposed physical interpretation could be rejected if measured intervals or energies became unavoidably complex, probabilities failed to remain

real and nonnegative, time evolution failed to preserve normalization, source-free solutions grew without bound, or predicted departures from the observed light cone and ordinary Lorentz behavior exceeded experimental limits. It remains to be determined whether a nontrivial complex-event dynamics can pass these basic tests or whether every consistent observable sector reduces to the real-paravector theory already developed.

8.10 FIXME: Dark-Sector and Cosmological Interpretations

No derivation of vacuum energy, accelerated cosmic expansion, missing gravitational mass, or any other dark-sector phenomenon is established in this manuscript. A gravitational, clustering, or observable-energy role is not implied by the existence of imaginary algebra components. Such interpretations remain contingent on tests that would have to be passed by a future model.

A consistent local gravitational equation and a conserved physical source must first be supplied by any such model. Ordinary local gravity must then be recovered while quantitative predictions are made for cosmic expansion, gravitational lensing, structure growth, and the motion of matter. A nearly constant large-scale contribution and a clustering matter-like contribution are physically different requirements and should not be inferred from the same algebraic term without separate derivations.

8.11 TODO: Limits of the Notation-Light Form

The compact notation is most effective when the relevant scalar and vector data are carried by one algebra element. In some extensions, maps with several independent inputs, information that is valid only on overlapping local regions, or global conditions that cannot be recovered from one local formula are instead required. That mathematical structure is not removed by avoidance of indices; only its recorded form is changed.

The conditions under which a paravector is sufficient, a linear map on paravectors is required, or several-input or global data must be introduced should be stated explicitly. Any new notation should be justified by a construction that cannot be expressed clearly with the existing functions and operators. The raw-text character of the framework is thereby preserved while notational economy is prevented from being mistaken for a reduction in physical content.

A Verification and Formula Status

Exact algebraic identities are separated in Table 48 from physical statements for which a stated representation or approximation is required. The accompanying symbolic report is generated by `tools/verify_sta_results.py`.

Table 48. Status of the principal constructions.

Statement	Standard correspondence	Status	Qualification
Eqs. (6) and (37)	Real Pauli algebra and three-dimensional real Clifford algebra [12, 14]	verified	Associative scalar–vector form with eight real degrees of freedom.
Eqs. (80)–(84)	Similarity transport and the standard double-sided Lorentz action	verified	Generic algebra elements are transported by similarity; real events are transported by congruence. The adjugate equals the inverse only after determinant-one normalization.
Eqs. (47) and (53)	Positive coefficient norm and ordinary matrix determinant	verified	The first is nonnegative and real; the second is generally complex and is the exact inverse divisor.
Eqs. (88) and (89)	Spatial rotations and Lorentz boosts [4, 5]	verified	For spatial rotors, the event congruence is reduced to similarity; for event boosts, the congruence form is retained.
Eq. (123)	Minkowski interval [6]	verified	Valid on real event paravectors with signature $(+, -, -, -)$.
Eqs. (151) and (154)	Maxwell electrodynamics [2, 15]	verified	Signs are fixed by $F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$ and the passive-boost convention.
Eq. (162)	Relativistic Lorentz force [15]	verified	The coordinate-time power and force equations are obtained by division by γ .
Eq. (164)	Electromagnetic energy density and flux	verified	T is the chosen-observer paravector; the full stress-energy object is the linear map on real paravectors.
Eqs. (172) and (178)	Klein–Gordon equation [7, 8]	verified	The frequency split is defined on the stated spatial operator domain.
Eqs. (187)–(190)	Conserved Klein–Gordon solution pairing	verified	Its integral is positive on the free positive-frequency sector; its local density is not a general position probability.
Eqs. (195) and (191)	Standard Dirac equation in two coupled two-entry blocks [10, 16]	verified	One Dirac field is represented by one fixed projected matrix column; two copies are represented by unrestricted algebra blocks.
Eqs. (197)–(212)	Four Dirac components, conserved current, and Born expectation rule	verified	The positive norm is conserved only for the complete two-block field; normalized outcome probabilities are supplied by Hermitian projectors.
Eq. (201)	Squared minimally coupled Dirac system	verified	The terms $+iqF^*$ and $+iqF$ are carried by the upper and lower blocks, respectively.
Eqs. (219)–(225)	Leading Pauli reduction [9, 17]	verified	The expansion is in inverse mass after the large/small energy-basis split; convection and magnetization terms are contained in the inherited current.
Eq. (238)	Free energy and helicity projectors	verified	Commutation of joint sectors is obtained for helicity (or any spin axis at rest), but not for an arbitrary fixed axis.

Python Verification of Main Algebraic and Operator Identities

The identities below were checked symbolically in Python using SymPy and the standard Pauli-matrix realization of the sigma basis. Algebraic identities were reduced directly in matrix form, while differential identities were evaluated on generic test functions $\psi(t, x, y, z)$ with symbolic potentials and field components. The verification script is `tools/verify_sta_results.py`.

Check	Status	Detail
Eqs. (2)–(3)	verified	The identity $\sigma_n \sigma_m = \delta_{nm} + \epsilon_{nmk} i \sigma_k$ is satisfied by all nine basis products in the Pauli realization.
Eq. (36)	verified	The product $(\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma})$ is reduced to $\mathbf{A} \cdot \mathbf{B} + i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}$ for symbolic components A_n, B_n .
Eqs. (37)–(53)	verified	Correct reduction is obtained for the complex paravector product, the adjugate identity $Z \text{adj}(Z) = \det(Z) = S^2 - \mathbf{V}^2$, and the Euclidean scalar part $\text{scalar}(ZZ^H) = S ^2 + \ \mathbf{V}\ ^2$.
Eqs. (50) and (116)	verified	For an arbitrary complex two-by-two matrix, intrinsic H is represented by the matrix conjugate transpose, intrinsic star by $\mathbf{T}^c \mathbf{T}^{-1}$, and star- H by $\text{adj}(\mathbf{M}) = \mathbf{T} \mathbf{M}^T \mathbf{T}^{-1}$. The result $\det(\mathbf{M}) \mathbf{I}$ is obtained from both adjugate products, and both $\text{adj}(\mathbf{M})^*$ and $\text{adj}(\mathbf{M}^*)$ are reduced to $\mathbf{M}^H = \mathbf{M}^{cT}$.
Eqs. (80)–(82)	verified	For generic complex $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$ and $T = f + \mathbf{g} \cdot \boldsymbol{\sigma}$, the stated scalar-preserving vector formula for $T^{-1} Z T$ is reproduced after multiplication through by $\det(T) = f^2 - \mathbf{g}^2$. The full complex-paravector similarity calculation is thereby checked.
Eqs. (89) and (154)	verified	For a boost along the first axis, standard time/longitudinal mixing is produced by the event congruence $X' = L^* X L^*$; the transverse coordinates are left real and unchanged, and $\det(X)$ is preserved. The event intertwiner $L X' = X L^*$ is also satisfied. That event boost is not produced by the rejected similarity $L^* X L$. Standard electric-magnetic mixing is produced by the field similarity $F' = L^* F L$, whereas a scalar part is introduced by the rejected double-star field sandwich.
Eqs. (65), (124), (128)	verified	The projector identities are satisfied by the idempotents $\Pi(\pm)$, the normalization $\det(U) = 1$ is obtained, and $E^2 - \mathbf{P}^2 = m^2$ is implied by $mU = E + \mathbf{P} \cdot \boldsymbol{\sigma}$.
Eqs. (149), (150), (151)	verified	The relations $S = \partial_t V + \partial_r \cdot \mathbf{A}$, $\mathbf{E} = -\partial_t \mathbf{A} - \partial_r V$, and $\mathbf{B} = \partial_r \times \mathbf{A}$ are obtained from the potential identity, while Gauss, Faraday, and Ampère-Maxwell component equations with the manuscript's signs are obtained from ∂F .
Eqs. (158) and (162)	verified	The component laws $V(\lambda) = V + \partial_t \lambda$ and $\mathbf{A}(\lambda) = \mathbf{A} - \partial_r \lambda$ are obtained from $\Phi(\lambda) = \Phi + \partial^* \lambda$. The expected power term $\gamma \mathbf{v} \cdot \mathbf{E}$ and force term $\gamma(\mathbf{E} + \mathbf{v} \times \mathbf{B})$ are obtained from the product FU .
Eqs. (179), (189), and (185)	verified	The expanded Klein-Gordon operator and the commutators $\text{com}(E, P_m) = iqE_m$ and $\text{com}(P_m, P_n) = iq(\partial_m A_n - \partial_n A_m)$ are reproduced on a generic test function. The mixed-current divergence is reduced to the difference of the two Klein-Gordon equations, and the ordered factorization $(E + \boldsymbol{\sigma} \cdot \mathbf{P})(E - \boldsymbol{\sigma} \cdot \mathbf{P}) = E^2 - \mathbf{P}^2 - iqF$ is obtained.
Eqs. (201), (220), and (229)	verified	The two squared equations are obtained by elimination of one Dirac component. After selection of the common overall sign in Eq. (201), $+iqF^*$ is carried by the upper component and $+iqF$ by the lower component. The exact Pauli identity $(\boldsymbol{\sigma} \cdot \mathbf{P})^2 = \mathbf{P}^2 - q \mathbf{B} \cdot \boldsymbol{\sigma}$ is satisfied on a test function. For uniform $\mathbf{B} = \ \mathbf{B}\ \mathbf{u}$, the relation $(\mathbf{B} \cdot \boldsymbol{\sigma}) \Pi(\pm) = \pm \ \mathbf{B}\ \Pi(\pm)$ is satisfied by the projectors.

Eqs. (191), (197), and (207)	verified	Exactly one two-entry column is retained by the fixed right projector. All four displayed Dirac component equations are reproduced by direct Pauli-matrix multiplication. The expressions $\rho(\Psi) = \begin{Bmatrix} a^* & b^* \end{Bmatrix} \begin{Bmatrix} a \\ b \end{Bmatrix} + \begin{Bmatrix} c^* & d^* \end{Bmatrix} \begin{Bmatrix} c \\ d \end{Bmatrix}$ and $\mathbf{j}(\Psi) = \begin{Bmatrix} c^* & d^* \end{Bmatrix} \boldsymbol{\sigma} \begin{Bmatrix} c \\ d \end{Bmatrix} - \begin{Bmatrix} a^* & b^* \end{Bmatrix} \boldsymbol{\sigma} \begin{Bmatrix} a \\ b \end{Bmatrix}$ are obtained from the opposite-order bilinears, while Hermiticity and rank one are obtained for the normalized one-block product.
Eq. (99)	verified	The identities $QQ^H = a^2 + \mathbf{b}^2$ and $Q^2 = a^2 - \mathbf{b}^2 - 2ia \mathbf{b} \cdot \boldsymbol{\sigma}$ are satisfied by the quaternion correspondence.

The literature comparison in Section A is complemented by these symbolic checks. Purely algebraic and operator-theoretic identities are thereby separated from statements whose interpretation is still determined by physical conventions.

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